Discrete Stochastic Processes

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Lecture 1: Introduction and probability review

1.1 Probability models and the real world

Probability is most useful when an experiment can be repeated under essentially the same conditions, as with games of chance. Even there, "essentially" hides real difficulties: a coin toss is determined by the initial velocity and orientation and by the surfaces involved, and successive tosses are linked through the coin and the person tossing it. Questions such as the probability of another catastrophic oil spill in the coming year are far harder to make precise.

Using probability therefore raises two separate problems.

1. *How to model a real-world problem.* Start with oversimplified models, use their mathematics to understand the real problem, and then add to and modify the models in stages. Usually no model is perfect: seek simplicity and distrust it.
2. *How to make a model free of hidden paradoxes.* Follow Kolmogorov's axioms (1933), which make probability theory a branch of measure theory. The axioms are needed to avoid paradoxes, but measure theory itself is not needed for the topics treated here.

1.2 Discrete stochastic processes

A stochastic process is a probability model whose sample points represent functions of time. It can often be viewed as a sequence of random variables evolving in time, or as a continuum of random variables, one for each real instant. A discrete stochastic process is one in which either time is discrete or the set of possible sample values is discrete; the precise boundary is unimportant. The processes studied are:

- *Counting processes*, in which each sample point is a sequence of arrival times. Special cases are Poisson processes and renewal processes.
- *Markov processes*, in which the future depends on the past only through the present state: finite-state Markov chains, countable-state Markov chains, and Markov processes with countable state spaces.
- *Random walks and martingales*,

together with mixtures of these that serve as standard models in communication systems and networks, computer systems, queueing, risk management, operations research, biology, control, and elsewhere.

1.3 The axioms of probability

Definition 1.1 [Probability model]. A probability model has three components: a sample space Ω, an arbitrary set of sample points; a collection of events, each a subset of Ω; and a probability measure assigning a number $\Pr\{A\} \in [0,1]$ to each event A. The events satisfy:

1. Ω is an event;
2. if $A_1, A_2, \ldots$ are events, then $\bigcup_{n=1}^{\infty} A_n$ is an event;
3. if A is an event, then its complement A^c is an event.

The probability measure satisfies:

1. $\Pr\{\Omega\} = 1$;
2. $\Pr\{A\} \geq 0$ for every event A;
3. if $A_1, A_2, \ldots$ are disjoint events, then

$$\Pr\left\{\bigcup_{n=1}^{\infty} A_n\right\} = \sum_{n=1}^{\infty} \Pr\{A_n\} = \lim_{m\to\infty} \sum_{n=1}^{m} \Pr\{A_n\}.$$

Not every subset need be an event. Usually each sample point is taken to be a singleton event; the remaining non-events are then weird, sometimes necessary, but usually ignorable. Several consequences follow at once.

- The empty set $\emptyset = \Omega^c$ is an event.
- If all sample points are singleton events, every finite or countable set is an event, being a finite or countable union of singletons.
- By De Morgan's law, $\left[\bigcup_n A_n\right]^c = \bigcap_n A_n^c$, so countable intersections of events are events. Hence all countable combinations of unions and intersections of events are events.
- $\Pr\{\emptyset\} = 0$, $\Pr\{A^c\} = 1 - \Pr\{A\}$, and $\Pr\{A\} \leq \Pr\{B\} \leq 1$ for events $A \subseteq B$.

Proposition 1.2 [Union bound]. *For a finite or countable collection of events,*

$$\Pr\left\{\bigcup_n A_n\right\} \leq \sum_n \Pr\{A_n\}.$$

Proof. For two events, write $A_1 \cup A_2 = A_1 \cup A_2 A_1^c$ as a disjoint union, where $A_2 A_1^c$ denotes $A_2 \cap A_1^c$. Then

$$\Pr\{A_1 \cup A_2\} = \Pr\{A_1\} + \Pr\{A_2 A_1^c\} \leq \Pr\{A_1\} + \Pr\{A_2\},$$

since $A_2 A_1^c \subseteq A_2$. In general, the sets $B_n = A_n \big(\bigcup_{m<n} A_m\big)^c$ are disjoint, have the same union as the A_n, and satisfy $B_n \subseteq A_n$; apply the third axiom. □

The axioms look routine and are ignored much of the time, but they are what justify infinite sums and limits. Random variables and expectations get most of the attention, yet the axioms make events and their probabilities the fundamental quantities.

1.4 Independent events and experiments

Definition 1.3 [Independent events]. Two events A_1 and A_2 are INDEPENDENT if $\Pr\{A_1 A_2\} = \Pr\{A_1\} \Pr\{A_2\}$.

Two probability models can be combined into one whose sample space is the Cartesian product $\Omega_1 \times \Omega_2$ and in which $\Pr\{AB\} = \Pr\{A\} \Pr\{B\}$ for every event A of the first model and B of the second. The original models are then INDEPENDENT in the combined model. If each model satisfies the axioms, so does the combined model, so complex models can be built from simpler ones.

1.5 Random variables and distribution functions

Definition 1.4 [Random variable]. A RANDOM VARIABLE (rv) X, or $X(\omega)$, is a function from Ω to $\mathbb{R}$ such that $\{\omega : X(\omega) \le a\}$ is an event for every $a \in \mathbb{R}$. Moreover, if $X_1, \ldots, X_n$ are rv's, then $\{\omega : X_1(\omega) \le a_1, \ldots, X_n(\omega) \le a_n\}$ is an event for all $a_1, \ldots, a_n \in \mathbb{R}$.

Every rv has a DISTRIBUTION FUNCTION $F_X(x) = \Pr\{X \le x\}$, a nondecreasing function rising from 0 at $-\infty$ to 1 at $+\infty$.

- If X takes only finitely or countably many values, it is DISCRETE and has a PROBABILITY MASS FUNCTION (PMF) $p_X(x) = \Pr\{X = x\}$.
- If $dF_X(x)/dx$ exists and is finite for all x, then X is CONTINUOUS with DENSITY $f_X(x) = dF_X(x)/dx$.
- If X has both discrete and continuous components, it is sometimes useful to view it as having a density with impulses.

The distribution function always exists. Since $\{X = x\} \subseteq \{X \le x\}$, at a jump of F_X at x the value $F_X(x)$ is the value at the top of the jump.

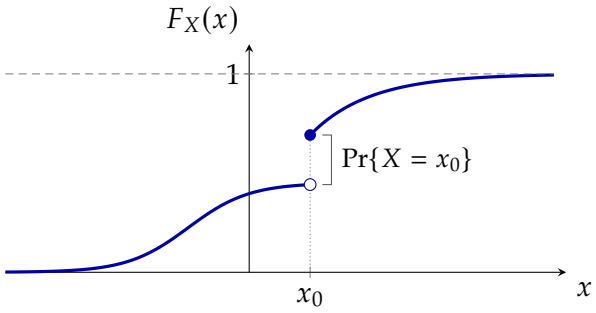


Figure 1: A schematic distribution function with a continuous part and a jump at x_0. The filled dot marks $F_X(x_0) = \Pr\{X \le x_0\}$, the top of the jump; the jump height is $\Pr\{X = x_0\}$.

Proposition 1.5 [Right continuity]. *Every distribution function is continuous from the right:*

$$\lim_{k\to\infty} F_X(x + 1/k) = F_X(x) \qquad \textit{for all } x.$$

Proof. Let $A_k = \{\omega : X(\omega) > x + 1/k\}$. Then $A_{k-1} \subseteq A_k$ for $k > 1$ and $\{\omega : X(\omega) > x\} = \bigcup_{k=1}^{\infty} A_k$. It suffices to show $\Pr\{\bigcup_k A_k\} = \lim_k \Pr\{A_k\}$, since then

$$\Pr\{X > x\} = \lim_{k\to\infty} \Pr\left\{X > x + \frac{1}{k}\right\},$$

and taking complements gives the claim. Let $B_1 = A_1$ and $B_k = A_k - A_{k-1}$ for $k > 1$. The B_k are disjoint, $\bigcup_k B_k = \bigcup_k A_k$, and $\bigcup_{m=1}^{k} B_m = A_k$. By the third axiom,

$$\Pr\left\{\bigcup_k A_k\right\} = \sum_{k=1}^{\infty} \Pr\{B_k\} = \lim_{k\to\infty} \sum_{m=1}^{k} \Pr\{B_m\} = \lim_{k\to\infty} \Pr\{A_k\}. \qquad \square$$

Source. Robert Gallager, MIT 6.262, *Discrete Stochastic Processes*, Lecture 1, Spring 2011. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 2: More review; the Bernoulli process

2.1 Expectations

The distribution function of a rv X often contains more detail than needed, and the expectation $\overline{X} = \mathbb{E}[X]$ is sometimes all that is required:

$$\mathbb{E}[X] = \sum_i a_i\, p_X(a_i) \qquad \text{for discrete } X,$$

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x)\, dx \qquad \text{for continuous } X,$$

$$\mathbb{E}[X] = \int_0^{\infty} F_X^c(x)\, dx \qquad \text{for arbitrary nonnegative } X,$$

$$\mathbb{E}[X] = -\int_{-\infty}^{0} F_X(x)\, dx + \int_0^{\infty} F_X^c(x)\, dx \qquad \text{for arbitrary } X,$$

where $F_X^c(x) = 1 - F_X(x) = \Pr\{X > x\}$ is the complementary distribution function. Almost as important is the standard deviation $\sigma_X = \sqrt{\mathbb{E}\big[(X - \overline{X})^2\big]}$.

To see why $\mathbb{E}[X] = \int_0^{\infty} F_X^c(x)\, dx$ for nonnegative X, take X discrete with values $0 \le a_1 < a_2 < \cdots$. The region under F_X^c splits into horizontal strips: strip i has height $p_X(a_i)$ and extends from 0 to a_i, so its area is $a_i p_X(a_i)$, and the total area is $\sum_i a_i p_X(a_i)$. If X has a density, the same argument applies to every Riemann sum for $\int x f_X(x)\, dx$, and thus to the limit. It is simpler and more fundamental to take $\int_0^{\infty} F_X^c(x)\, dx$ as the general definition of $\mathbb{E}[X]$ for nonnegative X; this is also useful in solving problems. For arbitrary X, apply this to the positive and negative parts: the negative part contributes $-\int_{-\infty}^{0} \Pr\{X \le x\}\, dx$.

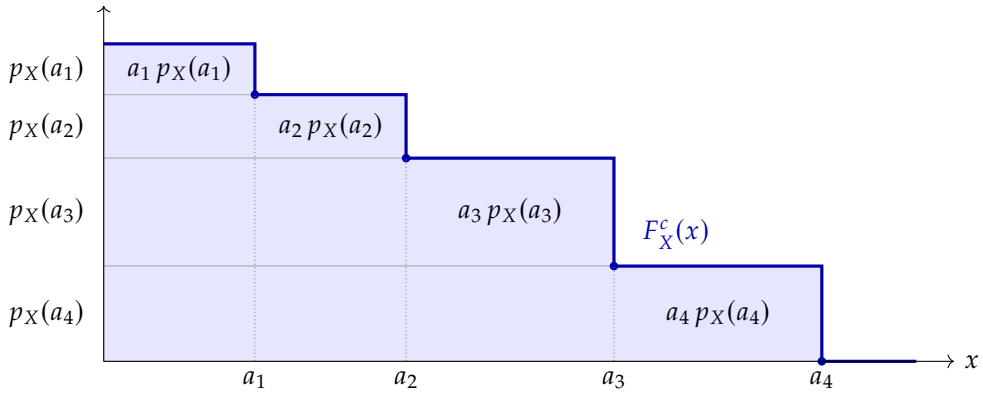


Figure 2: Schematic of $\mathbb{E}[X] = \int_0^{\infty} F_X^c(x)\, dx$ for a nonnegative discrete rv with four values. The area under the staircase F_X^c is the sum of the strips $a_i p_X(a_i)$.

2.2 Indicator random variables

Definition 2.1 [Indicator rv]. For every event A, the indicator rv $\mathbf{1}_A$ is defined by $\mathbf{1}_A(\omega) = 1$ for $\omega \in A$ and $\mathbf{1}_A(\omega) = 0$ otherwise.

The indicator is binary, with $p_{\mathbf{1}_A}(0) = 1 - \Pr\{A\}$ and $p_{\mathbf{1}_A}(1) = \Pr\{A\}$, so its distribution function steps from $1 - \Pr\{A\}$ at 0 up to 1 at 1. Hence

$$\mathbb{E}[\mathbf{1}_A] = \Pr\{A\}, \qquad \sigma_{\mathbf{1}_A} = \sqrt{\Pr\{A\}\big(1 - \Pr\{A\}\big)}.$$

Theorems about rv's can thus be applied to events.

2.3 Multiple random variables

A rv is not specified by its distribution function alone in any useful sense: the relationship between rv's matters. The joint distribution function of X and Y is

$$F_{XY}(x, y) = \Pr\big\{\{\omega : X(\omega) \le x\} \cap \{\omega : Y(\omega) \le y\}\big\}.$$

Definition 2.2 [Independent rv's]. The rv's $X_1, \ldots, X_n$ are independent if

$$F_{\vec{X}}(x_1, \ldots, x_n) = \prod_{m=1}^{n} F_{X_m}(x_m) \qquad \text{for all } x_1, \ldots, x_n.$$

The same product form holds for PMFs and densities. For discrete rv's, independence is more intuitive in terms of the conditional PMF $p_{X|Y}(x \mid y) = p_{XY}(x, y)/p_Y(y)$: X and Y are independent if $p_{X|Y}(x \mid y) = p_X(x)$ for all sample values x and y with $p_Y(y) > 0$. This essentially works for densities, although then $\Pr\{Y = y\} = 0$; it is not useful for distribution functions.

If $X_1, \ldots, X_n$ are independent, then every subset of them is independent: letting $x_m \to \infty$ in the product form removes X_m. The analogous statement fails for events if independence of $A_1, \ldots, A_n$ is defined only by $\Pr\{A_1 \cdots A_n\} = \prod_m \Pr\{A_m\}$; independence of events requires the product rule for every subcollection.

2.4 IID random variables and the extended model

Definition 2.3 [IID]. The rv's $X_1, \ldots, X_n$ are independent and identically distributed (IID) if $F_{\vec{X}}(x_1, \ldots, x_n) = \prod_{k=1}^{n} F_X(x_k)$ for all $x_1, \ldots, x_n$.

Given a probability model with sample space $\mathbb{R}$ and rv X, one can always form an extended model with sample space $\mathbb{R}^n$ in which $X_1, \ldots, X_n$ are IID copies of X, and let $n \to \infty$ to get a stochastic

process $X_1, X_2, \ldots$ of IID rv's. The object of study is the SAMPLE AVERAGE

$$\frac{S_n}{n} = \frac{X_1 + \cdots + X_n}{n}.$$

The laws of large numbers say that S_n/n essentially becomes deterministic as $n \to \infty$. If the extended model corresponds to repeated real-world experiments, S_n/n corresponds to the arithmetic average of the results; if $X = \mathbf{1}_A$, the sample average is the relative frequency of A.

A model can fail in two ways: the real-world trials may not be similar and isolated enough to match the IID extension, or the extension may be fine but the basic model wrong. Modeling uses science, symmetry, and analogy, and requires judgement. For example, two distinguishable dice (white and red) have 36 equiprobable outcomes (i, j), $1 \le i, j \le 6$. Two indistinguishable white dice have only 21 finest-grain outcomes, but these are not equiprobable and no one would use them as sample points; the right sample space is still the white/red one, with an "off-line" record of what can be distinguished. Neither the axioms nor experiment dictate this choice. The outcome of an experiment is a sample point, not a probability; testing a model needs many trials, modeled as one sample point of the extended model, and the laws of large numbers supply the test.

2.5 Laws of large numbers in pictures

Let $X_1, X_2, \ldots$ be IID with mean $\overline{X}$ and variance σ^2, and let $S_n = X_1 + \cdots + X_n$. Then

$$\mathbb{E}[S_n] = n\overline{X}, \quad \sigma^2_{S_n} = n\sigma^2; \qquad \mathbb{E}\left[\frac{S_n}{n}\right] = \overline{X}, \quad \sigma^2_{S_n/n} = \frac{\sigma^2}{n},$$

and $Z_n = (S_n - n\overline{X})/(\sqrt{n}\,\sigma)$ has mean 0 and variance 1.

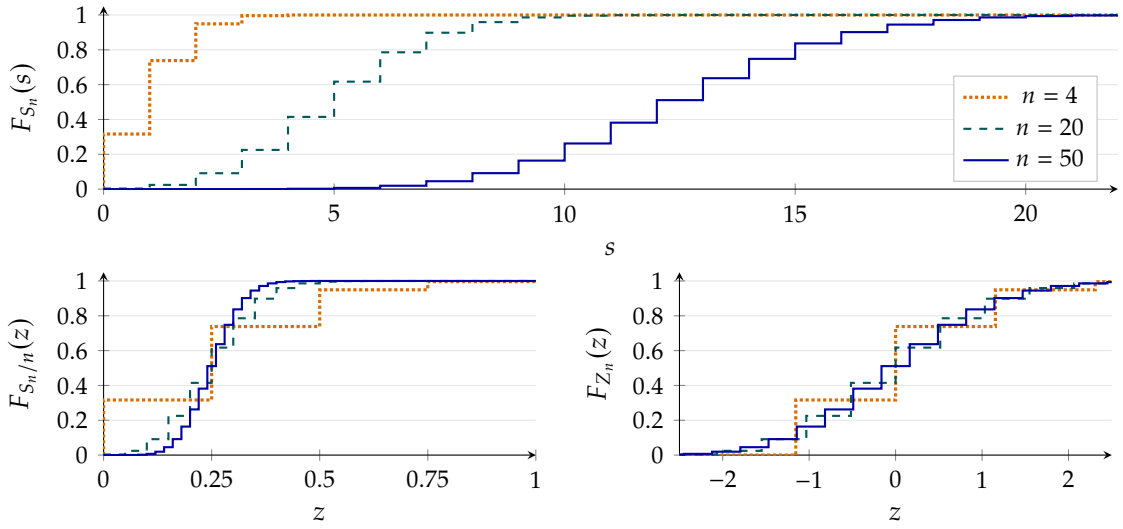


Figure 3: Exact distribution functions in the Bernoulli case $p_X(1) = 1/4$, $p_X(0) = 3/4$, for $n = 4, 20, 50$. Top: F_{S_n}. Bottom left: $F_{S_n/n}$, concentrating at $\overline{X} = 1/4$. Bottom right: F_{Z_n} for the normalized sum Z_n.

So the distribution of S_n moves with n and spreads like $\sqrt{n}$; that of S_n/n stays centered at $\overline{X}$ and concentrates like $1/\sqrt{n}$; that of Z_n keeps a fixed center and spread and, by the CENTRAL LIMIT THEOREM, approaches the normal distribution:

$$\lim_{n\to\infty} \Pr\left\{\frac{S_n - n\overline{X}}{\sqrt{n}\,\sigma} \le y\right\} = \int_{-\infty}^{y} \frac{1}{\sqrt{2\pi}} \exp\left(\frac{-x^2}{2}\right) dx.$$

2.6 The Bernoulli process

Definition 2.4 [BERNOULLI PROCESS]. A BERNOULLI PROCESS is a sequence $Y_1, Y_2, \ldots$ of IID binary rv's with $p_Y(1) = p > 0$ and $p_Y(0) = 1 - p = q > 0$. Let $S_n = Y_1 + \cdots + Y_n$.

The n-tuple of k ones followed by $n-k$ zeros has probability $p^k q^{n-k}$, and so does every n-tuple with k ones. For $p < 1/2$ this is largest at $k = 0$ and decreases in k. The number $\binom{n}{k}$ of such n-tuples increases in k for $k < n/2$ and then decreases. Altogether,

$$p_{S_n}(k) = \binom{n}{k} p^k q^{n-k}.$$

To see how this varies with k, take the ratio of successive terms:

$$\frac{p_{S_n}(k+1)}{p_{S_n}(k)} = \frac{n!}{(k+1)!\,(n-k-1)!} \frac{k!\,(n-k)!}{n!} \frac{p^{k+1}q^{n-k-1}}{p^k q^{n-k}} = \frac{n-k}{k+1}\frac{p}{q}. \tag{2.1}$$

This ratio is strictly decreasing in k. It is less than 1 for $k \ge pn$ (then $(n-k)p \le nqp < (k+1)q$) and greater than 1 for $k+1 \le pn$; for $k < pn < k+1$ it is close to 1. Thus $p_{S_n}(k)$ increases with k for $k < pn$ and decreases for $k > pn$, and its maximum is at $\lfloor pn \rfloor$ or $\lceil pn \rceil$.

2.7 The central limit theorem for the Bernoulli process

Use (2.1) for large n with k near pn. To simplify the algebra, assume pn is an integer and set $k = pn + i$ for relatively small i. Then

$$\frac{p_{S_n}(pn+i+1)}{p_{S_n}(pn+i)} = \frac{n-pn-i}{pn+i+1}\frac{p}{q} = \frac{nq-i}{pn+i+1}\frac{p}{q} = \frac{1-\frac{i}{nq}}{1+\frac{i+1}{np}}.$$

Using $\ln(1+x) = x - x^2/2 + \cdots$ for $|x| \ll 1$ and $1/p + 1/q = 1/(pq)$,

$$\ln\frac{p_{S_n}(pn+i+1)}{p_{S_n}(pn+i)} = \ln\left(1-\frac{i}{nq}\right) - \ln\left(1+\frac{i+1}{np}\right) = -\frac{i}{nq} - \frac{i}{np} - \frac{1}{np} + \cdots = -\frac{i}{npq} - \frac{1}{np} + \cdots,$$

where the neglected terms are of order i^2/n^2. So the logarithm of each unit increment is essentially linear in i. Writing an increment of j terms as a telescoping sum of unit increments and using

$1 + 2 + \cdots + (j-1) = j(j-1)/2$,

$$\ln \frac{p_{S_n}(pn+j)}{p_{S_n}(pn)} = \sum_{i=0}^{j-1} \ln \frac{p_{S_n}(pn+i+1)}{p_{S_n}(pn+i)} = -\frac{j(j-1)}{2npq} - \frac{j}{np} + \cdots \approx \frac{-j^2}{2npq}.$$

The terms linear in j are dropped because they are of the same order as a single unit increment. Hence

$$p_{S_n}(pn+j) \approx p_{S_n}(pn) \exp\Big(\frac{-j^2}{2npq}\Big),$$

for j both positive and negative: a quantized Gaussian with unknown scale factor $p_{S_n}(pn)$. Choosing that factor so the values sum to one gives the discrete PMF form of the central limit theorem,

$$\boxed{p_{S_n}(pn+j) \approx \frac{1}{\sqrt{2\pi npq}} \exp\Big(\frac{-j^2}{2npq}\Big)}.$$

Source. Robert Gallager, MIT 6.262, *Discrete Stochastic Processes*, Lecture 2, Spring 2011. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 3: Laws of large numbers, convergence

3.1 Probability models and repeated trials

Probability models are natural for real-world situations that are repeatable, with trials that have the same initial conditions, are essentially isolated from each other, have a fixed set of possible outcomes, and have essentially random individual outcomes. For any model, the extended model for an n-tuple or a sequence of IID repetitions is well defined. In it, relative frequencies and sample averages "become deterministic" and can be compared with those of the repeated real experiment. The laws of large numbers make "become deterministic" precise. They operate only within the extended model, but they are the only truly experimental way to compare a model with reality.

3.2 Markov, Chebyshev, and Chernoff bounds

Bounds play an unusually large role in probability, both because limit theorems use them and because probability applications are inherently imprecise.

Theorem 3.1 [Markov inequality]. *If Y is a nonnegative rv with expectation $\mathbb{E}[Y]$, then for every $y > 0$,*

$$\Pr\{Y \geq y\} \leq \frac{\mathbb{E}[Y]}{y}.$$

Proof. $\mathbb{E}[Y]$ is the area under F_Y^c. Since $F_Y^c(u) \geq \Pr\{Y \geq y\}$ for $0 \leq u < y$, the rectangle of width y and height $\Pr\{Y \geq y\}$ lies under the curve (Figure 4), so $y \Pr\{Y \geq y\} \leq \mathbb{E}[Y]$. □

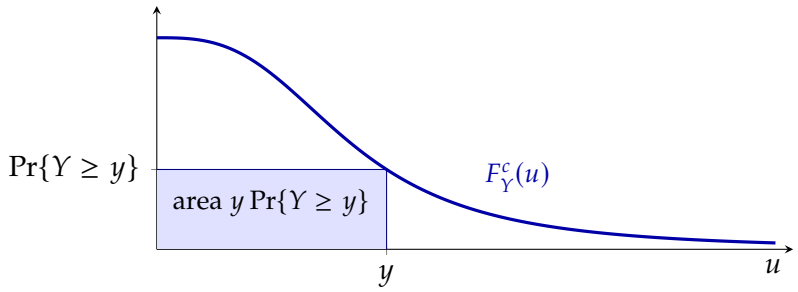


Figure 4: Schematic proof of the Markov inequality. The shaded rectangle has area $y \Pr\{Y \geq y\}$ and lies under F_Y^c, whose total area is $\mathbb{E}[Y]$.

The Markov bound is usually very loose. It holds with equality when Y is binary with values 0 and y, and it decreases only as $1/y$.

Theorem 3.2 [Chebyshev inequality]. *If Z has mean $\overline{Z}$ and variance σ_Z^2, then for every $\epsilon > 0$,*

$$\Pr\{|Z - \overline{Z}| \geq \epsilon\} \leq \frac{\sigma_Z^2}{\epsilon^2}. \tag{3.1}$$

Proof. Let $Y = (Z - \overline{Z})^2$, so $\mathbb{E}[Y] = \sigma_Z^2$. By Markov, for every $y > 0$, $\Pr\{\sqrt{Y} \geq \sqrt{y}\} = \Pr\{Y \geq y\} \leq \sigma_Z^2/y$. Since $\sqrt{Y} = |Z - \overline{Z}|$, setting $\epsilon = \sqrt{y}$ gives (3.1). □

Chebyshev requires a variance, but decreases as $1/\epsilon^2$ in the distance ϵ from the mean.

Theorem 3.3 [Chernoff bound]. *For every $z > 0$ and every $r > 0$ such that the moment generating function $g_Z(r) = \mathbb{E}\left[e^{rZ}\right]$ exists,*

$$\Pr\{Z \geq z\} \leq g_Z(r) \exp(-rz).$$

Proof. Let $Y = e^{rZ}$, so $\mathbb{E}[Y] = g_Z(r)$. Markov gives $\Pr\{Y \geq y\} \leq g_Z(r)/y$ for $y > 0$. Since $r > 0$, $\{Z \geq z\} = \{e^{rZ} \geq e^{rz}\}$, so $\Pr\{Z \geq z\} \leq g_Z(r)/e^{rz}$. □

The Chernoff bound decreases exponentially in z and is useful for large deviations from the mean.

3.3 The weak law of large numbers

Let $X_1, X_2, \ldots$ be IID with mean $\overline{X}$ and variance σ^2, and $S_n = X_1 + \cdots + X_n$. As pictured in Lecture 2, the distribution of S_n has mean $n\overline{X}$ and standard deviation $\sqrt{n}\,\sigma$, while S_n/n has mean $\overline{X}$ and standard deviation $\sigma/\sqrt{n}$. Thus

$$\mathrm{Var}\left(\frac{S_n}{n}\right) = \mathbb{E}\left[\left(\frac{S_n}{n} - \overline{X}\right)^2\right] = \frac{\sigma^2}{n}, \tag{3.2}$$

and in particular $\lim_{n\to\infty} \mathbb{E}\left[(S_n/n - \overline{X})^2\right] = 0$. The equation (3.2) says more than this limit, since it gives the variance and the $1/n$ rate, but the limit is a standard form of convergence.

Definition 3.4 [Convergence in mean square]. A sequence of rv's $Y_1, Y_2, \ldots$ converges in mean square to a rv Y if $\lim_{n\to\infty} \mathbb{E}\left[(Y_n - Y)^2\right] = 0$.

Mean-square convergence does not directly say how likely $|S_n/n - \overline{X}|$ is to exceed ϵ. Chebyshev applied to (3.2) does:

$$\Pr\left\{\left|\frac{S_n}{n} - \overline{X}\right| \geq \epsilon\right\} \leq \frac{\sigma^2}{n\epsilon^2} \qquad \text{for every } \epsilon > 0. \tag{3.3}$$

So accuracy ϵ holds with probability at least $1 - \sigma^2/(n\epsilon^2)$, which can be made as close to 1 as desired by increasing n.

Theorem 3.5 [Weak law of large numbers]. *If $X_1, X_2, \ldots$ are IID with $\mathbb{E}[|X|] < \infty$, then*

$$\lim_{n\to\infty} \Pr\left\{\left|\frac{S_n}{n} - \overline{X}\right| \geq \epsilon\right\} = 0 \qquad \textit{for every } \epsilon > 0.$$

The argument above proves this when the X_i have finite variance; it remains true when only $\mathbb{E}[|X|] < \infty$. Equivalently, by the definition of a limit, for every $\epsilon > 0$ and $\delta > 0$,

$$\Pr\left\{\left|\frac{S_n}{n} - \overline{X}\right| \geq \epsilon\right\} \leq \delta \qquad \text{for all large enough } n. \tag{3.4}$$

This says less about the speed of convergence than (3.3), but holds without a variance.

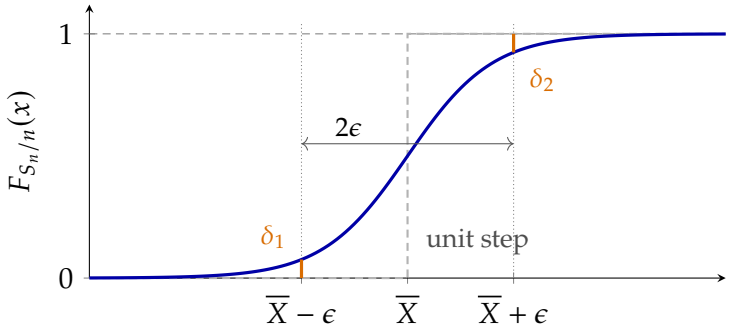


Figure 5: Schematic of the weak law. Outside the window $[\overline{X} - \epsilon, \overline{X} + \epsilon]$, $F_{S_n/n}$ is within δ_1 of 0 on the left and within δ_2 of 1 on the right, where $\delta_1 + \delta_2 = \delta \leq \sigma^2/(n\epsilon^2)$.

In other words, $\Pr\{S_n/n \leq x\}$ approaches a unit step at $\overline{X}$ as $n \to \infty$. For fixed ϵ, $\delta \to 0$ as $n \to \infty$; if $\sigma < \infty$, at least as fast as $\sigma^2/(n\epsilon^2)$, and otherwise possibly more slowly.

Definition 3.6 [Convergence in probability]. A sequence of rv's $Y_1, Y_2, \ldots$ converges in probability to a rv Y if for every $\epsilon > 0$ and $\delta > 0$, $\Pr\{|Y_n - Y| \geq \epsilon\} \leq \delta$ for all large enough n.

Thus S_n/n converges to $\overline{X}$ in probability whenever $\mathbb{E}[|X|] < \infty$. More generally, if $Y_n \to Y$ in mean square, then Chebyshev's argument (Markov applied to $(Y_n - Y)^2$) gives $\Pr\{|Y_n - Y| \geq \epsilon\} \leq \mathbb{E}[(Y_n - Y)^2]/\epsilon^2 \to 0$, so *mean-square convergence implies convergence in probability*. The converse fails, since the weak law needs no variance. Convergence in probability means that the distribution function of $Y_n - Y$ approaches a unit step at 0.

3.4 The central limit theorem and convergence in distribution

The normalized sum $Z_n = (S_n - n\overline{X})/(\sqrt{n}\,\sigma)$ has mean 0 and variance 1 for every n, and it also becomes Gaussian:

$$\lim_{n\to\infty} \Pr\left\{\frac{S_n - n\overline{X}}{\sqrt{n}\,\sigma} \leq y\right\} = \int_{-\infty}^{y} \frac{1}{\sqrt{2\pi}} \exp\left(\frac{-x^2}{2}\right) dx = \Phi(y).$$

Lecture 2 derived this for the Bernoulli case; the general proof (by Fourier transforms) is messy and not insightful. The CLT concerns the distribution function F_{S_n}, not the PMF or density.

Definition 3.7 [Convergence in distribution]. A sequence of rv's $Z_1, Z_2, \ldots$ converges in distribution to Z if $\lim_{n\to\infty} F_{Z_n}(z) = F_Z(z)$ at every z where F_Z is continuous.

The CLT says that $(S_n - n\overline{X})/(\sqrt{n}\,\sigma)$ converges in distribution to a normal rv with distribution function Φ. Convergence in distribution is almost a misnomer: the rv's need not get close to each other in any ordinary sense. Any IID sequence converges in distribution, since all its terms already have the same distribution.

Theorem 3.8. *Convergence in probability implies convergence in distribution. Hence mean-square convergence implies convergence in probability, which implies convergence in distribution.*

Proof. Suppose $Y_n \to Y$ in probability and fix $\epsilon > 0$. If $Y_n \le y$ then either $Y \le y + \epsilon$ or $|Y_n - Y| > \epsilon$; if $Y \le y - \epsilon$ then either $Y_n \le y$ or $|Y_n - Y| > \epsilon$. Hence

$$F_Y(y-\epsilon) - \Pr\{|Y_n - Y| > \epsilon\} \le F_{Y_n}(y) \le F_Y(y+\epsilon) + \Pr\{|Y_n - Y| > \epsilon\}.$$

Let $n \to \infty$, then $\epsilon \to 0$: at a continuity point y of F_Y, both bounds tend to $F_Y(y)$. □

This creates an apparent paradox: the CLT says something very strong about how S_n/n approaches $\overline{X}$, yet convergence in distribution is a very weak form of convergence. The resolution is that the rv's converging in distribution are $(S_n - n\overline{X})/(\sqrt{n}\,\sigma)$, while those converging in probability to 0 are $(S_n - n\overline{X})/n$, a squashed version of them. For $0 < \sigma < \infty$, for example, the CLT gives $\lim_{n\to\infty} \Pr\{(S_n - n\overline{X})/n \le 0\} = 1/2$, which cannot be deduced from the weak law.

3.5 Convergence with probability 1

A rv is a far more complicated object than a number, so there are many types of convergence of a sequence of rv's. A very important one has a deceptively simple definition.

Definition 3.9 [Convergence with probability 1]. A sequence of rv's $Z_1, Z_2, \ldots$ converges with probability 1 (WP1) to a rv Z if

$$\Pr\Big\{\omega \in \Omega : \lim_{n\to\infty} Z_n(\omega) = Z(\omega)\Big\} = 1.$$

Each sample point ω maps into a sequence of real numbers $Z_1(\omega), Z_2(\omega), \ldots$. Some of these sequences have a limit, and for some that limit is $Z(\omega)$. Convergence WP1 means that the set of ω for which the sequence converges to $Z(\omega)$ is an event of probability 1. It is often simpler to study whether $Y_n = Z_n - Z$ converges to 0 WP1.

Theorem 3.10 [Strong law of large numbers]. *If $X_1, X_2, \ldots$ are IID with $\mathbb{E}[|X|] < \infty$, then $\{S_n/n;\ n \geq 1\}$ converges to $\overline{X}$ WP1; that is, all sample paths of $\{S_n/n;\ n \geq 1\}$ converge to $\overline{X}$ except for a set of probability 0.*

These are the same conditions under which the weak law holds. The strong law is considerably easier to work with than the weak law in the study of renewal processes, but feeling at home with it, and with these sets of probability 1, takes some investment of time.

Source. Robert Gallager, MIT 6.262, *Discrete Stochastic Processes*, Lecture 3, Spring 2011. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 4: Poisson (the perfect arrival process)

4.1 Review: modes of convergence

For a sequence $Z_1, Z_2, \ldots$ of rv's and a rv Z:

- $Z_n \to Z$ *in distribution* if $\lim_{n\to\infty} F_{Z_n}(z) = F_Z(z)$ at each z where F_Z is continuous;
- $Z_n \to Z$ *in probability* if $\lim_{n\to\infty} \Pr\{|Z_n - Z| > \epsilon\} = 0$ for every $\epsilon > 0$ (equivalently, for every $\epsilon, \delta > 0$, $\Pr\{|Z_n - Z| > \epsilon\} \le \delta$ for all large enough n);
- $Z_n \to Z$ *in mean square* if $\lim_{n\to\infty} \mathbb{E}\left[|Z_n - Z|^2\right] = 0$.

Mean-square convergence implies convergence in probability, which implies convergence in distribution. For IID $X_1, X_2, \ldots$ with mean $\overline{X}$ and $S_n = \sum_{i=1}^{n} X_i$: the weak law says S_n/n converges in probability (hence in distribution) to the deterministic rv $\overline{X}$; if the variance σ^2 is finite, the CLT says $Z_n = (S_n - n\overline{X})/(\sigma\sqrt{n})$ converges in distribution to $\mathcal{N}(0,1)$.

4.2 Arrival processes

Definition 4.1 [Arrival process]. An arrival process is a sequence of increasing rv's $0 < S_1 < S_2 < \cdots$, where $S_{i-1} < S_i$ means that $X_i = S_i - S_{i-1}$ is a positive rv, that is, $F_{X_i}(0) = 0$. The S_i are the arrival epochs and the differences $X_1 = S_1$, $X_i = S_i - S_{i-1}$ for $i \ge 2$ are the interarrival times. For each $t > 0$, $N(t)$ is the number of arrivals in $(0, t]$, and $\{N(t);\ t > 0\}$ is the arrival counting process.

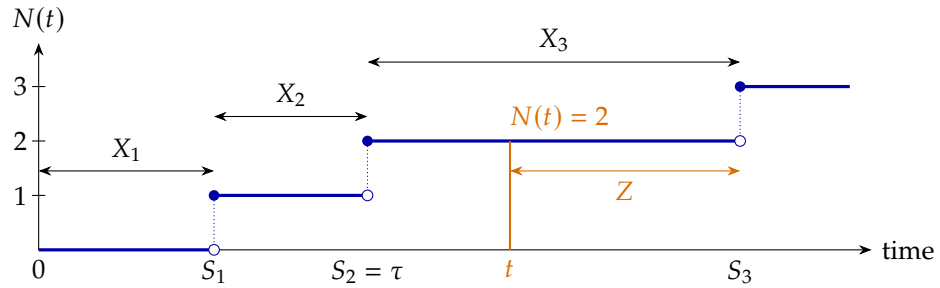


Figure 6: A sample path of an arrival process: arrival epochs S_i, interarrival times X_i, and the counting process $N(t)$. Here $N(t) = 2$ and $S_2 = \tau$, and Z is the interval from t to the next arrival.

A sample path is a sequence of sample values $S_1 = s_1$, $S_2 = s_2, \ldots$, and corresponds to a staircase function with $N(t) = n$ for $S_n \le t < S_{n+1}$. The process can be viewed as the ensemble, with joint probability distributions, of such staircase functions. A sample path is specified equally well by the values of $N(t)$ for all t, by the S_i, or by the X_i, so an arrival process can be described by whichever

of the three is convenient. The key relation between them is

$$\{S_n \le t\} = \{N(t) \ge n\} \qquad \text{for all } n \ge 1,\ t > 0. \tag{4.1}$$

Indeed, if $S_n = \tau$ for some $\tau \le t$, then $N(\tau) = n$ and $N(t) \ge n$.

General arrival processes are too complicated to say much about, so we specialize.

Definition 4.2 [Renewal and Poisson processes]. A renewal process is an arrival process whose interarrival times $X_1, X_2, \ldots$ are IID. A Poisson process is a renewal process in which each X_i is exponential, $F_X(x) = 1 - \exp(-\lambda x)$ for $x \ge 0$, where the fixed parameter $\lambda > 0$ is the rate.

4.3 Exponential rv's and memorylessness

Definition 4.3 [Memoryless rv]. A rv X is memoryless if X is positive and, for all real $t > 0$ and $x > 0$,

$$\Pr\{X > t + x\} = \Pr\{X > t\} \Pr\{X > x\}. \tag{4.2}$$

An exponential rv is memoryless, since $\exp(-\lambda(t + x)) = \exp(-\lambda t)\exp(-\lambda x)$.

Theorem 4.4. *A rv X is memoryless if and only if it is exponential.*

Proof. It remains to show that a memoryless X is exponential. Let $h(x) = \ln \Pr\{X > x\}$. If $\Pr\{X > x\} = 0$ for some $x > 0$, then (4.2) gives $\Pr\{X > x/2^k\} = 0$ for all k, so $\Pr\{X > 0\} = 0$, contradicting positivity. Hence h is finite, nonincreasing, and satisfies $h(t + x) = h(t) + h(x)$. Additivity gives $h(x) = xh(1)$ for rational $x > 0$, and monotonicity extends this to all $x > 0$. Thus $\Pr\{X > x\} = e^{-\lambda x}$ with $\lambda = -h(1)$, and $\lambda > 0$ since $\Pr\{X > x\} \to 0$. □

In conditional form, $\Pr\{X > t + x \mid X > t\} = \Pr\{X > x\}$. If service times in a checkout line are exponential and you have already waited 15 minutes for the person in front, the remaining service time is exponential, the same as when service started: it does not remember the elapsed time.

Theorem 4.5 [Residual time]. *For a Poisson process of rate λ and any $t > 0$, let Z be the interval from t to the next arrival after t. Then $F_Z^c(z) = \exp(-\lambda z)$ for all $z > 0$. Moreover, Z is independent of $N(t)$, and, given $N(t) = n$, Z is independent of $S_1, \ldots, S_n$ and of $\{N(\tau);\ 0 < \tau < t\}$.*

Proof. Condition on $N(t) = n$ and $S_n = \tau$, the number of arrivals in $(0, t]$ and the epoch of the most recent one (Figure 6 with $n = 2$). This event is $\{S_n = \tau,\ X_{n+1} > t - \tau\}$, and then $Z = X_{n+1} - (t - \tau)$. Since X_{n+1} is independent of $X_1, \ldots, X_n$, memorylessness gives

$$\Pr\{Z > z \mid N(t) = n,\ S_n = \tau\} = \Pr\{X_{n+1} > t - \tau + z \mid X_{n+1} > t - \tau\} = \exp(-\lambda z).$$

The same holds conditional on any values of $X_1, \ldots, X_n$ consistent with $N(t) = n$. The conditional distribution does not vary with the conditioning values, so Z is exponential and statistically independent of $N(t)$, of $S_{N(t)}$ (the epoch of the last arrival up to and including t), and of the whole history before t. □

This extends memorylessness to the entire Poisson process. Starting at any $t > 0$, the first interarrival Z is exponential with rate λ and independent of everything before t, and the subsequent interarrival times $X_{N(t)+2}, X_{N(t)+3}, \ldots$ are IID exponential and independent of Z and of the past. So the arrivals after t form a Poisson process whose counting process is $N(t') - N(t)$, $t' > t$. It is a Poisson process shifted to start at t: for each t', $N(t') - N(t)$ has the same distribution as $N(t' - t)$, and the same holds for joint distributions. This new process is independent of $\{N(\tau);\ 0 < \tau \leq t\}$.

4.4 Stationary and independent increments

Write $\widetilde{N}(t, t') = N(t') - N(t)$ for the number of arrivals in the increment $(t, t']$.

Definition 4.6 [Stationary increments]. A counting process $\{N(t);\ t > 0\}$ has the stationary increment property if $\widetilde{N}(t, t')$ has the same distribution as $N(t' - t)$ for all $t' > t > 0$.

Thus the distribution of the number of arrivals in an interval depends on its length but not on its starting time.

Definition 4.7 [Independent increments]. A counting process $\{N(t);\ t > 0\}$ has the independent increment property if for every $0 < t_1 < t_2 < \cdots < t_n$ the rv's $N(t_1), \widetilde{N}(t_1, t_2), \ldots, \widetilde{N}(t_{n-1}, t_n)$ are independent.

Equivalently, the numbers of arrivals in non-overlapping intervals are independent.

Theorem 4.8. *Poisson processes have stationary and independent increments.*

Proof. Stationarity is the shift property above. For independence, $\widetilde{N}(t_{i-1}, t_i)$ is a function of the process after t_{i-1}, which is independent of $\{N(\tau);\ \tau \leq t_{i-1}\}$, and therefore of $N(t_1), \widetilde{N}(t_1, t_2), \ldots, \widetilde{N}(t_{i-2}, t_{i-1})$. Induct on i. □

4.5 The Erlang and Poisson distributions

The density of $S_2 = X_1 + X_2$ is the convolution

$$f_{S_2}(t) = \int_0^t \left[\lambda e^{-\lambda x}\right]\left[\lambda e^{-\lambda(t-x)}\right] dx = \lambda^2 t e^{-\lambda t}.$$

Iterating, convolving $f_{S_{n-1}}$ with $\lambda e^{-\lambda t}$ at each step, gives the Erlang density

$$\boxed{f_{S_n}(t) = \frac{\lambda^n t^{n-1} \exp(-\lambda t)}{(n-1)!}}, \qquad t \geq 0.$$

Indeed, if this holds for $n-1$, then

$$\int_0^t \frac{\lambda^{n-1} x^{n-2} e^{-\lambda x}}{(n-2)!} \lambda e^{-\lambda(t-x)}\, dx = \frac{\lambda^n e^{-\lambda t}}{(n-2)!} \int_0^t x^{n-2}\, dx = \frac{\lambda^n t^{n-1} e^{-\lambda t}}{(n-1)!}.$$

It pays to look at the joint density. Since the X_i are IID,

$$f_{\vec{X}^n}(x_1, \ldots, x_n) = \lambda^n \exp(-\lambda x_1 - \cdots - \lambda x_n) = \lambda^n \exp(-\lambda s_n), \qquad s_n = \sum_{i=1}^n x_i.$$

The map from $(x_1, \ldots, x_n)$ to $(s_1, \ldots, s_n)$ is linear with unit Jacobian, so

$$f_{\vec{S}^n}(s_1, \ldots, s_n) = \lambda^n \exp(-\lambda s_n) \qquad \text{for } 0 < s_1 < s_2 < \cdots < s_n.$$

The joint density depends only on s_n: given that the nth arrival is at s_n, the other $n-1$ arrivals are uniformly distributed in $(0, s_n)$, subject to the ordering. Integrating out $s_1, \ldots, s_{n-1}$ over a region of volume $s_n^{n-1}/(n-1)!$ again gives the Erlang density.

Theorem 4.9 [Poisson PMF]. *For a Poisson process of rate λ, $N(t)$ has the* Poisson PMF

$$\boxed{p_{N(t)}(n) = \frac{(\lambda t)^n \exp(-\lambda t)}{n!}}, \qquad n \geq 0.$$

Proof. Compute $\Pr\{t < S_{n+1} \leq t + \delta\}$ in two ways and let $\delta \to 0$. From the density of S_{n+1},

$$\Pr\{t < S_{n+1} \leq t + \delta\} = f_{S_{n+1}}(t)\big(\delta + o(\delta)\big),$$

where $\lim_{\delta \to 0} o(\delta)/\delta = 0$. Alternatively, the event occurs if $N(t) = n$ and at least one arrival falls in $(t, t+\delta]$, or if $N(t) < n$ and at least two arrivals fall in $(t, t+\delta]$. By independent and stationary increments, using $\Pr\{N(\delta) \geq 1\} = 1 - e^{-\lambda\delta} = \lambda\delta + o(\delta)$ and the Erlang bound $\Pr\{N(\delta) \geq 2\} = \Pr\{S_2 \leq \delta\} = o(\delta)$,

$$\Pr\{t < S_{n+1} \leq t + \delta\} = p_{N(t)}(n)\big(\lambda\delta + o(\delta)\big) + o(\delta).$$

Equating and letting $\delta \to 0$ gives $p_{N(t)}(n) = f_{S_{n+1}}(t)/\lambda$, which is the Poisson PMF. □

The Poisson PMF depends on λ and t only through λt, the mean number of arrivals in an interval of length t. This must be so, since changing the unit of time changes λ inversely. Also $N(t) = N(t_1) + \widetilde{N}(t_1, t)$ for any $0 < t_1 < t$, a sum of two independent rv's with the Poisson PMFs for

t_1 and $t - t_1$. The same holds for any k disjoint intervals, which is one reason the Poisson counting process is so easy to work with.

4.6 Alternative definitions of a Poisson process

An arrival process for which $N(t)$ has the Poisson PMF with parameter λt for every t need not be a Poisson process: marginal PMFs alone are not enough, and the joint distributions must also be those of the Poisson process.

Theorem 4.10. *If an arrival process has the stationary and independent increment properties and $N(t)$ has the Poisson PMF for a given λ and all $t > 0$, then the process is Poisson.*

Proof sketch. Stationary and independent increments, together with the Poisson marginals, determine the joint distribution of the numbers of arrivals over any finite set of disjoint intervals, and it is that of a Poisson process. These joint distributions determine the process. □

Stationary and independent increments alone do not suffice: they allow BULK PROCESSES, in which simultaneous arrivals are possible. A Poisson process also satisfies the INCREMENTAL CONDITION

$$\Pr\{\widetilde{N}(t, t+\delta) = n\} = \begin{cases} 1 - \lambda\delta + o(\delta) & n = 0, \\ \lambda\delta + o(\delta) & n = 1, \\ o(\delta) & n \geq 2. \end{cases} \tag{4.3}$$

Theorem 4.11. *If an arrival process has the stationary and independent increment properties and satisfies (4.3), then it is a Poisson process.*

4.7 Relation to the Bernoulli process

The Bernoulli process is often viewed as a discrete-time Poisson process, but its binary rv's $Y_1, Y_2, \ldots$ have no direct analog in the Poisson process. Viewed as an arrival process, the Bernoulli process has an arrival at time n if and only if $Y_n = 1$. Then $S_n = Y_1 + \cdots + Y_n$ is the number of arrivals up to and including time n, so $\{S_n;\ n \geq 1\}$ is the analog of the counting process $\{N(t);\ t > 0\}$, not of the arrival epochs.

The interarrival times are the intervals between successive 1's: $X_1 = k$ if $Y_i = 0$ for $1 \leq i \leq k-1$ and $Y_k = 1$, so

$$p_{X_1}(k) = p(1-p)^{k-1}, \qquad k \geq 1,$$

and the subsequent interarrival times are IID with X_1. Thus the interarrival times are GEOMETRIC. The counting process is defined only at integer times, but over integer intervals it has the stationary and independent increment properties.

A SHRUNKEN BERNOULLI PROCESS has its slots at spacing δ rather than 1. Shrink δ while holding the arrival probability per unit time $p/\delta = \lambda$ fixed. Then the geometric interarrival times become exponential, $\Pr\{X > x\} = (1 - \lambda\delta)^{\lfloor x/\delta \rfloor} \to e^{-\lambda x}$, and the Bernoulli counting process converges to the Poisson counting process of rate λ.

Source. Robert Gallager, MIT 6.262, *Discrete Stochastic Processes*, Lecture 4, Spring 2011. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 5: Poisson combining and splitting

5.1 Review of Poisson processes

A Poisson process is an arrival process with IID exponential interarrival times. It can be represented by its arrival epochs $S_1, S_2, \ldots$, its interarrival times $X_1, X_2, \ldots$, or its counting process $\{N(t);\, t > 0\}$, with $\{S_n \le t\} = \{N(t) \ge n\}$ and $N(t) = n$ for $S_n \le t < S_{n+1}$. The interarrival times are memoryless: $\Pr\{X_i > t + x \mid X_i > t\} = \Pr\{X_i > x\} = \exp(-\lambda x)$ for $x, t > 0$. Given $N(t) = n$ and $S_n = \tau$, we have $X_{n+1} > t - \tau$, and the residual time $Z = X_{n+1} - (t - \tau)$ satisfies

$$\Pr\big\{Z > z \mid N(t), S_{N(t)}\big\} = \exp(-\lambda z).$$

So Z is independent of $\{N(\tau);\, \tau \le t\}$, Z is the first interarrival time of the Poisson process $\{N(t') - N(t);\, t' > t\}$, and that process is independent of $\{N(\tau);\, \tau \le t\}$.

Consequently, for any $0 < t_1 < t_2 < \cdots < t_k$, the increments $\{N(t);\, 0 < t \le t_1\}$, $\{\widetilde{N}(t_1, t);\, t_1 < t \le t_2\}, \ldots, \{\widetilde{N}(t_{k-1}, t);\, t_{k-1} < t \le t_k\}$ are stationary and independent Poisson counting processes over their intervals. The PMF $p_{N(t)}(n) = (\lambda t)^n \exp(-\lambda t)/n!$ is a function only of the mean λt. Since $N(t_1)$ and $\widetilde{N}(t_1, t)$ are independent and Poisson, and their sum $N(t)$ is Poisson, this illustrates the general fact that *sums of independent Poisson rv's are Poisson, with the means adding*.

Either of the two theorems of Lecture 4 serves as an alternative definition: an arrival process with stationary and independent increments is Poisson if $N(t)$ has the Poisson PMF for all $t > 0$, or if it satisfies the incremental condition

$$\Pr\big\{\widetilde{N}(t, t+\delta) = n\big\} = \begin{cases} 1 - \lambda\delta + o(\delta) & n = 0, \\ \lambda\delta + o(\delta) & n = 1, \\ o(\delta) & n \ge 2. \end{cases} \tag{5.1}$$

5.2 Combining independent Poisson processes

Definition 5.1 [Independent processes]. Two Poisson processes $\{N_1(t);\, t > 0\}$ and $\{N_2(t);\, t > 0\}$ are independent if for all $t_1, \ldots, t_n$ the rv's $N_1(t_1), \ldots, N_1(t_n)$ are independent of $N_2(t_1), \ldots, N_2(t_n)$.

Theorem 5.2 [Combining]. *If $\{N_1(t);\, t > 0\}$ and $\{N_2(t);\, t > 0\}$ are independent Poisson processes of rates λ_1 and λ_2, and $N(t) = N_1(t) + N_2(t)$ for all $t > 0$, then $\{N(t);\, t > 0\}$ is a Poisson process of rate $\lambda = \lambda_1 + \lambda_2$.*

Proof. The combined process inherits stationary and independent increments from the two independent processes. In any increment, $\widetilde{N}(t, t+\delta) = \widetilde{N}_1(t, t+\delta) + \widetilde{N}_2(t, t+\delta)$, so

$$\begin{aligned} p_{\widetilde{N}(t,t+\delta)}(1) &= p_{\widetilde{N}_1(t,t+\delta)}(1)\, p_{\widetilde{N}_2(t,t+\delta)}(0) + p_{\widetilde{N}_1(t,t+\delta)}(0)\, p_{\widetilde{N}_2(t,t+\delta)}(1) \\ &= [\delta\lambda_1 + o(\delta)][1 - \delta\lambda_2 + o(\delta)] + [1 - \delta\lambda_1 + o(\delta)][\delta\lambda_2 + o(\delta)] = \delta(\lambda_1 + \lambda_2) + o(\delta). \end{aligned}$$

Similarly, $\widetilde{N}(t, t+\delta) = 0$ exactly when both increments are 0, with probability $1-(\lambda_1+\lambda_2)\delta+o(\delta)$, and the incremental condition (5.1) holds with rate $\lambda_1 + \lambda_2$. □

It is analytically cleaner to use the Poisson distribution directly: $\widetilde{N}_1(t, t+\delta)$ and $\widetilde{N}_2(t, t+\delta)$ are independent Poisson rv's, so their sum $\widetilde{N}(t, t+\delta)$ is Poisson with mean $\lambda\delta$. The sum of many small independent arrival processes tends to be close to Poisson even when the small processes are not: the independence between processes overcomes the dependence between successive arrivals within each process.

5.3 Splitting a Poisson process

Each arrival of a Poisson process $\{N(t);\, t > 0\}$ of rate λ is switched to $\{N_1(t);\, t > 0\}$ with probability p and otherwise to $\{N_2(t);\, t > 0\}$. The switch is a Bernoulli process independent of $\{N(t);\, t > 0\}$: a p-biased coin is flipped independently at each arrival.

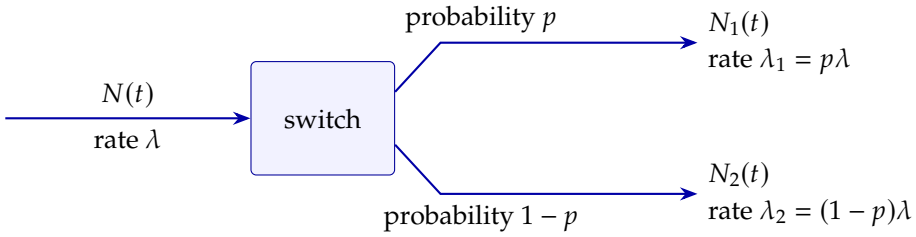


Figure 7: Splitting a Poisson process of rate λ by independent coin flips into processes of rates $p\lambda$ and $(1-p)\lambda$.

Theorem 5.3 [Splitting]. *The split processes are independent Poisson processes of rates $\lambda_1 = p\lambda$ and $\lambda_2 = (1-p)\lambda$.*

Proof. Each split process clearly has stationary and independent increments, and satisfies the incremental condition with rate $p\lambda$ or $(1-p)\lambda$, so each is Poisson. The incremental condition does not make independence of the two processes clear, since both processes must sometimes have arrivals in the same small increment; that information is hidden in the $o(\delta)$ terms. Directly,

for one interval,

$$\Pr\{N_1(t) = m,\ N_2(t) = k\} = \frac{(\lambda t)^{m+k} e^{-\lambda t}}{(m+k)!} \binom{m+k}{m} p^m (1-p)^k$$
$$= \frac{(p\lambda t)^m e^{-p\lambda t}}{m!} \frac{((1-p)\lambda t)^k e^{-(1-p)\lambda t}}{k!},$$

which factors; by independent increments the same factorization holds jointly over any set of disjoint intervals. □

Combining and splitting are often used together: view separate independent Poisson processes as one combined process, then split it again by binary choices between the processes.

Example: *Last-come first-served queue.* Arrivals are Poisson with rate λ, and service times are independent and exponential with rate μ. A new arrival starts service immediately, interrupting the customer in service. View services as a Poisson process of rate μ, either ignoring it when there is nothing to serve or imagining a low-priority task served at rate μ when the system is otherwise idle. The arrival process plus the service process is Poisson of rate $\lambda + \mu$, and each of its events is independently a service completion with probability $\mu/(\lambda + \mu)$. So an arrival completes service before being interrupted with probability $\mu/(\lambda + \mu)$. Given that it is interrupted, it suffers no further interruption exactly when two services (the interrupting job's and then its own) finish before the next arrival, with probability $\mu^2/(\lambda + \mu)^2$.

5.4 Non-homogeneous Poisson processes

In optical transmission, a photon stream is reasonably modeled as a Poisson process, and modulating its power turns the steady photon rate into a variable rate $\lambda(t)$. The number of photons in $(t, t+\delta]$ is modeled as a Poisson rv whose mean is δ times the average rate over $(t, t+\delta]$. In the small-increment model,

$$\Pr\{\widetilde{N}(t, t+\delta) = n\} = \begin{cases} 1 - \delta\lambda(t) + o(\delta) & n = 0, \\ \lambda(t)\delta + o(\delta) & n = 1, \\ o(\delta) & n \geq 2. \end{cases}$$

This model shows that the number of arrivals in each increment is Poisson,

$$\Pr\{\widetilde{N}(t, \tau) = n\} = \frac{[\widetilde{m}(t, \tau)]^n \exp[-\widetilde{m}(t, \tau)]}{n!}, \qquad \widetilde{m}(t, \tau) = \int_t^\tau \lambda(u)\, du.$$

Combining and splitting non-homogeneous processes works as in the homogeneous case, but the interarrival times are no longer independent exponentials. The remaining results are for homogeneous processes.

5.5 Conditional arrival densities

Many useful results describe $(0, t]$ conditional on $N(t)$. First condition on $N(t) = 1$. The arrival lies in $(s_1, s_1 + \delta]$ if there is no arrival in $(0, s_1]$, one in $(s_1, s_1 + \delta]$, and none in $(s_1 + \delta, t]$, so

$$f_{S_1|N(t)}(s_1 \mid 1) = \lim_{\delta \to 0} \frac{p_{N(s_1)}(0)\, p_{\widetilde{N}(s_1, s_1+\delta)}(1)\, p_{\widetilde{N}(s_1+\delta, t)}(0)}{\delta\, p_{N(t)}(1)}$$
$$= \lim_{\delta \to 0} \frac{e^{-\lambda s_1} \lambda\delta e^{-\lambda\delta} e^{-\lambda(t-s_1-\delta)}}{\delta\lambda t e^{-\lambda t}} = \frac{1}{t}.$$

This does not depend on s_1: S_1 is uniform over $(0, t]$. For $N(t) = 2$ and $0 < s_1 < s_2 < t$, the same computation with two δ-slots gives

$$f(s_1, s_2 \mid 2) = \lim_{\delta \to 0} \frac{e^{-\lambda s_1} \lambda\delta e^{-\lambda\delta} e^{-\lambda(s_2-s_1-\delta)} \lambda\delta e^{-\lambda\delta} e^{-\lambda(t-s_2-\delta)}}{\delta^2 p_{N(t)}(2)} = \frac{2}{t^2},$$

again independent of s_1, s_2, and λ. In general the exponents always sum to λt, and

$$\boxed{f_{\vec{S}^{(n)}|N(t)}\big(\vec{s}^{(n)} \mid n\big) = \lim_{\delta \to 0} \frac{(\delta\lambda)^n \exp(-\lambda t)}{\delta^n p_{N(t)}(n)} = \frac{n!}{t^n}} \qquad \text{for } 0 < s_1 < \cdots < s_n < t.$$

This is a uniform n-dimensional density over the constraint region $0 < s_1 < \cdots < s_n < t$, which therefore has volume $t^n/n!$.

To see why $n!$ appears, let $U_1, \ldots, U_n$ be IID and uniform over $(0, t]$, and let $S_1, \ldots, S_n$ be their ORDER STATISTICS: $S_1 = \min(U_1, \ldots, U_n)$, and S_k is the kth smallest. The cube of volume t^n where $\vec{U}^n$ has nonzero density splits into $n!$ regions, one for each ordering of $u_1, \ldots, u_n$; by symmetry each has volume $t^n/n!$, and the region where $\vec{S}^n$ is nonzero is one of them. Since $\vec{S}^n$ has the same density whether it is the conditional density of n arrival epochs given $N(t) = n$ or the order statistics of n uniform rv's, results about either apply to the other.

Example: *The first arrival given $N(t) = n$.* Viewing S_1 as the minimum of $U_1, \ldots, U_n$,

$$\Pr\Big\{\min_{1 \le i \le n} U_i > s_1\Big\} = \prod_{i=1}^{n} \Pr\{U_i > s_1\} = \Big(1 - \frac{s_1}{t}\Big)^n, \qquad F^c_{S_1|N(t)}(s_1 \mid n) = \Big(1 - \frac{s_1}{t}\Big)^n$$

for $0 \le s_1 \le t$. Integrating the complementary distribution function gives $\mathbb{E}[S_1 \mid N(t) = n] = \int_0^t (1 - s/t)^n \, ds = t/(n+1)$.

5.6 Conditional interarrival times

Return to $N(t) = 2$ and look at the interarrival times, with $x_1 = s_1$ and $x_2 = s_2 - s_1$. The region $0 < s_1 < s_2 < t$ has area $t^2/2$, which explains the density $2/t^2$. The map to (x_1, x_2) is linear with unit Jacobian: a $\delta \times \delta$ box in the (s_1, s_2) plane maps to a parallelogram of the same area. Its image is

the region $x_1, x_2 > 0$, $x_1 + x_2 < t$, again of area $t^2/2$, so $f(x_1, x_2 \mid N(t) = 2) = 2/t^2$ there.

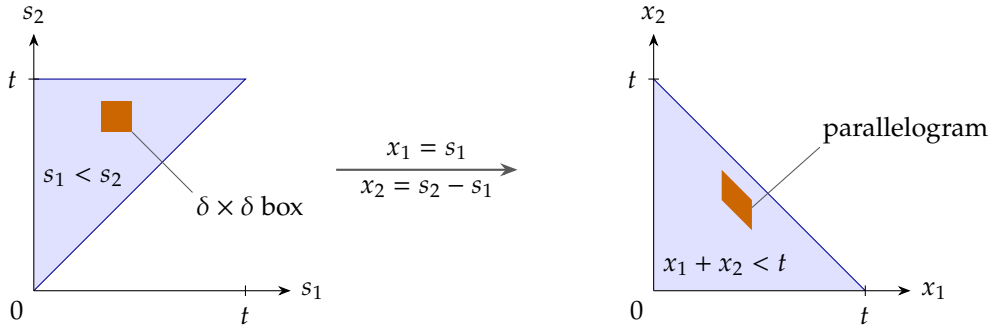


Figure 8: Given $N(t) = 2$, the epochs are uniform on the triangle $0 < s_1 < s_2 < t$ (left), and the interarrival times are uniform on the triangle $x_1, x_2 > 0$, $x_1 + x_2 < t$ (right). Both have area $t^2/2$; the unit-Jacobian map sends a small square to a parallelogram of equal area.

Let $X^*_{N(t)+1} = t - S_{N(t)}$, the time from the last arrival up to t; for $N(t) = 2$ write $x_3^* = t - s_2$. Then

$$f(x_1, x_2 \mid N(t) = 2) = \frac{2}{t^2} \qquad \text{for } x_1, x_2, x_3^* > 0,\ x_1 + x_2 + x_3^* = t.$$

This is symmetric in x_1, x_2, x_3^*, so any two of X_1, X_2, X_3^* can replace X_1, X_2 here. Since $X_1 = S_1$,

$$F^c_{X_1|N(t)}(x_1 \mid 2) = \left(1 - \frac{x_1}{t}\right)^2, \qquad \mathbb{E}[X_1 \mid N(t) = 2] = \frac{t}{3},$$

and by symmetry the same formulas hold for X_2 and X_3^*. As a check, $\mathbb{E}[X_1 + X_2 + X_3^* \mid N(t) = 2] = t$.

The same holds for $N(t) = n$, with $X^*_{n+1} = t - S_{N(t)}$:

$$F^c_{X_1|N(t)}(x \mid n) = F^c_{X^*_{n+1}|N(t)}(x \mid n) = \left(1 - \frac{x}{t}\right)^n, \qquad \mathbb{E}[X_1 \mid N(t) = n] = \mathbb{E}\left[X^*_{n+1} \mid N(t) = n\right] = \frac{t}{n+1}.$$

Averaging over $N(t)$,

$$\mathbb{E}\left[X^*_{N(t)+1}\right] = \sum_{n=0}^{\infty} \frac{t}{n+1} \frac{(\lambda t)^n e^{-\lambda t}}{n!} = \sum_{n=0}^{\infty} \frac{(\lambda t)^{n+1} e^{-\lambda t}}{\lambda (n+1)!} = \frac{1 - e^{-\lambda t}}{\lambda}.$$

Note. An apparent paradox. The mean interarrival time is $1/\lambda$. But the mean time from any given t to the next arrival is $1/\lambda$, and the mean time back to the previous arrival (or to 0) is $(1 - e^{-\lambda t})/\lambda$. So the interval containing t has mean length $(2 - e^{-\lambda t})/\lambda$, nearly twice $1/\lambda$ for large t. Heuristically, fix a sample path and choose t uniformly over a large interval far from 0: long interarrival intervals cover proportionally more of the time axis, so t tends to land in one.

Source. Robert Gallager, MIT 6.262, *Discrete Stochastic Processes*, Lecture 5, Spring 2011. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 6: From Poisson to Markov

6.1 Joint conditional densities for Poisson

Let a Poisson process of rate λ have interarrival intervals $X_1, X_2, \ldots$, arrival epochs $S_n = X_1 + \cdots + X_n$, and counting process $N(t)$. The interarrivals are IID exponential, so the joint density of $X_1, \ldots, X_n$ and S_{n+1} is

$$f_{X_1 \cdots X_n S_{n+1}}(x_1, \ldots, x_n, s_{n+1}) = \lambda^{n+1} e^{-\lambda s_{n+1}}, \qquad 0 < x_1, \ldots, x_n, \ \sum_k x_k < s_{n+1}.$$

Proposition 6.1 [Uniform conditional density]. *For $0 < x_1, \ldots, x_n$ and $\sum_{k=1}^{n} x_k < t$,*

$$f_{X_1 \cdots X_n | S_{n+1}}(x_1, \ldots, x_n \mid t) = \frac{\lambda^{n+1} e^{-\lambda t}}{\lambda^{n+1} t^n e^{-\lambda t}/n!} = \frac{n!}{t^n}, \tag{6.1}$$

$$f_{X_1 \cdots X_n | N(t)}(x_1, \ldots, x_n \mid n) = \frac{\lambda^n e^{-\lambda t}}{(\lambda t)^n e^{-\lambda t}/n!} = \frac{n!}{t^n}, \tag{6.2}$$

and both densities are 0 elsewhere. Thus, given either condition, the interarrivals are uniformly distributed over the constraint region.

Proof. For (6.1), divide the joint density at $s_{n+1} = t$ by the Erlang density $\lambda^{n+1} t^n e^{-\lambda t}/n!$ of S_{n+1}. For (6.2), the event $N(t) = n$ with $X_k \approx x_k$ requires $X_{n+1} > t - s_n$, where $s_n = \sum_k x_k$, so the joint density is $\lambda^n e^{-\lambda s_n} e^{-\lambda(t - s_n)} = \lambda^n e^{-\lambda t}$. Divide by $\Pr\{N(t) = n\} = (\lambda t)^n e^{-\lambda t}/n!$. □

The two results agree for a reason. Conditioning $X_1, \ldots, X_n$ on both $N(t) = n$ and $S_{n+1} = t_1$ for any $t_1 > t$ leaves (6.2) unchanged, and letting $t_1 \to t$ gives (6.1). So $n!/t^n$ is the density conditional on n arrivals in the open interval $(0, t)$, unaffected by future arrivals.

The density and the constraint region are unchanged by any permutation of $x_1, \ldots, x_n$. Hence, given $N(t) = n$, every X_k with $1 \le k \le n$ has the same marginal distribution as $X_1 = S_1$:

$$\Pr\{X_k > x \mid N(t) = n\} = \frac{(t - x)^n}{t^n}, \qquad 0 \le x \le t.$$

Indeed, the part of the region with $x_1 > x$ is a translate of the region for $t - x$, whose volume is $(t - x)^n/n!$; multiply by $n!/t^n$.

The last interval X_{n+1} is not covered by (6.1): under $S_{n+1} = t$ the $n + 1$ variables satisfy $\sum_{k=1}^{n+1} X_k = t$, so they have no $(n + 1)$-dimensional density. The same uniform density applies to every subset of n of the $n + 1$ variables, and the constraint is symmetric in all $n + 1$, so X_{n+1} has the same marginal as the others. The same holds under $N(t) = n$ with the final interval $X^*_{n+1} = t - S_n$ in place of X_{n+1}.

6.2 Finite-state Markov chains

Any integer-time process $\{X_n; n \geq 0\}$ is characterized by the probabilities $\Pr\{X_n = j \mid X_{n-1} = i, X_{n-2} = k, \ldots, X_0 = m\}$ for all $n \geq 1$ and all states.

Definition 6.2 [Finite-state Markov chain]. A finite-state Markov chain is an integer-time process $\{X_n; n \geq 0\}$ in which each rv X_n takes values in a fixed finite set, usually $\{1, 2, \ldots, M\}$, and

$$\Pr\{X_n = j \mid X_{n-1} = i, X_{n-2} = k, \ldots, X_0 = m\} = P_{ij}$$

for all $n \geq 1$ and all states $i, j, k, \ldots, m$, where P_{ij} depends only on i and j. The initial PMF p_{X_0} is arbitrary.

The definition says two things: X_n depends on the past only through X_{n-1}, and the transition probabilities P_{ij} do not depend on n. (Some authors call this a homogeneous chain and allow P_{ij} to vary with n in general.) The rv's X_n are dependent, but only in this simple way: the state X_n summarizes everything in the past that is relevant for the future. The chain is completely described by $\{P_{ij}; 1 \leq i, j \leq M\}$ and p_{X_0}; often the initial state is fixed and treated as a parameter, and the chain is identified with $\{P_{ij}\}$.

The transition probabilities can be viewed as a directed graph, with an arc from i to j exactly when $P_{ij} > 0$, or as the matrix $[P] = [P_{ij}]$. The graph shows which transitions are possible; the matrix suits algebraic and asymptotic questions.

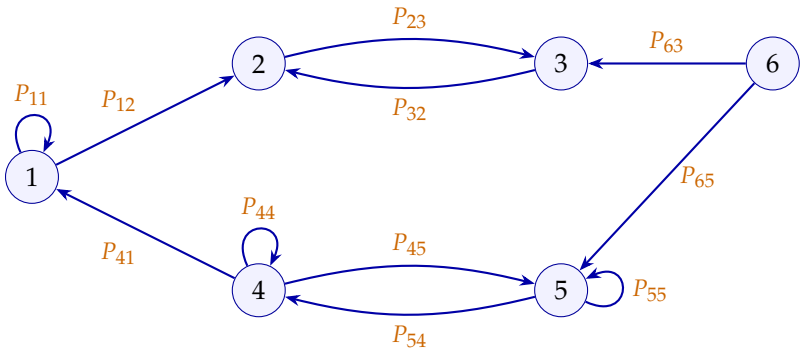


Figure 9: A six-state chain; each arc (i, j) is labeled with $P_{ij} > 0$. Its classes are $\{2, 3\}$, $\{4, 5\}$, $\{1\}$, and $\{6\}$. States 2 and 3 are recurrent; 1, 4, 5, and 6 are transient.

6.3 Classification of states

Definition 6.3 [Walks, paths, cycles]. An n-step walk is an ordered string of states $(i_0, i_1, \ldots, i_n)$, $n \geq 1$, with an arc from i_{m-1} to i_m for each $1 \leq m \leq n$. A path is a walk with no repeated states. A cycle is a walk whose last state equals its first and which repeats no other state.

In Figure 9, $(4,4,1,2,3,2)$ and $(4,1,2,3)$ are walks, $(4,1,2,3)$ and $(6,3,2)$ are paths, and $(2,3,2)$ and $(5,5)$ are cycles. Whether $(2,3,2)$ and $(3,2,3)$ count as the same cycle makes no difference.

Definition 6.4 [Accessibility]. State j is accessible from i, written $i \to j$, if a walk exists from i to j.

Let $P_{ij}^n = \Pr\{X_n = j \mid X_0 = i\}$. If (i,k,j) is a walk, then $P_{ij}^2 \geq P_{ik}P_{kj} > 0$; similarly, an n-step walk from i to j gives $P_{ij}^n > 0$. Thus $i \to j$ exactly when $P_{ij}^n > 0$ for some $n \geq 1$, and if $i \not\to j$ then $P_{ij}^n = 0$ for all $n \geq 1$: there is no possibility of ever reaching j from i. Concatenating walks shows that $i \to j$ and $j \to k$ imply $i \to k$.

Definition 6.5 [Communicating states and classes]. States i and j communicate, written $i \leftrightarrow j$, if $i \to j$ and $j \to i$. A class C is a nonempty set of states such that each $i \in C$ communicates with every other $j \in C$ and with no $j \notin C$.

Communication is transitive, and if $i \leftrightarrow j$ there is a closed walk through both i and j. In Figure 9 the classes are $C_1 = \{2,3\}$, $C_2 = \{4,5\}$, $C_3 = \{1\}$, and $C_4 = \{6\}$. The set $\{6\}$ is a class because 6 communicates with no other state; the definition does not require a state to communicate with itself, and no walk returns to 6.

Definition 6.6 [Recurrent and transient states]. A state i is recurrent if $j \to i$ for every j with $i \to j$; that is, no state from which there is no return can be entered. Otherwise i is transient.

In Figure 9, states 2 and 3 are recurrent. States 4 and 5 are transient, since $4 \to 1$ and $5 \to 1$ but $1 \not\to 4, 5$; states 1 and 6 are transient as well.

Theorem 6.7 [Recurrence is a class property]. *The states in a class are either all recurrent or all transient.*

Proof. Let i be recurrent and $\mathcal{S}_i = \{j : i \to j\}$. By recurrence, $j \to i$ for all $j \in \mathcal{S}_i$, so $i \leftrightarrow j$ exactly when $j \in \mathcal{S}_i$, and $\mathcal{S}_i$ is the class of i. If $j \in \mathcal{S}_i$ and $j \to k$, then $i \to k$, so $k \to i \to j$; hence j is recurrent. So one recurrent state makes its whole class recurrent, and a class with a transient state is entirely transient. □

6.4 Periodic states and classes

Definition 6.8 [Period]. The period of state i is $d(i) = \gcd\{n \geq 1 : P_{ii}^n > 0\}$. State i is aperiodic if $d(i) = 1$ and periodic with period $d(i)$ if $d(i) > 1$.

In Figure 10, $P_{44}^n > 0$ for $n = 4, 6, 8, 10, \ldots$, so $d(4) = 2$, and $P_{77}^n > 0$ for $n = 6, 10, 12, 14, \ldots$, so $d(7) = 2$.

Theorem 6.9 [Period is a class property]. *All states in the same class have the same period.*

Proof. Let $i \leftrightarrow j$, with walks of lengths a from i to j and b from j to i. Then $P_{ii}^{a+b} > 0$, and $P_{ii}^{a+n+b} > 0$ whenever $P_{jj}^{n} > 0$. So $d(i)$ divides $a+b$ and $a+n+b$, hence every such n, and $d(i) \mid d(j)$. By symmetry $d(i) = d(j)$. □

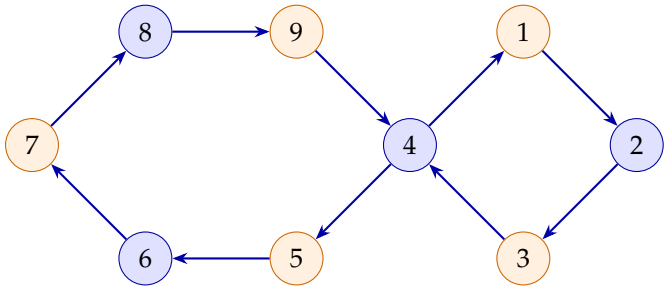


Figure 10: A recurrent class of period 2. Returns to 4 use the cycles of lengths 4 and 6 through 4. Blue and orange mark the two subclasses $\{2,4,6,8\}$ and $\{1,3,5,7,9\}$.

A periodic class with period $d > 1$ can be partitioned into subclasses $\mathcal{S}_1, \ldots, \mathcal{S}_d$ such that for $1 \le \ell < d$ and $i \in \mathcal{S}_\ell$, $P_{ij} > 0$ only for $j \in \mathcal{S}_{\ell+1}$, and for $i \in \mathcal{S}_d$, $P_{ij} > 0$ only for $j \in \mathcal{S}_1$. Starting in a given subclass, the state cycles through the d subclasses in order.

6.5 Ergodic Markov chains

Definition 6.10 [Ergodic class and chain]. A class that is both recurrent and aperiodic is ergodic. A Markov chain consisting of a single class that is ergodic is an ergodic Markov chain.

Ergodic chains gradually lose their memory of where they started: P_{ij}^{n} tends to a limit $\pi_j > 0$ as $n \to \infty$, independent of the starting state i. This result is basic to all finite-state chains and is proved in Lecture 7. A first step is the much weaker statement that $P_{ij}^{n} > 0$ for all large enough n, which is a combinatorial rather than a probabilistic question.

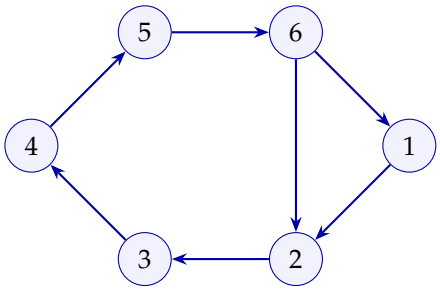


Figure 11: An ergodic six-state chain whose only cycles have lengths 5 and 6.

In Figure 11, starting in state 2, the next four states are forced $(3,4,5,6)$. From 6 there are two choices, after which the state is one of two possible states for several steps, then one of three, and so

on, until every state is possible.

Theorem 6.11 [Positivity of high powers]. *For an ergodic Markov chain with M states, $P^n_{ij} > 0$ for all i, j and all $n \geq (M-1)^2 + 1$.*

For the chain of Figure 11, $M = 6$ and the bound is 26, and it cannot be lowered: every closed walk from 1 to 1 has length $6c + 5a$ with $c \geq 1$ and $a \geq 0$, and 25 is not of this form, so $P^{25}_{11} = 0$.

Source. Robert Gallager, MIT 6.262, *Discrete Stochastic Processes*, Lecture 6, Spring 2011. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 7: Finite-state Markov chains; the matrix approach

7.1 The transition matrix and its powers

Since X_n depends on the past only through X_{n-1}, the joint probability of $X_0, \dots, X_n$ is determined by p_{X_0} and the transition probabilities:

$$\Pr\{X_0 = x_0, X_1 = x_1, \dots, X_n = x_n\} = p_{X_0}(x_0)\, P_{x_0 x_1} P_{x_1 x_2} \cdots P_{x_{n-1} x_n}.$$

The transition probabilities form the transition matrix $[P] = [P_{ij}]$, an $M \times M$ matrix with nonnegative entries and rows summing to 1.

Condition only on the initial state and let $P^n_{ij} = \Pr\{X_n = j \mid X_0 = i\}$. For $n = 2$,

$$P^2_{ij} = \sum_k \Pr\{X_2 = j \mid X_1 = k, X_0 = i\} \Pr\{X_1 = k \mid X_0 = i\} = \sum_k P_{ik} P_{kj},$$

which is the (i, j) entry of $[P][P] = [P^2]$. Iterating, $P^n_{ij} = \sum_k P^{n-1}_{ik} P_{kj}$, so P^n_{ij} is the (i, j) entry of $[P^n]$, the nth power of $[P]$. High powers are computed quickly by forming $[P^2], [P^4], [P^8], \dots$ by repeated squaring and multiplying the required ones. Since $[P^{m+n}] = [P^m][P^n]$, the Chapman–Kolmogorov equations hold:

$$P^{m+n}_{ij} = \sum_k P^m_{ik} P^n_{kj}.$$

7.2 Convergence and steady-state vectors

A central question is whether the effect of the initial state dies out, that is, whether $\lim_{n\to\infty} P^n_{ij} = \pi_j$ for all i, j, with π_j depending only on j. If so, multiplying by P_{jk} and summing over j gives

$$\pi_k = \lim_{n\to\infty} P^{n+1}_{ik} = \lim_{n\to\infty} \sum_j P^n_{ij} P_{jk} = \sum_j \pi_j P_{jk}.$$

In matrix terms, if $\lim_n [P^n]$ exists and has every row equal to $\vec{\pi}$, then $\vec{\pi} = \vec{\pi}[P]$.

Definition 7.1 [Steady-state vector]. A probability vector is a vector $\vec{\pi} = (\pi_1, \dots, \pi_M)$ with $\pi_i \geq 0$ and $\sum_i \pi_i = 1$. A probability vector is a steady-state vector for $[P]$ if $\vec{\pi} = \vec{\pi}[P]$.

This does not reduce the question to the study of $\vec{\pi} = \vec{\pi}[P]$. Surprisingly, $\lim_n [P^n]$ is relatively simple to study, while the solution set of $\vec{\pi} = \vec{\pi}[P]$ is more complicated. That equation always has at least one probability-vector solution, often more, but a solution does not imply that $\lim_n [P^n]$ exists (see the first example below).

7.3 Ergodic Markov chains

Convergence of $[P^n]$ to a matrix with equal rows $\vec{\pi}$ means that for each column j, $\lim_n P^n_{ij} = \pi_j$ for every i.

Theorem 7.2 [Convergence for ergodic chains]. *Let $[P]$ be the transition matrix of an ergodic finite-state Markov chain. Then for each j, $\max_i P^n_{ij}$ is nonincreasing in n, $\min_i P^n_{ij}$ is nondecreasing in n, and*

$$\lim_{n\to\infty} \max_i P^n_{ij} = \lim_{n\to\infty} \min_i P^n_{ij} = \pi_j > 0,$$

with exponential convergence in n.

The key is the monotonicity of the maxima and minima, which holds for every finite-state chain with an almost trivial proof.

Lemma 7.3 [Monotone extremes]. *Let $[P]$ be the transition matrix of an arbitrary finite-state Markov chain. Then for each j, $\max_i P^n_{ij}$ is nonincreasing in n and $\min_i P^n_{ij}$ is nondecreasing in n.*

Proof. For any states i, j and any n,

$$P^{n+1}_{ij} = \sum_k P_{ik} P^n_{kj} \le \sum_k P_{ik} \max_\ell P^n_{\ell j} = \max_\ell P^n_{\ell j}.$$

This holds for the maximizing i, so $\max_i P^{n+1}_{ij} \le \max_\ell P^n_{\ell j}$. Replacing maxima by minima and reversing the inequalities gives $\min_i P^{n+1}_{ij} \ge \min_\ell P^n_{\ell j}$. □

Example: Take $M = 2$ with $P_{12} = P_{21} = 1$. Then P^n_{12} alternates $1, 0, 1, \ldots$ and P^n_{22} alternates $0, 1, 0, \ldots$. The maximum of the two is always 1 and the minimum always 0: monotone, but the limits differ. Here $\vec{\pi} = (1/2, 1/2)$ satisfies $\vec{\pi} = \vec{\pi}[P]$ although $[P^n]$ does not converge.

Example: Take the ergodic chain with $P_{12} = P_{21} = 3/4$. Then $P^n_{12} = \frac{3}{4}, \frac{3}{8}, \frac{9}{16}, \ldots$ and $P^n_{22} = \frac{1}{4}, \frac{5}{8}, \frac{7}{16}, \ldots$. Each sequence oscillates while approaching $1/2$, but $\max(P^n_{12}, P^n_{22}) = \frac{3}{4}, \frac{5}{8}, \frac{9}{16}, \ldots$ decreases to $1/2$ and $\min(P^n_{12}, P^n_{22}) = \frac{1}{4}, \frac{3}{8}, \frac{7}{16}, \ldots$ increases to $1/2$.

Before proving the theorem in general, specialize to $[P] > 0$, meaning $P_{ij} > 0$ for all i, j.

Lemma 7.4 [Strictly positive matrices]. *Let $[P] > 0$ be the transition matrix of a finite-state Markov chain and $\alpha = \min_{i,j} P_{ij}$. Then for all states j and all $n \ge 1$,*

$$\max_i P^{n+1}_{ij} - \min_i P^{n+1}_{ij} \le \left(\max_\ell P^n_{\ell j} - \min_\ell P^n_{\ell j}\right)(1 - 2\alpha),$$

$$\max_\ell P^n_{\ell j} - \min_\ell P^n_{\ell j} \le (1 - 2\alpha)^n,$$

$$\lim_{n\to\infty} \max_\ell P^n_{\ell j} = \lim_{n\to\infty} \min_\ell P^n_{\ell j} > 0.$$

Proof. Tighten the proof of Lemma 7.3. For given n and j, let $\ell_{\min}$ minimize $P^n_{\ell j}$ over ℓ. Then

$$\begin{aligned} P^{n+1}_{ij} = \sum_k P_{ik} P^n_{kj} &\le \sum_{k \ne \ell_{\min}} P_{ik} \max_\ell P^n_{\ell j} + P_{i\ell_{\min}} \min_\ell P^n_{\ell j} \\ &= \max_\ell P^n_{\ell j} - P_{i\ell_{\min}} \Big(\max_\ell P^n_{\ell j} - \min_\ell P^n_{\ell j} \Big) \\ &\le \max_\ell P^n_{\ell j} - \alpha \Big(\max_\ell P^n_{\ell j} - \min_\ell P^n_{\ell j} \Big). \end{aligned}$$

This bounds $\max_i P^{n+1}_{ij}$. Interchanging max with min and $\le$ with $\ge$,

$$\min_i P^{n+1}_{ij} \ge \min_\ell P^n_{\ell j} + \alpha \Big(\max_\ell P^n_{\ell j} - \min_\ell P^n_{\ell j} \Big),$$

and subtracting gives the first claim. Each row has at least two entries, each at least α, so $\min_\ell P_{\ell j} \ge \alpha$ and $\max_\ell P_{\ell j} \le 1 - \alpha$; hence $\max_\ell P_{\ell j} - \min_\ell P_{\ell j} \le 1 - 2\alpha$, and iterating gives the second claim. By Lemma 7.3, $\max_\ell P^n_{\ell j}$ is a nonincreasing nonnegative sequence and so has a limit, and likewise for the minima; the second claim makes the limits equal, and they are at least $\min_\ell P_{\ell j} \ge \alpha > 0$. □

Proof of Theorem 7.2. By the positivity theorem of Lecture 6, $[P^h] > 0$ for $h = (M-1)^2 + 1$. Apply Lemma 7.4 to $[P^h]$ with $\alpha = \min_{i,j} P^h_{ij} > 0$; only the positivity of α matters. Then

$$\lim_{m \to \infty} \max_\ell P^{hm}_{\ell j} = \lim_{m \to \infty} \min_\ell P^{hm}_{\ell j} = \pi_j > 0.$$

By Lemma 7.3, $\max_\ell P^n_{\ell j}$ is nonincreasing in n, so it has the same limit as its subsequence $\max_\ell P^{hm}_{\ell j}$; the same argument applies to the minima. Moreover the difference $\max_\ell P^n_{\ell j} - \min_\ell P^n_{\ell j}$ is nonincreasing in n, so it is at most $(1-2\alpha)^{\lfloor n/h \rfloor}$, which is the exponential convergence. □

So for ergodic chains $\lim_n P^n_{ij} = \pi_j$ for all i, where $\vec{\pi}$ is a probability vector. By the computation above, $\vec{\pi}$ is a steady-state vector, and it is the unique one: if $\vec{\pi}'$ is any steady-state vector, then $\vec{\pi}' = \vec{\pi}'[P^n] \to \vec{\pi}' \vec{e} \vec{\pi} = \vec{\pi}$, where $\vec{e} = (1, \ldots, 1)^{\mathsf{T}}$.

7.4 Ergodic unichains

Definition 7.5 [Ergodic unichain]. An ergodic unichain is a finite-state chain containing a single ergodic class together with an arbitrary set of transient states.

If a state forms a singleton transient class, there is a fixed probability, say α, of leaving it at each step, and the probability of remaining in it for more than n steps is $(1-\alpha)^n$. The probability of remaining in an arbitrary set of transient states also decays to 0 exponentially in n. Each transient

state has a path, of length at most M, to a recurrent state; if $\beta > 0$ is the smallest probability over transient starting states of reaching the recurrent class within M steps, the probability of still being transient after kM steps is at most $(1-\beta)^k$. So the ergodic class is eventually entered, and steady state within that class is then reached. For every state j,

$$\lim_{n\to\infty} \max_i P_{ij}^n = \lim_{n\to\infty} \min_i P_{ij}^n = \pi_j,$$

where now $\pi_j = 0$ for each transient state and $\pi_j > 0$ for each recurrent state.

7.5 Other finite-state Markov chains

Consider a chain with several ergodic classes $C_1, \ldots, C_m$. The classes do not communicate and should be analyzed separately. Analyzed as a whole, $[P]$ has m linearly independent steady-state vectors, one nonzero on each class. $[P^n]$ still converges, but its rows are not all equal: there are m sets of rows, and the limiting row for a state of class k is nonzero only on the states of class k.

Next consider a recurrent chain that is periodic with period d. It separates into d subclasses with a cyclic rotation between them. In $[P^d]$ each subclass becomes an ergodic class $C_1, \ldots, C_d$, so $\lim_n [P^{nd}]$ exists. A steady state is reached within each subclass, but the chain keeps rotating from one subclass to the next.

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Lecture 8: Markov eigenvalues and eigenvectors

8.1 Ergodic unichains in matrix form

For an ergodic finite-state chain, $\lim_n P^n_{ij} = \pi_j$, where $\vec{\pi} = (\pi_1, \ldots, \pi_M)$ is a strictly positive probability vector. Multiplying by P_{jk} and summing over j, $\pi_k = \lim_n \sum_j P^n_{ij} P_{jk} = \sum_j \pi_j P_{jk}$, so $\vec{\pi}$ is a steady-state vector: $\vec{\pi} = \vec{\pi}[P]$ and $\vec{\pi} \geq 0$. In matrix terms,

$$\lim_{n\to\infty} [P^n] = \vec{e}\vec{\pi},$$

where $\vec{e} = (1, 1, \ldots, 1)^{\mathsf{T}}$ is a column vector and $\vec{\pi}$ is a row vector.

The same holds for ergodic unichains, except that the steady-state vector is positive on the ergodic states and 0 on the transient states. Listing the t transient states first,

$$[P] = \begin{bmatrix} [P_{\mathcal{T}}] & [P_{\mathcal{TR}}] \\ [0] & [P_{\mathcal{R}}] \end{bmatrix}, \qquad [P_{\mathcal{T}}] = \begin{bmatrix} P_{11} & \cdots & P_{1t} \\ \vdots & & \vdots \\ P_{t1} & \cdots & P_{tt} \end{bmatrix}.$$

Each transient state eventually makes a transition, through $[P_{\mathcal{TR}}]$, to a recurrent state, and the recurrent class then reaches steady state as before.

8.2 Linear algebra facts

Definition 8.1 [Eigenvalues and eigenvectors]. A complex number λ is an eigenvalue of a real square matrix $[A]$, and a complex vector $\vec{v} \neq 0$ is a right eigenvector of $[A]$, if $[A]\vec{v} = \lambda\vec{v}$, or equivalently $[A - \lambda I]\vec{v} = 0$. A row vector $\vec{u} \neq 0$ with $\vec{u}[A] = \lambda\vec{u}$ is a left eigenvector.

Every stochastic matrix satisfies $\sum_j P_{ij} = 1$, that is, $[P]\vec{e} = \vec{e}$. So $\lambda = 1$ is an eigenvalue of every stochastic matrix, with right eigenvector $\vec{e}$.

A square matrix $[A]$ is singular if $[A]\vec{v} = 0$ for some $\vec{v} \neq 0$. This happens exactly when the columns of $[A]$ are linearly dependent, exactly when the rows are, and exactly when $\det[A] = 0$. Hence the following are equivalent: λ is an eigenvalue of $[P]$; $[P - \lambda I]$ is singular; $\det[P - \lambda I] = 0$; $[P]\vec{v} = \lambda\vec{v}$ for some $\vec{v} \neq 0$; and $\vec{u}[P] = \lambda\vec{u}$ for some $\vec{u} \neq 0$.

In particular $[P - I]$ is singular, so there is a row vector $\vec{\pi} \neq 0$ with $\vec{\pi}[P] = \vec{\pi}$. This does not show that a probability vector solves this equation; that is shown below for every chain, and for ergodic unichains it is already known.

The determinant of an $M \times M$ matrix is

$$\det[A] = \sum_{\mu} \pm \prod_{i=1}^{M} A_{i,\mu(i)},$$

the sum being over all permutations μ of $1, \dots, M$, with + for even and − for odd permutations. Consequently $\det[P - \lambda I]$ is a polynomial of degree M in λ, and $[P]$ has M eigenvalues, the roots of $\det[P - \lambda I] = 0$. If k of the roots equal λ, the eigenvalue λ has ALGEBRAIC MULTIPLICITY k.

8.3 Markov chains with two states

The left and right eigenvector equations are

$$\begin{aligned} \pi_1 P_{11} + \pi_2 P_{21} &= \lambda \pi_1, & P_{11}\nu_1 + P_{12}\nu_2 &= \lambda \nu_1, \\ \pi_1 P_{12} + \pi_2 P_{22} &= \lambda \pi_2, & P_{21}\nu_1 + P_{22}\nu_2 &= \lambda \nu_2. \end{aligned}$$

With $P_{11} = 1 - P_{12}$ and $P_{22} = 1 - P_{21}$,

$$\det[P - \lambda I] = (P_{11} - \lambda)(P_{22} - \lambda) - P_{12}P_{21} = (\lambda - 1)\big(\lambda - (1 - P_{12} - P_{21})\big),$$

so $\lambda_1 = 1$ and $\lambda_2 = 1 - P_{12} - P_{21}$. If $P_{12} = P_{21} = 0$ the chain has two recurrent classes and $\lambda = 1$ has multiplicity 2; otherwise it has multiplicity 1. If $P_{12} = P_{21} = 1$ the chain is periodic and $\lambda_2 = -1$; otherwise $|\lambda_2| < 1$.

Assume from now on that $P_{12} > 0$ or $P_{21} > 0$. Then the eigenvectors are

$$\begin{aligned} \vec{\pi}^{(1)} &= \left(\frac{P_{21}}{P_{12}+P_{21}}, \frac{P_{12}}{P_{12}+P_{21}}\right), & \vec{\nu}^{(1)} &= (1,1)^{\mathsf{T}}, \\ \vec{\pi}^{(2)} &= (1,-1), & \vec{\nu}^{(2)} &= \left(\frac{P_{12}}{P_{12}+P_{21}}, \frac{-P_{21}}{P_{12}+P_{21}}\right)^{\mathsf{T}}, \end{aligned}$$

and they satisfy $\vec{\pi}^{(i)}\vec{\nu}^{(j)} = \delta_{ij}$.

Proposition 8.2 [BIORTHOGONALITY]. *If $\vec{\pi}^{(i)}[P] = \lambda_i \vec{\pi}^{(i)}$ and $[P]\vec{\nu}^{(j)} = \lambda_j \vec{\nu}^{(j)}$ with $\lambda_i \neq \lambda_j$, then $\vec{\pi}^{(i)}\vec{\nu}^{(j)} = 0$.*

Proof. $\lambda_i \vec{\pi}^{(i)}\vec{\nu}^{(j)} = \vec{\pi}^{(i)}[P]\vec{\nu}^{(j)} = \lambda_j \vec{\pi}^{(i)}\vec{\nu}^{(j)}$, and $\lambda_i \neq \lambda_j$. □

Normalizing either vector makes $\vec{\pi}^{(i)}\vec{\nu}^{(i)} = 1$. The right eigenvector equations combine into $[P][U] = [U][\Lambda]$, where

$$[\Lambda] = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}, \qquad [U] = \begin{bmatrix} \nu_1^{(1)} & \nu_1^{(2)} \\ \nu_2^{(1)} & \nu_2^{(2)} \end{bmatrix}.$$

Since $\vec{\pi}^{(i)}\vec{\nu}^{(j)} = \delta_{ij}$, the matrix with rows $\vec{\pi}^{(1)}, \vec{\pi}^{(2)}$ times $[U]$ is $[I]$. So $[U]$ is invertible, $[U^{-1}]$ has rows

$\vec{\pi}^{(1)}$ and $\vec{\pi}^{(2)}$, $[P] = [U][\Lambda][U^{-1}]$, and $[P^2] = [U][\Lambda][U^{-1}][U][\Lambda][U^{-1}] = [U][\Lambda^2][U^{-1}]$. Similarly, for every $n \geq 1$,

$$[P^n] = [U][\Lambda^n][U^{-1}]. \tag{8.1}$$

Splitting $[\Lambda^n] = [\Lambda_1^n] + \cdots + [\Lambda_M^n]$, where $[\Lambda_i^n]$ has λ_i^n in position (i,i) and zeros elsewhere, gives

$$[P^n] = \sum_{i=1}^{M} \lambda_i^n \, \vec{\nu}^{(i)} \vec{\pi}^{(i)}. \tag{8.2}$$

For $M = 2$ the steady-state vector is $\vec{\pi} = \vec{\pi}^{(1)}$, and

$$\vec{\nu}^{(1)}\vec{\pi} = \begin{bmatrix} \pi_1 & \pi_2 \\ \pi_1 & \pi_2 \end{bmatrix}, \qquad \vec{\nu}^{(2)}\vec{\pi}^{(2)} = \begin{bmatrix} \pi_2 & -\pi_2 \\ -\pi_1 & \pi_1 \end{bmatrix}, \qquad [P^n] = \begin{bmatrix} \pi_1 + \pi_2\lambda_2^n & \pi_2 - \pi_2\lambda_2^n \\ \pi_1 - \pi_1\lambda_2^n & \pi_2 + \pi_1\lambda_2^n \end{bmatrix}.$$

This solution is exact: $[P^n]$ converges to $\vec{e}\vec{\pi}$ geometrically at rate $|\lambda_2|$. For $P_{12} = P_{21} = 3/4$, $\lambda_2 = -1/2$ and $P_{12}^n = \frac{1}{2} - \frac{1}{2}(-\frac{1}{2})^n$, which reproduces the oscillating sequence $\frac{3}{4}, \frac{3}{8}, \frac{9}{16}, \ldots$ of Lecture 7.

8.4 Distinct eigenvalues for $M > 2$ states

Now let $[P]$ have M distinct eigenvalues. Each λ_i has a right eigenvector $\vec{\nu}^{(i)}$ and a left eigenvector $\vec{\pi}^{(i)}$, with $\vec{\pi}^{(i)}\vec{\nu}^{(j)} = 0$ for $j \neq i$ by Proposition 8.2; scale so that $\vec{\pi}^{(i)}\vec{\nu}^{(i)} = 1$. Let $[U]$ have columns $\vec{\nu}^{(1)}, \ldots, \vec{\nu}^{(M)}$ and $[V]$ rows $\vec{\pi}^{(1)}, \ldots, \vec{\pi}^{(M)}$. Then $[V][U] = I$, so $[V] = [U^{-1}]$; the right eigenvectors are linearly independent and span M-space, and so are the left ones. As before, $[P][U] = [U][\Lambda]$ gives $[P] = [U][\Lambda][U^{-1}]$, $[P^n] = [U][\Lambda^n][U^{-1}]$, and the expansion (8.2). Since each row of $[P]$ sums to 1, $\vec{e}$ is a right eigenvector of eigenvalue 1.

Theorem 8.3 [Steady-state vectors exist]. *Let $\vec{\pi}$ be a left eigenvector of $[P]$ for eigenvalue 1. Then $(|\pi_1|, \ldots, |\pi_M|)$, normalized to sum 1, is a steady-state vector. If the eigenvalue 1 is simple (as when all eigenvalues are distinct), then $\vec{\pi}$ itself, normalized to $\vec{\pi}\vec{e} = 1$, is a steady-state vector.*

Proof. For every j, $\pi_j = \sum_k \pi_k P_{kj}$, so

$$|\pi_j| \leq \sum_k |\pi_k| P_{kj}. \tag{8.3}$$

Summing over j gives $\sum_j |\pi_j| \leq \sum_k |\pi_k|$, which holds with equality, so (8.3) holds with equality for each j. Thus $(|\pi_1|, \ldots, |\pi_M|)$ is a nonnegative nonzero solution of the steady-state equation; normalize it to sum 1. If 1 is a simple eigenvalue, its left eigenvectors are all multiples of one another, so $\vec{\pi} = c\,(|\pi_1|, \ldots, |\pi_M|)$ for a constant c, and $\vec{\pi}\vec{e} = 1$ forces $c = 1/\sum_j |\pi_j| > 0$. □

Theorem 8.4 [Eigenvalues lie in the unit disk]. *Every eigenvalue λ_ℓ of a stochastic matrix satisfies $|\lambda_\ell| \leq 1$.*

Proof. A left eigenvector $\vec{\pi}^{(\ell)}$ of $[P]$ with eigenvalue λ_ℓ is a left eigenvector of $[P^n]$ with eigenvalue λ_ℓ^n. Thus for all j,

$$\lambda_\ell^n \pi_j^{(\ell)} = \sum_i \pi_i^{(\ell)} P_{ij}^n, \qquad |\lambda_\ell^n|\,|\pi_j^{(\ell)}| \le \sum_i |\pi_i^{(\ell)}| P_{ij}^n.$$

Let $\beta > 0$ be the largest of $|\pi_j^{(\ell)}|$ over j. For the maximizing j, $|\lambda_\ell^n|\beta \le \sum_i \beta P_{ij}^n \le \beta M$. So $|\lambda_\ell|^n \le M$ for all n, which forces $|\lambda_\ell| \le 1$. □

Both theorems hold for all finite-state chains. With M distinct eigenvalues, $[P^n] = \sum_i \lambda_i^n \vec{\nu}^{(i)} \vec{\pi}^{(i)}$. For an ergodic unichain one eigenvalue is 1 and the others are strictly less than 1 in magnitude: since $[P^n]$ converges (Lecture 7), $\lambda_i^n = \vec{\pi}^{(i)}[P^n]\vec{\nu}^{(i)}$ converges, which for $\lambda_i \neq 1$ requires $|\lambda_i| < 1$. So the rate at which $[P^n]$ approaches $\vec{e}\vec{\pi}$ is set by the second largest eigenvalue in magnitude. If $[P]$ is a periodic unichain with period d, there are d eigenvalues equally spaced around the unit circle, and $[P^n]$ does not converge.

8.5 Repeated eigenvalues with a full set of eigenvectors

Suppose next that some eigenvalue has multiplicity $k > 1$ but has k linearly independent eigenvectors. The left eigenvectors of that eigenvalue can be chosen biorthonormal to its right eigenvectors. With $[U]$ the matrix of columns $\vec{\nu}^{(1)}, \ldots, \vec{\nu}^{(M)}$, $[U]$ is invertible, $[U^{-1}]$ has rows $\vec{\pi}^{(1)}, \ldots, \vec{\pi}^{(M)}$, and again $[P^n] = \sum_{i=1}^M \lambda_i^n \vec{\nu}^{(i)} \vec{\pi}^{(i)}$.

Example: A chain consisting of ℓ ergodic sets of states. Each set gives an eigenvalue 1 whose right eigenvector is 1 on that set and 0 elsewhere, and a steady-state vector that is nonzero only on that set. $[P^n]$ converges to a block-diagonal matrix in which, within each ergodic set, all rows are the same.

8.6 The Jordan form

An eigenvalue of algebraic multiplicity $k \ge 2$ may have fewer than k linearly independent eigenvectors. Then $[P] = [U][\Lambda][U^{-1}]$ is replaced by a Jordan form $[P] = [U][J][U^{-1}]$, where $[J]$ carries the eigenvalues on the main diagonal and ones on the next diagonal up where needed for the deficient eigenvectors, as in

$$[J] = \begin{bmatrix} \lambda_1 & 0 & 0 & 0 & 0 \\ 0 & \lambda_1 & 0 & 0 & 0 \\ 0 & 0 & \lambda_2 & 1 & 0 \\ 0 & 0 & 0 & \lambda_2 & 0 \\ 0 & 0 & 0 & 0 & \lambda_2 \end{bmatrix}.$$

Example: Let

$$[P] = \begin{bmatrix} 1/2 & 1/2 & 0 \\ 0 & 1/2 & 1/2 \\ 0 & 0 & 1 \end{bmatrix}.$$

The eigenvalues are 1 and $1/2$, the latter with algebraic multiplicity 2 but only one eigenvector up to scaling. Here $P^n_{11} = 2^{-n}$, $P^n_{12} = n2^{-n}$, and $P^n_{13} = 1 - (n+1)2^{-n}$, so $[P^n]$ approaches its steady state $\vec{e}\,(0, 0, 1)$ as $n(1/2)^n$.

For stochastic $[P]$, the eigenvalue 1 always has as many linearly independent eigenvectors as its algebraic multiplicity.

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Lecture 9: Markov rewards and dynamic programming

9.1 Eigenvalues of block-triangular chains

The determinant of an $M \times M$ matrix is

$$\det[A] = \sum_{\mu} \pm \prod_{i=1}^{M} A_{i,\mu(i)}, \tag{9.1}$$

summed over all permutations μ of $1, \ldots, M$.

Proposition 9.1 [Block-triangular determinants]. *If $[A_{\mathcal{T}}]$ is $t \times t$ and*

$$[A] = \begin{bmatrix} [A_{\mathcal{T}}] & [A_{\mathcal{TR}}] \\ [0] & [A_{\mathcal{R}}] \end{bmatrix}, \qquad \textit{then} \qquad \det[A] = \det[A_{\mathcal{T}}] \det[A_{\mathcal{R}}].$$

Proof. For a product in (9.1) to be nonzero, $\mu(i) > t$ whenever $i > t$. Then μ maps $\{t+1, \ldots, M\}$ onto itself, and hence $\mu(i) \le t$ when $i \le t$. So the contributing permutations factor into a permutation of $1, \ldots, t$ and one of $t+1, \ldots, M$, and the sum factors accordingly. □

For a unichain with the t transient states listed first,

$$\det[P - \lambda I] = \det[P_{\mathcal{T}} - \lambda I_t] \det[P_{\mathcal{R}} - \lambda I_r],$$

so the eigenvalues of $[P]$ are the t eigenvalues of $[P_{\mathcal{T}}]$ together with the r eigenvalues of $[P_{\mathcal{R}}]$. If $\vec{\pi}$ is a left eigenvector of $[P_{\mathcal{R}}]$, then $(0, \ldots, 0, \pi_1, \ldots, \pi_r)$ is a left eigenvector of $[P]$ with the same eigenvalue:

$$(\vec{0} \mid \vec{\pi}) \begin{bmatrix} [P_{\mathcal{T}}] & [P_{\mathcal{TR}}] \\ [0] & [P_{\mathcal{R}}] \end{bmatrix} = \lambda (\vec{0} \mid \vec{\pi}).$$

The left eigenvectors of $[P_{\mathcal{T}}]$ are more complicated and not very interesting.

The same argument applies with two recurrent classes,

$$[P] = \begin{bmatrix} [P_{\mathcal{T}}] & [P_{\mathcal{TR}}] & [P_{\mathcal{TR}'}] \\ [0] & [P_{\mathcal{R}}] & [0] \\ [0] & [0] & [P_{\mathcal{R}'}] \end{bmatrix}, \qquad \det[P - \lambda I] = \det[P_{\mathcal{T}} - \lambda I_t] \det[P_{\mathcal{R}} - \lambda I_r] \det[P_{\mathcal{R}'} - \lambda I_{r'}].$$

The eigenvalues of $[P]$ are the t from $[P_{\mathcal{T}}]$, the r from $[P_{\mathcal{R}}]$, and the r' from $[P_{\mathcal{R}'}]$. A left eigenvector $\vec{\pi}$ of $[P_{\mathcal{R}}]$ gives the left eigenvector $(\vec{0}, \vec{\pi}, \vec{0})$ of $[P]$, and one of $[P_{\mathcal{R}'}]$ gives $(\vec{0}, \vec{0}, \vec{\pi})$.

9.2 Rewards for Markov chains

Suppose each state i carries a given REWARD r_i. With X_n the state at time n, the reward at time n is the rv $R(X_n)$ mapping $X_n = i$ to r_i. Only expected rewards matter here; for example, $\mathbb{E}[R(X_n) \mid X_0 = i] = \sum_j r_j P^n_{ij}$. The expected AGGREGATE REWARD over the n steps from m to $m+n-1$, given $X_m = i$, is

$$v_i(n) = \mathbb{E}\big[R(X_m) + \cdots + R(X_{m+n-1}) \mid X_m = i\big] = r_i + \sum_j P_{ij} r_j + \cdots + \sum_j P^{n-1}_{ij} r_j,$$

which does not depend on m. If the chain is an ergodic unichain, the successive terms tend to the steady-state GAIN per step

$$g = \sum_j \pi_j r_j,$$

independent of the starting state. So $v_i(n)$ is a transient depending on i plus ng. The transient is important, particularly when $g = 0$.

9.3 Expected first-passage times

In an arbitrary unichain, consider the expected number of steps, starting from a given state $i \neq 1$, until some given recurrent state, say 1, is first entered. This is a reward problem: assign one unit of reward to each successive state until state 1 is entered.

Modify the chain by setting $P_{11} = 1$, and set $r_1 = 0$ and $r_i = 1$ for $i \neq 1$, so the reward stops when state 1 is entered. For each sample path starting from $i \neq 1$, the probability of the initial segment up to the first entry to 1 is unchanged, so the expected FIRST-PASSAGE TIME is unchanged. The modified chain is an ergodic unichain whose single recurrent state 1 is a TRAPPING STATE. If state 1 is first entered at time ℓ, the aggregate reward over times $0, \ldots, n-1$ is ℓ for all $n \geq \ell$. So the expected first-passage time from i is $v_i = \lim_{n\to\infty} v_i(n)$.

There is a neat way to find all of these at once. Start at $X_0 = i \neq 1$. There is a unit reward at time 0, and given $X_1 = j$ the remaining expected reward is v_j. Hence

$$v_i = 1 + \sum_j P_{ij} v_j \quad (i \neq 1), \qquad v_1 = 0,$$

or in vector form, with $P_{11} = 1$,

$$\vec{v} = \vec{r} + [P]\vec{v}, \qquad \vec{r} = (0, 1, 1, \ldots, 1)^{\mathsf{T}}, \qquad v_1 = 0.$$

If $\vec{v}$ satisfies $\vec{v} = \vec{r} + [P]\vec{v}$, so does $\vec{v} + \alpha\vec{e}$; the condition $v_1 = 0$ resolves this ambiguity. The solution with $v_1 = 0$ is unique: the difference $\vec{w}$ of two solutions satisfies $\vec{w} = [P]\vec{w}$, and since 1 is a simple eigenvalue of $[P]$, $\vec{w} = \alpha\vec{e}$; then $w_1 = 0$ gives $\vec{w} = 0$.

9.4 Aggregate rewards with a final reward

Often the interest is in the aggregate reward over n steps, from m to $m+n-1$, followed by a final reward u_j for $X_{m+n} = j$. This flexibility is particularly valuable in dynamic programming. The aggregate expected reward including the final reward is

$$v_i(n, \vec{u}) = r_i + \sum_j P_{ij} r_j + \cdots + \sum_j P_{ij}^{n-1} r_j + \sum_j P_{ij}^n u_j,$$

or in vector form

$$\vec{v}(n, \vec{u}) = \vec{r} + [P]\vec{r} + \cdots + [P^{n-1}]\vec{r} + [P^n]\vec{u}.$$

Equivalently, $\vec{v}(n, \vec{u}) = \vec{r} + [P]\vec{v}(n-1, \vec{u})$ with $\vec{v}(0, \vec{u}) = \vec{u}$.

9.5 Dynamic programming

Consider a finite set of states $1, \ldots, M$ in discrete time. At each time the decision maker observes the state, say j, and chooses one of a finite set of alternatives. Alternative k consists of a current reward $r_j^{(k)}$ and a set of transition probabilities $\{P_{jl}^{(k)}; 1 \le l \le M\}$ to the next state.

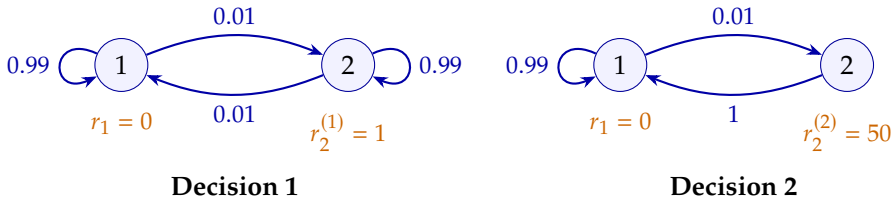


Figure 12: Two alternatives in state 2; state 1 has only one. Decision 2 seeks instant gratification, decision 1 long-term gratification.

In Figure 12, always using decision 1 gives a symmetric chain with $\vec{\pi} = (1/2, 1/2)$ and gain $g = 1/2$; always using decision 2 gives $\vec{\pi} = (100/101, 1/101)$ and gain $g = 50/101 < 1/2$. Over a single step, though, decision 2 earns 50 rather than 1.

The process of random transitions and state-based decisions starts at time m in a given state and continues until time $m+n-1$. After the nth decision, made at time $m+n-1$, there is a final transition based on that decision, and at time $m+n$ there is a final reward $\vec{u} = (u_1, \ldots, u_M)$ based on the final state. The objective of DYNAMIC PROGRAMMING is to find both the optimal decision at each time and the maximal expected aggregate reward $v_i^*(n, \vec{u})$ for each starting state i and each number of steps n. It is best to start with a single step and proceed to more steps, which amounts to starting at the end and working back to the beginning. The algorithm is due to Richard Bellman.

One step. Starting at time m in state i, decision k gives reward $r_i^{(k)}$ at time m and then, through $P_{ij}^{(k)}$, the expected final reward $\sum_j u_j P_{ij}^{(k)}$ at time $m+1$. Choosing k to maximize the total,

$$v_i^*(1,\vec{u}) = \max_k \Big[r_i^{(k)} + \sum_j P_{ij}^{(k)} u_j \Big].$$

Two steps. The key fact is that an optimal decision at time $m+1$ can be selected using only the state j at time $m+1$, and it is optimal whatever decision was made at time m. Given $X_{m+1} = j$, the maximal expected reward over times $m+1$ and $m+2$ is $\max_k \big[r_j^{(k)} + \sum_l P_{jl}^{(k)} u_l \big] = v_j^*(1,\vec{u})$. Hence the maximal expected aggregate reward over $m, m+1, m+2$ given $X_m = i$ is

$$v_i^*(2,\vec{u}) = \max_k \Big[r_i^{(k)} + \sum_j P_{ij}^{(k)} v_j^*(1,\vec{u}) \Big].$$

General n. The same reasoning gives, for $n = 3$ and in general,

$$v_i^*(n,\vec{u}) = \max_k \Big[r_i^{(k)} + \sum_j P_{ij}^{(k)} v_j^*(n-1,\vec{u}) \Big], \qquad v_j^*(0,\vec{u}) = u_j. \tag{9.2}$$

The maximizing k in (9.2) is the optimal decision in state i when n steps remain. For a given n, the algorithm computes $v_j^*(h,\vec{u})$ for all states j and all $h \le n$, starting at $h = 1$.

Algorithm 9.1: Dynamic programming

Input: Rewards $r_i^{(k)}$, transition probabilities $P_{ij}^{(k)}$, final reward $\vec{u}$, horizon n.
Output: Optimal rewards $v_i^*(h,\vec{u})$ and decisions $k_i^*(h)$ for $1 \le h \le n$.

```
v*_j(0, u⃗) ← u_j for every state j
for h ← 1 to n do
    for each state i do
        k*_i(h) ← arg max_k [ r_i^(k) + Σ_j P_ij^(k) v*_j(h − 1, u⃗) ]
        v*_i(h, u⃗) ← max_k [ r_i^(k) + Σ_j P_ij^(k) v*_j(h − 1, u⃗) ]
return {v*_i(h, u⃗)} and {k*_i(h)}
```

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Lecture 10: Renewals and the strong law of large numbers (SLLN)

10.1 Renewal processes

An arrival process can be specified by its arrival epochs $\{S_1, S_2, \ldots\}$, by its interarrival intervals $\{X_1, X_2, \ldots\}$, or by its counting process $\{N(t); t > 0\}$, where $N(t) = n$ for $S_n \leq t < S_{n+1}$.

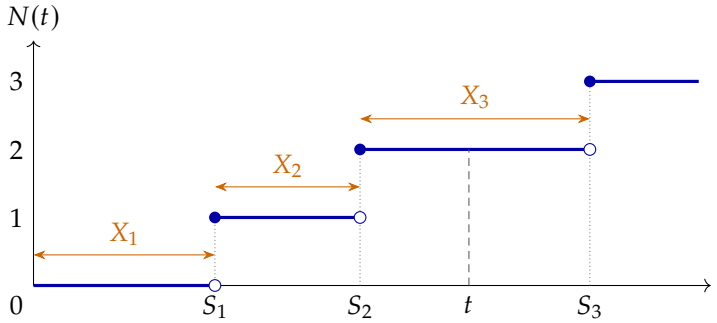


Figure 13: A sample path of an arrival process: epochs S_n, interarrival intervals X_n, and counting process $N(t)$, here with $N(t) = 2$. Schematic; the epochs are illustrative.

Definition 10.1 [Renewal process]. A renewal process is an arrival process whose interarrival intervals $X_1, X_2, \ldots$ are IID.

Renewal processes are characterized by "starting over" at each renewal. Intuitively, this lets a complex process be broken into intervals between renewals. Renewal theory treats the gross characteristics, such as how many renewals occur per unit time and laws of large numbers about long-term behavior; the local characteristics can then be studied without worrying about long-term interactions.

Example: In a Markov chain, successive visits to a given recurrent state are renewals. That the intervals between visits are independent is almost obvious and is checked carefully later. A non-obvious consequence of renewal theory is that the expected recurrence time of state i is $1/\pi_i$; this could be derived from Markov chains directly, but the renewal argument is cleaner. The theory of Markov chains with a countably infinite state space will also come from renewal processes.

Example: A $G/G/m$ queue has IID interarrival times with a general distribution, IID service times with a general distribution, and m servers. Assume the queue is empty for $t < 0$ and an arrival occurs at $t = 0$. A very complicated interaction between arrivals and departures follows until the queue finally empties. After an interval that depends on this busy period, a new arrival occurs, and it can be taken as a renewal.

10.2 Convergence with probability 1

Definition 10.2 [Convergence WP1]. A sequence of rv's $\{Z_n; n \geq 1\}$ converges with probability 1 (WP1) to a rv Z if $\Pr\{\omega \in \Omega : \lim_{n\to\infty} Z_n(\omega) = Z(\omega)\} = 1$.

Theorem 10.3 [A sufficient condition for convergence WP1]. *Let $\{Y_n; n \geq 1\}$ be rv's satisfying $\sum_{n=1}^{\infty} \mathbb{E}[|Y_n|] < \infty$. Then $\{Y_n; n \geq 1\}$ converges to 0 WP1, that is,* $\Pr\{\omega \in \Omega : \lim_{n\to\infty} Y_n(\omega) = 0\} = 1$.

Proof. For any $\alpha > 0$ and any finite $m \geq 1$, the Markov inequality gives

$$\Pr\Big\{\sum_{n=1}^{m} |Y_n| > \alpha\Big\} \leq \frac{\mathbb{E}\big[\sum_{n=1}^{m} |Y_n|\big]}{\alpha} = \frac{\sum_{n=1}^{m} \mathbb{E}[|Y_n|]}{\alpha} \leq \frac{\sum_{n=1}^{\infty} \mathbb{E}[|Y_n|]}{\alpha}.$$

Let $A_m = \{\omega : \sum_{n=1}^{m} |Y_n(\omega)| > \alpha\}$. Since $|Y_n(\omega)| \geq 0$, $A_m \subseteq A_{m+1}$, so $\Pr\{A_m\}$ is a nondecreasing bounded sequence, and by the nesting property of the probability axioms,

$$\Pr\Big\{\bigcup_{m=1}^{\infty} A_m\Big\} = \lim_{m\to\infty} \Pr\{A_m\} \leq \frac{\sum_{n=1}^{\infty} \mathbb{E}[|Y_n|]}{\alpha}.$$

For a given ω, the partial sums $\sum_{n=1}^{m} |Y_n(\omega)|$ are nondecreasing in m. Either they are all at most α, in which case $\sum_{n=1}^{\infty} |Y_n(\omega)| \leq \alpha$ and $\omega \notin \bigcup_m A_m$, or some partial sum exceeds α, in which case $\sum_{n=1}^{\infty} |Y_n(\omega)| > \alpha$ and $\omega \in \bigcup_m A_m$. Hence

$$\Pr\Big\{\omega : \sum_{n=1}^{\infty} |Y_n(\omega)| \leq \alpha\Big\} \geq 1 - \frac{\sum_{n=1}^{\infty} \mathbb{E}[|Y_n|]}{\alpha}.$$

If $\sum_n |Y_n(\omega)| \leq \alpha$, the series converges, so its tails $\sum_{n=m}^{\infty} |Y_n(\omega)|$ tend to 0 and $\lim_n |Y_n(\omega)| = 0$. Therefore $\Pr\{\omega : \lim_n |Y_n(\omega)| = 0\} \geq 1 - \sum_n \mathbb{E}[|Y_n|]/\alpha$ for every α, and letting $\alpha \to \infty$ gives probability 1. □

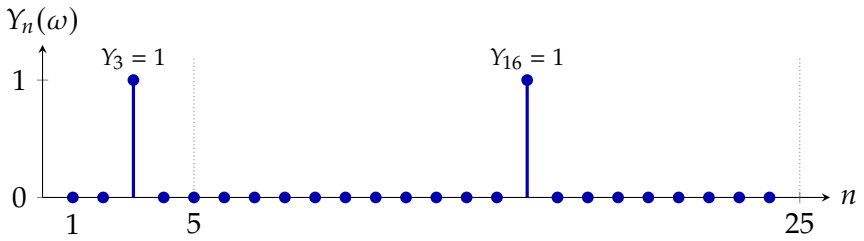


Figure 14: A sample path, for $n < 25$, of the sequence defined below: a single 1 in $[1, 5)$ and a single 1 in $[5, 25)$, at positions chosen uniformly (here $n = 3$ and $n = 16$).

Example: (Convergence in probability but not WP1.) For each $j \geq 0$, let $Y_n = 1$ for one equiprobable choice of $n \in [5^j, 5^{j+1})$ and $Y_n = 0$ for the other n in that range. For every ω and j, $Y_n(\omega)$ is 1 for some $n \in [5^j, 5^{j+1})$ and 0 elsewhere in that range, so $Y_n(\omega)$ converges for no ω: the sequence does

not converge WP1. Here $\mathbb{E}[|Y_n|] = 1/(5^{j+1} - 5^j)$ for $5^j \le n < 5^{j+1}$, so $\sum_n \mathbb{E}[|Y_n|] = \sum_j 1 = \infty$ and Theorem 10.3 does not apply. However, $\lim_n \Pr\{|Y_n| > \epsilon\} = 0$ for every $\epsilon > 0$, so the sequence converges in probability.

10.3 The strong law of large numbers

Theorem 10.4 [Strong law of large numbers]. *Let $\{X_n; n \ge 1\}$ be IID rv's with $\mathbb{E}[|X|] < \infty$, and let $S_n = \sum_{m=1}^n X_m$. Then*

$$\Pr\Big\{\omega \in \Omega : \lim_{n\to\infty} \frac{S_n(\omega)}{n} = \overline{X}\Big\} = 1.$$

This is stated more tersely as $\Pr\{\lim_n S_n/n = \overline{X}\} = 1$, more tersely yet as $\lim_n S_n/n = \overline{X}$ WP1, and still more tersely as $S_n/n \xrightarrow{\text{WP1}} \overline{X}$. The meaning is complicated, and the terse forms sometimes conceal it. The strong law, like convergence WP1 in general, is quite different from the other forms of convergence: it concerns whole sample paths from $n = 1$ to ∞, which makes it harder to speak of a rate of convergence. It connects directly to the ordinary convergence of a sequence of numbers, applied to each sample path; this is what makes it powerful for renewal processes. Most of the work in proving it is done by Theorem 10.3.

Proof assuming $\overline{X} = 0$ and $\mathbb{E}[X^4] < \infty$. Expand

$$\mathbb{E}[S_n^4] = \mathbb{E}\Big[\Big(\sum_{i=1}^n X_i\Big)\Big(\sum_{j=1}^n X_j\Big)\Big(\sum_{k=1}^n X_k\Big)\Big(\sum_{\ell=1}^n X_\ell\Big)\Big] = \sum_{i=1}^n\sum_{j=1}^n\sum_{k=1}^n\sum_{\ell=1}^n \mathbb{E}[X_iX_jX_kX_\ell].$$

In a term with $i = 1$ and $j, k, \ell > 1$, X_1 is independent of $X_jX_kX_\ell$, so $\mathbb{E}[X_1X_jX_kX_\ell] = \mathbb{E}[X_1]\mathbb{E}[X_jX_kX_\ell] = 0$. Similarly, every term in which one index differs from all the others is 0. Two kinds of nonzero terms remain: the n terms with $i = j = k = \ell$, and the terms in which the indices form two distinct pairs, of which there are $n(n-1)$ with $i = j$, $k = \ell$, another $n(n-1)$ with $i = k$, $j = \ell$, and another $n(n-1)$ with $i = \ell$, $j = k$. Thus

$$\mathbb{E}[S_n^4] = n\mathbb{E}[X^4] + 3n(n-1)\big(\mathbb{E}[X^2]\big)^2.$$

Since $\mathbb{E}[X^4]$ is the second moment of the rv X^2, $(\mathbb{E}[X^2])^2 \le \mathbb{E}[X^4]$, and

$$\mathbb{E}[S_n^4] \le \big[n + 3n(n-1)\big]\mathbb{E}[X^4] \le 3n^2\mathbb{E}[X^4], \qquad \sum_{n=1}^\infty \frac{\mathbb{E}[S_n^4]}{n^4} \le 3\mathbb{E}[X^4]\sum_{n=1}^\infty \frac{1}{n^2} < \infty.$$

By Theorem 10.3 with $Y_n = S_n^4/n^4$, $\Pr\{\omega : \lim_n S_n^4(\omega)/n^4 = 0\} = 1$. For every ω with $\lim_n S_n^4(\omega)/n^4 = 0$, also $\lim_n |S_n(\omega)/n| = 0$. Hence $\Pr\{\omega : \lim_n S_n(\omega)/n = 0\} = 1$. For a nonzero mean, apply this to $X_n - \overline{X}$. □

The step from $\{S_n^4(\omega)/n^4\}$ to $\{|S_n(\omega)/n|\}$, made sample path by sample path, is the key to much of

the usefulness of the strong law.

10.4 Strange aspects of the strong law

Consider the Bernoulli process with $p_X(1) = p$. Then $\mathbb{E}[X] = p$, and by the theorem the set of sample paths $\{\omega : \lim_n S_n/n = p\}$ has probability 1. Now consider another Bernoulli process with $p_X(1) = p' \neq p$. Then $\Pr\{\omega : \lim_n S_n/n = p\} = 0$ and, with no change of events, $\Pr\{\omega : \lim_n S_n/n = p'\} = 1$. There are uncountably many choices of p, so there are uncountably many disjoint events, each of probability 1 under its own p. They partition the sample space, together with the sample paths on which S_n/n does not converge. Nothing is wrong here, but these events are peculiar and must be treated carefully.

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Lecture 11: Renewals: strong law and rewards

Convergence with probability 1 (WP1) is a statement about entire sample paths. This makes it easy to pass from the arrival epochs S_n of a renewal process to its counting process $N(t)$, and to time averages of functions of the sample path such as the residual life.

11.1 Convergence WP1 and the strong law

Theorem 11.1. *Let $\{Y_n;\ n \geq 1\}$ be rv's with $\sum_{n=1}^{\infty} \mathbb{E}[|Y_n|] < \infty$. Then $\Pr\{\omega : \lim_{n\to\infty} Y_n(\omega) = 0\} = 1$.*

Proof. By monotone convergence, $\mathbb{E}\left[\sum_n |Y_n|\right] = \sum_n \mathbb{E}[|Y_n|] < \infty$, so the nonnegative rv $\sum_n |Y_n|$ is finite WP1. For every ω at which this sum is finite, its terms satisfy $|Y_n(\omega)| \to 0$. □

Finiteness of the series means its tails vanish, $\lim_{m\to\infty} \sum_{n\geq m} \mathbb{E}[|Y_n|] = 0$. This is stronger than $\lim_n \mathbb{E}[|Y_n|] = 0$, which only implies that $Y_n \to 0$ in probability. The stronger requirement is what lets us say something about whole sample paths.

The usefulness of Theorem 11.1 depends on the choice of the Y_n. For the strong law with $\overline{X} = 0$ and $\mathbb{E}[X^4] < \infty$, take $Y_n = S_n^4/n^4$, where $S_n = X_1 + \cdots + X_n$. Expanding S_n^4, only the terms X_i^4 and $X_i^2 X_j^2$ have nonzero mean, so $\mathbb{E}[S_n^4] = n\mathbb{E}[X^4] + 3n(n-1)\sigma^4 \sim n^2$ and $\sum_n \mathbb{E}[S_n^4]/n^4 < \infty$. Hence $S_n^4/n^4 \to 0$ WP1. Whenever $a_n = S_n^4(\omega)/n^4 \to 0$, also $|S_n(\omega)|/n = a_n^{1/4} \to 0$, so $\Pr\{\lim_n S_n/n = 0\} = 1$.

More generally, $\{Z_n;\ n \geq 1\}$ converges WP1 to a constant α if $\Pr\{\lim_n Z_n(\omega) = \alpha\} = 1$, equivalently if $Z_n - \alpha \to 0$ WP1. If the X_n are IID with mean $\overline{X}$, the $X_n - \overline{X}$ are IID with mean 0; for renewal processes, whose interarrival intervals are positive, it is more convenient to keep the mean in. The step from S_n^4/n^4 to $|S_n|/n$ is a special case of the following.

Theorem 11.2. *Suppose $Z_n \to \alpha$ WP1 and f is a real function continuous at α. Then $f(Z_n) \to f(\alpha)$ WP1.*

Proof. For each ω with $\lim_n Z_n(\omega) = \alpha$, continuity of f at α gives $\lim_n f(Z_n(\omega)) = f(\alpha)$, the corresponding fact for a sequence of numbers. These ω form a set of probability 1. □

For example, $f(x) = x + \beta$ shows that $Z_n + \beta \to \alpha + \beta$, a trivial change of mean, and $f(x) = x^{1/4}$ on $x \geq 0$ gives the step above.

11.2 The strong law for renewal processes

Let $X_1, X_2, \ldots$ be the IID positive inter-renewal intervals of a renewal process, with arrival epochs $S_n = X_1 + \cdots + X_n$ and counting process $N(t) = \max\{n : S_n \le t\}$. Since $X > 0$, $\overline{X} > 0$. If $\overline{X} < \infty$, the SLLN gives $S_n/n \to \overline{X}$ WP1, and $f(x) = 1/x$ is continuous at $\overline{X}$, so

$$\Pr\left\{\omega : \lim_{n\to\infty} \frac{n}{S_n(\omega)} = \frac{1}{\overline{X}}\right\} = 1.$$

Theorem 11.3 [Strong law for renewal processes]. *For a renewal process with $\overline{X} < \infty$,*

$$\Pr\left\{\omega : \lim_{t\to\infty} \frac{N(t,\omega)}{t} = \frac{1}{\overline{X}}\right\} = 1.$$

Proof. Since $S_{N(t)} \le t < S_{N(t)+1}$, for $N(t) \ge 1$ (Figure 15),

$$\frac{N(t)}{S_{N(t)+1}} < \frac{N(t)}{t} \le \frac{N(t)}{S_{N(t)}}.$$

Also $N(t) \to \infty$ WP1: each S_n is a finite rv and $\{N(t) \ge n\} = \{S_n \le t\}$, so for each n, $N(t) \ge n$ for all large t, except on a set of probability 0. Fix ω outside the union of this null set and the one where $n/S_n \not\to 1/\overline{X}$. As t increases, $N(t)/S_{N(t)}$ runs through the same values as n/S_n, so it converges to $1/\overline{X}$. Similarly

$$\frac{N(t)}{S_{N(t)+1}} = \frac{N(t)+1}{S_{N(t)+1}} \cdot \frac{N(t)}{N(t)+1} \longrightarrow \frac{1}{\overline{X}} \cdot 1.$$

The ratio $N(t)/t$ lies between two quantities with the same limit. □

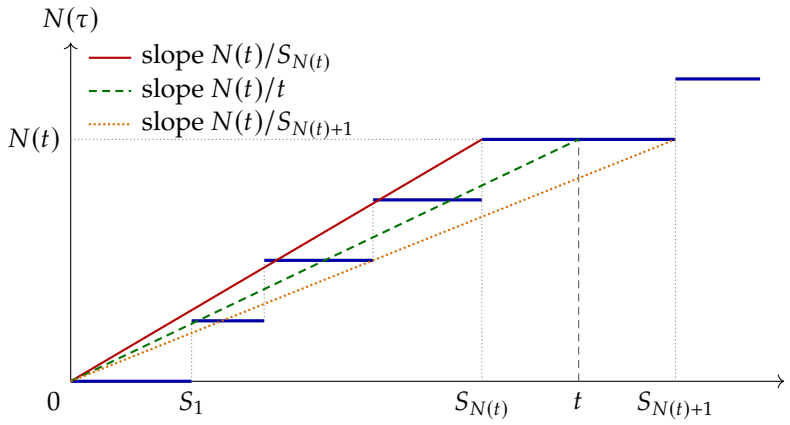


Figure 15: Schematic sample path of $N(t)$. The three lines from the origin to height $N(t)$ have slopes $N(t)/S_{N(t)} \ge N(t)/t > N(t)/S_{N(t)+1}$.

11.3 Central limit theorem for renewals

Let X have variance $\sigma^2 < \infty$. By the CLT,

$$\Pr\{S_n \le t\} \approx \Phi(\alpha) \qquad \text{for } t = n\overline{X} + \alpha\sigma\sqrt{n},$$

where Φ is the standard normal distribution function. Since $\{S_n \le t\} = \{N(t) \ge n\}$, also $\Pr\{N(t) \ge n\} \approx \Phi(\alpha)$. Geometrically, a horizontal deviation $\alpha\sigma\sqrt{n}$ of S_n from the line of slope $1/\overline{X}$ corresponds to a vertical deviation $\alpha\sigma\sqrt{n}/\overline{X}$. Solving for n, and using $n \approx t/\overline{X}$ inside the square root,

$$n = \frac{t}{\overline{X}} - \frac{\alpha\sigma\sqrt{n}}{\overline{X}} \approx \frac{t}{\overline{X}} - \frac{\alpha\sigma\sqrt{t}}{\overline{X}^{3/2}}, \qquad \text{so} \qquad \Pr\left\{\frac{N(t) - t/\overline{X}}{\sigma\sqrt{t}\,\overline{X}^{-3/2}} \le -\alpha\right\} \approx 1 - \Phi(\alpha) = \Phi(-\alpha).$$

Theorem 11.4 [CLT FOR RENEWAL PROCESSES]. *If $\sigma^2 < \infty$, then $N(t)$ is asymptotically Gaussian with mean $t/\overline{X}$ and standard deviation $\sigma\sqrt{t}\,\overline{X}^{-3/2}$: for every y,*

$$\lim_{t\to\infty} \Pr\left\{\frac{N(t) - t/\overline{X}}{\sigma\sqrt{t}\,\overline{X}^{-3/2}} \le y\right\} = \Phi(y).$$

11.4 Time-average residual life

Definition 11.5 [RESIDUAL LIFE]. The RESIDUAL LIFE of a renewal process at time t is the time until the next renewal, $Y(t) = S_{N(t)+1} - t$.

It is how long one waits for a bus, if bus arrivals formed a renewal process. For each sample point, $Y(t)$ is a function of t determined by the sample path of renewals, so it can be viewed as a reward assigned to the renewal process. We study its time average $\frac{1}{t}\int_0^t Y(\tau)\,d\tau$.

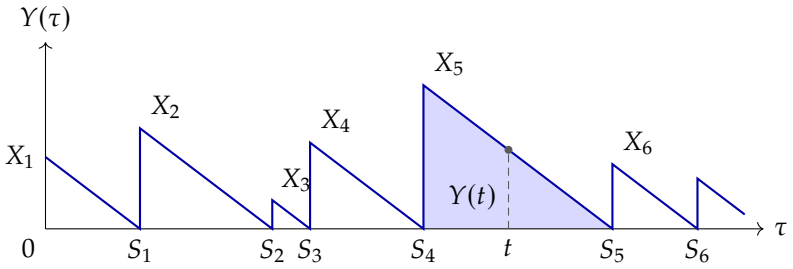


Figure 16: Schematic sample function of the residual life. On $[S_{n-1}, S_n)$ it falls from X_n to 0 with slope -1, so the shaded triangle has area $X_n^2/2$.

A sample function of $Y(t)$ is a sequence of isosceles right triangles, one starting at each arrival epoch (Figure 16). The triangle over $[S_{n-1}, S_n)$ has area $X_n^2/2$, and the integral up to t includes all the

complete triangles up to $S_{N(t)}$ and part of the next one. Hence

$$\frac{1}{2t}\sum_{n=1}^{N(t)} X_n^2 \le \frac{1}{t}\int_0^t Y(\tau)\,d\tau \le \frac{1}{2t}\sum_{n=1}^{N(t)+1} X_n^2.$$

Theorem 11.6 [Time-average residual life]. *If $\overline{X} < \infty$, then*

$$\lim_{t\to\infty}\frac{1}{t}\int_0^t Y(\tau)\,d\tau = \frac{\mathbb{E}[X^2]}{2\mathbb{E}[X]} \qquad \text{WP1}.$$

Proof. Write the lower bound as

$$\frac{\sum_{n=1}^{N(t)} X_n^2}{2t} = \frac{\sum_{n=1}^{N(t)} X_n^2}{N(t)}\cdot\frac{N(t)}{2t}.$$

The X_n^2 are IID, so by the SLLN their sample average converges to $\mathbb{E}[X^2]$ WP1; since $N(t)\to\infty$ WP1, the first factor converges to $\mathbb{E}[X^2]$. By Theorem 11.3 the second converges to $1/(2\overline{X})$. The upper bound has the same limit, replacing $N(t)$ by $N(t)+1$ and using $(N(t)+1)/t \to 1/\overline{X}$. □

Write $\overline{Y}_{ta} = \mathbb{E}[X^2]/(2\mathbb{E}[X])$. If X is almost deterministic, $\mathbb{E}[X^2] \approx \overline{X}^2$ and $\overline{Y}_{ta} \approx \mathbb{E}[X]/2$. If X is exponential, $\mathbb{E}[X^2] = 2\overline{X}^2$ and $\overline{Y}_{ta} = \mathbb{E}[X]$, as memorylessness predicts.

Example: *Rare long intervals.* Let $p_X(\epsilon) = 1-\epsilon$ and $p_X(1/\epsilon) = \epsilon$ for small $\epsilon > 0$. Then $\mathbb{E}[X] = \epsilon(1-\epsilon)+1 \approx 1$ and $\mathbb{E}[X^2] = \epsilon^2(1-\epsilon)+1/\epsilon \approx 1/\epsilon$, so $\overline{Y}_{ta} \approx 1/(2\epsilon)$. About $1/\epsilon$ short intervals of length ϵ separate consecutive long intervals, so the expected time between long intervals is only about 1. Yet almost all time is spent inside long intervals of length $1/\epsilon$, where the residual life averages $1/(2\epsilon)$.

Source. Robert Gallager, MIT 6.262, *Discrete Stochastic Processes*, Lecture 11, Spring 2011. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 12: Renewal rewards, stopping trials, and Wald's equality

The time-average argument for residual life extends to any reward that depends only on the position within the current inter-renewal interval. Stopping trials then formalize stopping a sequence at a time chosen by the sample path itself, and Wald's equality computes the mean of the sum up to that time.

12.1 Strong law, age, and duration

For a renewal process with $0 < \overline{X} < \infty$, $N(t)/t \to 1/\overline{X}$ WP1, so the rate of renewals over the infinite time horizon is $1/\overline{X}$. Convergence WP1 implies convergence in probability, so this also gives the weak law for renewals:

$$\lim_{t\to\infty} \Pr\left\{\left|\frac{N(t)}{t} - \frac{1}{\overline{X}}\right| > \epsilon\right\} = 0 \qquad \text{for all } \epsilon > 0.$$

For each sample point ω, the residual life $Y(t,\omega)$ is a sample function, and $\frac{1}{t}\int_0^t Y(\tau,\omega)\,d\tau \to \mathbb{E}[X^2]/(2\mathbb{E}[X])$ WP1. This limit is infinite if $\mathbb{E}[X^2] = \infty$; think of $p_X(\epsilon) = 1-\epsilon$, $p_X(1/\epsilon) = \epsilon$ as $\epsilon \to 0$. Two similar processes are the age and the duration,

$$Z(t) = t - S_{N(t)}, \qquad \widetilde{X}(t) = S_{N(t)+1} - S_{N(t)}.$$

The age rises with slope 1 from 0 to X_n over the nth interval, again a triangle of area $X_n^2/2$, while the duration equals X_n throughout that interval, a rectangle of area X_n^2. The same sandwich argument gives, WP1,

$$\lim_{t\to\infty} \frac{1}{t}\int_0^t Z(\tau)\,d\tau = \frac{\mathbb{E}[X^2]}{2\mathbb{E}[X]}, \qquad \lim_{t\to\infty} \frac{1}{t}\int_0^t \widetilde{X}(\tau)\,d\tau = \frac{\mathbb{E}[X^2]}{\mathbb{E}[X]}.$$

12.2 Time averages for renewal rewards

Residual life, age, and duration are examples of rewards assigned to a renewal process. In general the reward $R(t)$ at time t is restricted to be a function of the inter-renewal period containing t; in the simplest form, $R(t) = \mathcal{R}\big(Z(t), \widetilde{X}(t)\big)$ for a function $\mathcal{R}(z,x)$. The reward accumulated over the nth interval is

$$R_n(\omega) = \int_{S_{n-1}(\omega)}^{S_n(\omega)} R(t,\omega)\,dt.$$

Interval 1 runs from 0 to S_1 with $Z(t) = t$; in interval n, $S_{N(t)} = S_{n-1}$, so $Z(t) = t - S_{n-1}$ and $\widetilde{X}(t) = X_n$. Thus

$$R_n = \int_{S_{n-1}}^{S_n} \mathcal{R}(t - S_{n-1}, X_n)\, dt = \int_0^{X_n} \mathcal{R}(z, X_n)\, dz,$$

a function of X_n alone. The R_n are therefore IID, with

$$\mathbb{E}[R_n] = \int_{x=0}^{\infty} \int_{z=0}^{x} \mathcal{R}(z, x)\, dz\, dF_X(x).$$

Theorem 12.1 [Renewal-reward theorem]. *If $\overline{X} < \infty$ and $\mathbb{E}[R_n]$ exists, then*

$$\lim_{t\to\infty} \frac{1}{t} \int_0^t R(\tau)\, d\tau = \frac{\mathbb{E}[R_n]}{\overline{X}} \qquad \text{WP1.}$$

Proof. First let $\mathcal{R} \geq 0$. As for residual life,

$$\frac{1}{t} \sum_{n=1}^{N(t)} R_n \leq \frac{1}{t} \int_0^t R(\tau)\, d\tau \leq \frac{1}{t} \sum_{n=1}^{N(t)+1} R_n,$$

and each bound is a sample average of IID rv's times $N(t)/t$ or $(N(t)+1)/t$. By the SLLN, $N(t) \to \infty$, and the strong law for renewals, both bounds converge to $\mathbb{E}[R_n]/\overline{X}$ WP1. For general $\mathcal{R}$, apply this to its positive and negative parts separately. □

With $\mathcal{R}(z, x) = x - z$, z, or x, the theorem recovers the residual-life, age, and duration averages, since $\int_0^x (x - z)\, dz = \int_0^x z\, dz = x^2/2$ and $\int_0^x x\, dz = x^2$.

Example: *Moments of the age.* For the kth moment of the age, take $\mathcal{R}(Z(t), \widetilde{X}(t)) = Z^k(t)$. Then

$$\mathbb{E}[R_n] = \int_{x=0}^{\infty} \int_{z=0}^{x} z^k\, dz\, dF_X(x) = \int_0^{\infty} \frac{x^{k+1}}{k+1}\, dF_X(x) = \frac{\mathbb{E}[X^{k+1}]}{k+1},$$

so

$$\lim_{t\to\infty} \frac{1}{t} \int_0^t Z^k(\tau)\, d\tau = \frac{\mathbb{E}[X^{k+1}]}{(k+1)\overline{X}} \quad \text{WP1.}$$

12.3 Stopping trials

Often we want to analyze an initial segment of a process, not over $(0, t]$ for a fixed t, but up to a time selected by the sample path itself. Since that time is a rv depending on the process up to that time, the definition must avoid circularity. We consider only discrete-time processes $\{X_n;\ n \geq 1\}$.

Let J be a positive integer rv describing when to stop. At trial 1, X_1 is observed and a decision whether to stop is made from X_1; if we stop, $J = 1$. At each later trial n, if stopping has not occurred,

X_n is observed and a decision is made from $X_1, \ldots, X_n$; if we stop, $J = n$.

Definition 12.2 [Stopping trial]. A stopping trial (or stopping time) J for $\{X_n;\, n \geq 1\}$ is a positive integer-valued rv such that, for each $n \geq 1$, the indicator rv $\mathbb{I}_{\{J=n\}}$ is a function of $X_1, \ldots, X_n$. A possibly defective stopping trial is the same except that J may be a defective rv, that is, $\Pr\{J < \infty\} < 1$ is allowed.

We visualize conducting trials until the event $\{J = n\}$ occurs; it is conceptually simpler to stop observing after trial J but keep conducting trials, so the whole sequence remains defined. The events $\{J = 1\}, \{J = 2\}, \ldots$ are disjoint. Examples: a gambler plays until broke; flip a coin until 10 successive heads appear; test a hypothesis with repeated trials until one hypothesis is sufficiently probable a posteriori; observe renewals until $S_n \geq 100$.

If each X_i has finitely many possible values, any possibly defective stopping trial is a rooted tree: each sample path stops at a terminal node (Figure 17).

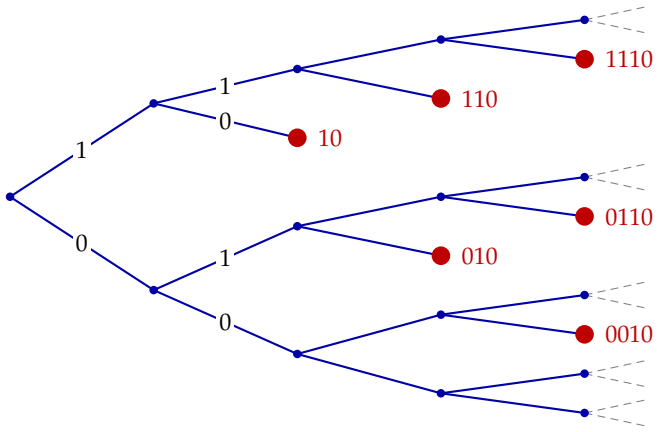


Figure 17: Binary X_i, stopping when the pattern $(1, 0)$ first occurs, drawn to depth 4. Upper branches are $X_i = 1$ and lower branches $X_i = 0$. Large dots are stopping nodes, labeled by their sample sequence; dashed stubs continue.

12.4 Wald's equality

Theorem 12.3 [Wald's equality]. *Let $\{X_n;\, n \geq 1\}$ be IID with mean $\overline{X}$. If J is a stopping trial for $\{X_n\}$ with $\mathbb{E}[J] < \infty$, then $S_J = X_1 + \cdots + X_J$ satisfies*

$$\mathbb{E}[S_J] = \overline{X}\,\mathbb{E}[J].$$

Proof. Write $S_J = \sum_{n\geq 1} X_n \mathbb{I}_{\{J\geq n\}}$. The key step is that X_n and $\mathbb{I}_{\{J\geq n\}}$ are independent. Indeed, $\mathbb{I}_{\{J\geq n\}} = 1 - \mathbb{I}_{\{J<n\}}$, and $\mathbb{I}_{\{J<n\}} = \sum_{k<n} \mathbb{I}_{\{J=k\}}$ is a function of $X_1, \ldots, X_{n-1}$, which are independent

of X_n. Hence

$$\mathbb{E}[S_J] = \sum_n \mathbb{E}\big[X_n \mathbb{I}_{\{J \geq n\}}\big] = \sum_n \mathbb{E}[X_n]\,\mathbb{E}\big[\mathbb{I}_{\{J \geq n\}}\big] = \overline{X} \sum_n \Pr\{J \geq n\} = \overline{X}\,\mathbb{E}[J].$$

Exchanging expectation and sum is justified because the same computation with $|X_n|$ gives $\sum_n \mathbb{E}[|X_n|\mathbb{I}_{\{J \geq n\}}] = \mathbb{E}[|X|]\,\mathbb{E}[J] < \infty$. □

This is surprising, since X_n is certainly not independent of $\mathbb{I}_{\{J=n\}}$, nor of $\mathbb{I}_{\{J=n+1\}}$. The resolution is that, given $J \geq n$, the trial at which stopping occurs depends on X_n, but whether $J \geq n$ occurs depends only on $X_1, \ldots, X_{n-1}$.

In many applications Wald gives one equation in two unknowns. Frequently $\mathbb{E}[S_J]$ is easy to find, and the equation then yields $\mathbb{E}[J]$. The next example shows, among other things, why $\mathbb{E}[J] < \infty$ is required.

12.5 Stop when you're ahead

Toss a coin with heads probability p, betting \$1 on each toss: win on heads and lose on tails, so $X_n = \pm 1$ with $\overline{X} = 2p - 1$. Stop when the winnings reach \$1: $J = \min\{n : S_n = 1\}$.

Case $p > 1/2$. Without stopping, the winnings grow without bound by the SLLN, passing through 1, so J is a rv, and $S_J = 1$ WP1. Also $\mathbb{E}[J] < \infty$: Wald applies to the bounded stopping trial $\min(J, n)$, and $S_{\min(J,n)} \leq 1$ gives $\mathbb{E}[\min(J, n)] \leq 1/\overline{X}$; let $n \to \infty$. Hence

$$\mathbb{E}[J] = \frac{\mathbb{E}[S_J]}{\overline{X}} = \frac{1}{2p - 1}.$$

To verify this another way, $J = 1$ with probability p. Otherwise $S_1 = -1$, and reaching 1 requires first rising from -1 to 0 (taking $\mathbb{E}[J]$ steps on average) and then from 0 to 1 (another $\mathbb{E}[J]$). So $\mathbb{E}[J] = 1 + (1 - p)\,2\mathbb{E}[J]$, which again gives $1/(2p - 1)$.

Case $p < 1/2$. It is still possible to win and stop: for example, $J = 1$ with probability p and $J = 3$ with probability $p^2(1 - p)$. It is also possible to head south forever. Let $\theta = \Pr\{J < \infty\}$. Given $S_1 = -1$, the event $\{J < \infty\}$ requires $S_m - S_1 = 1$ for some m and then $S_n - S_m = 1$ for some $n > m$; these are independent events, each of probability θ. Hence

$$\theta = p + (1 - p)\theta^2,$$

with solutions $\theta = p/(1 - p)$ and $\theta = 1$. The root $\theta = 1$ is impossible: let $\theta_k = \Pr\{J \leq k\}$, so $\theta_0 = 0$ and the same decomposition gives $\theta_k \leq p + (1 - p)\theta_{k-1}^2$. If $\theta_{k-1} \leq p/(1 - p)$, then $\theta_k \leq p + p^2/(1-p) = p/(1-p)$, so by induction every $\theta_k \leq p/(1-p)$, and $\theta = \lim_k \theta_k = p/(1-p) < 1$. Thus J is defective and Wald's equality is inapplicable.

Case $p = 1/2$. Now $\theta = \frac{1}{2} + \frac{1}{2}\theta^2$, that is, $(\theta - 1)^2 = 0$, so $J < \infty$ WP1. But $\mathbb{E}[J] = \infty$, consistent with $1/(2p - 1) \to \infty$ as $p \downarrow 1/2$: if $\mathbb{E}[J]$ were finite, Wald would give $1 = \mathbb{E}[S_J] = 0 \cdot \mathbb{E}[J] = 0$. You make your \$1 with probability 1 in a fair game, and can repeat the feat, but it takes infinite expected time and requires access to infinite capital.

Source. Robert Gallager, MIT 6.262, *Discrete Stochastic Processes*, Lecture 12, Spring 2011. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 13: Little, M/G/1, ensemble averages

Wald's equality gives the expected number of renewals in finite time and hence the elementary renewal theorem. A mild generalization of stopping trials shows that busy periods of a G/G/1 queue start a renewal process, which makes Little's theorem a renewal-reward statement and leads to the Pollaczek-Khinchin formula for M/G/1.

13.1 The elementary renewal theorem

Recall that a stopping trial J for $\{X_n;\, n \geq 1\}$ is a positive integer rv with each $\mathbb{I}_{\{J=n\}}$ a function of $X_1, \ldots, X_n$. A possibly defective stopping trial may be defective; in many applications it is not initially obvious whether it is. Wald's equality says $\mathbb{E}[S_J] = \overline{X}\,\mathbb{E}[J]$ for IID X_n and $\mathbb{E}[J] < \infty$.

For a renewal counting process and a given t, let J be the smallest n with $S_n > t$. We stop at trial n exactly when $S_{n-1} \leq t < S_n$, which is determined by $X_1, \ldots, X_n$, so J is a stopping trial for the inter-renewal intervals. Since $N(t)$ counts the arrivals in $(0, t]$, $J = N(t) + 1$ is the index of the first arrival after t.

$\mathbb{E}[N(t)]$ is finite. Since $X > 0$ WP1, $\mathbb{E}[e^{-X}] < 1$, and the Markov inequality gives $\Pr\{S_n \leq t\} = \Pr\{e^{-S_n} \geq e^{-t}\} \leq e^t \left(\mathbb{E}[e^{-X}]\right)^n$, so $\mathbb{E}[N(t)] = \sum_{n\geq 1} \Pr\{S_n \leq t\} < \infty$. Thus $\mathbb{E}[J] < \infty$ and

$$\mathbb{E}\left[S_{N(t)+1}\right] = \overline{X}\,\mathbb{E}[J] = \overline{X}\left(\mathbb{E}[N(t)] + 1\right).$$

This relates two unknown quantities, but $S_{N(t)+1} > t$ already gives a simple bound:

$$\mathbb{E}[N(t)] = \frac{\mathbb{E}[S_{N(t)+1}]}{\overline{X}} - 1 > \frac{t}{\overline{X}} - 1. \tag{13.1}$$

Theorem 13.1 [Elementary renewal theorem]. *For a renewal counting process $\{N(t);\, t > 0\}$ with mean inter-renewal interval $\overline{X}$,*

$$\lim_{t\to\infty} \frac{\mathbb{E}[N(t)]}{t} = \frac{1}{\overline{X}}.$$

Proof. By (13.1), $\liminf_t \mathbb{E}[N(t)]/t \geq 1/\overline{X}$. For the upper bound, truncate: for $b > 0$ let $\breve{X}_i = \min(X_i, b)$, with arrival epochs $\breve{S}_n$ and counting process $\breve{N}(t)$. Since $\breve{X}_i \leq X_i$, $\breve{S}_n \leq S_n$ and so $N(t) \leq \breve{N}(t)$. Since each truncated interval is at most b, $\breve{S}_{\breve{N}(t)+1} \leq t + b$, and Wald for the truncated process gives

$$\mathbb{E}[\breve{X}]\left(\mathbb{E}[\breve{N}(t)] + 1\right) \leq t + b, \qquad \text{so} \qquad \frac{\mathbb{E}[N(t)]}{t} \leq \frac{\mathbb{E}[\breve{N}(t)]}{t} \leq \frac{1 + b/t}{\mathbb{E}[\breve{X}]}.$$

Let $b = \sqrt{t}$. As $t \to \infty$, $b/t \to 0$ and $\mathbb{E}[\min(X, \sqrt{t})] \uparrow \overline{X}$ by monotone convergence, so $\limsup_t \mathbb{E}[N(t)]/t \le 1/\overline{X}$. □

This is convergence of $N(t)/t$ in the mean, one more type of convergence, alongside the strong and weak laws for renewals.

13.2 Generalized stopping trials

Definition 13.2 [Generalized stopping trial]. A generalized stopping trial J for a sequence of pairs of rv's $(X_1, V_1), (X_2, V_2), \ldots$ is a positive integer rv such that, for each $n \ge 1$, $\mathbb{I}_{\{J=n\}}$ is a function of $X_1, V_1, \ldots, X_n, V_n$.

Then $\mathbb{I}_{\{J<n\}} = 1 - \mathbb{I}_{\{J\ge n\}}$ is a function of $X_1, V_1, \ldots, X_{n-1}, V_{n-1}$. Wald's equality $\mathbb{E}[S_J] = \overline{X}\,\mathbb{E}[J]$, with $S_J = X_1 + \cdots + X_J$, still holds by the same proof if the X_i are IID and each X_i is independent of $(X_1, V_1, \ldots, X_{i-1}, V_{i-1})$: this is exactly what makes X_n independent of $\mathbb{I}_{\{J\ge n\}}$. Each V_i may also be a vector of rv's.

13.3 The G/G/1 queue

In a G/G/1 queue, customer 0 arrives at time 0 and customers $1, 2, \ldots$ arrive at epochs $s_1 < s_2 < \cdots$ with IID interarrival intervals $X_1, X_2, \ldots$. A single server serves customers in order of arrival with IID service times $V_0, V_1, \ldots$, independent of the arrivals. Customer i waits W_i^q in queue and then is served for V_i.

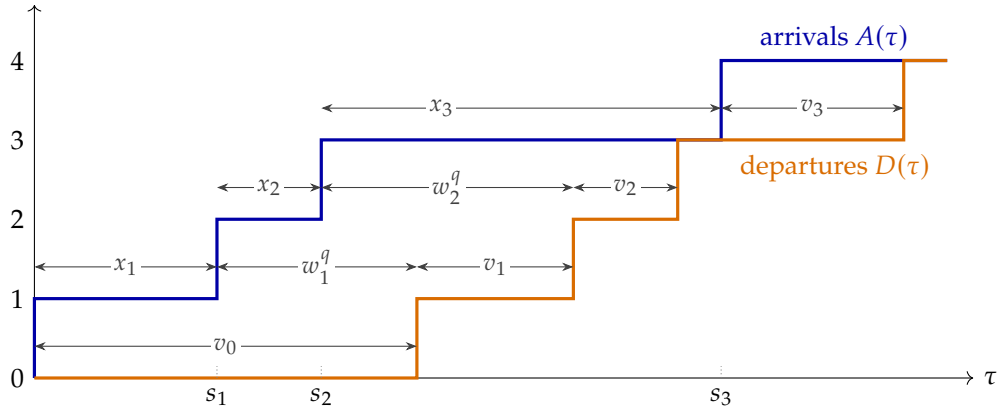


Figure 18: A schematic G/G/1 sample path. Row i shows customer i: the interarrival interval x_i before its arrival s_i, its queueing delay w_i^q, and its service v_i. Customer 3 arrives to an empty system and starts a new busy period, so $J = 3$.

Let J be the index of the first arrival that starts a new busy period, that is, finds the system empty. Use the pairs $(X_1, V_0), (X_2, V_1), \ldots$. Whether customer n finds the system empty depends only

on the arrival epochs up to s_n and the services of customers $0, \ldots, n-1$, so $\mathbb{I}_{\{J=n\}}$ is a function of $X_1, V_0, \ldots, X_n, V_{n-1}$; in Figure 18, $\{J = 3\}$ is determined by $(X_1, V_0, X_2, V_1, X_3, V_2)$. Thus J is a generalized stopping trial and Wald's equality holds for it. Moreover, the new arrivals $X_{J+1}, X_{J+2}, \ldots$ and services $V_J, V_{J+1}, \ldots$ are independent of the old and identically distributed to them, so the new busy period evolves like the old.

Hence the intervals between the starts of successive busy periods form a renewal process. We then have one renewal process embedded in another: call the first the ARRIVAL PROCESS and the second, whose renewals occur at arrivals to an empty system and which embodies both arrivals and services, the RENEWAL PROCESS. The same analysis applies to G/G/m and many other queueing systems.

13.4 Little's theorem

Consider a queueing system whose arrival process is a renewal process, with an arrival at time 0. The service can be almost anything; to be specific, take a G/G/1 queue. Assume that the system empties out eventually WP1 and restarts at the next arrival, so the restart epochs form a renewal process. Let $A(\tau)$ and $D(\tau)$ count the arrivals and departures in $[0, \tau]$, let $L(\tau) = A(\tau) - D(\tau)$ be the number in the system, and let W_i be the time customer i spends in the system.

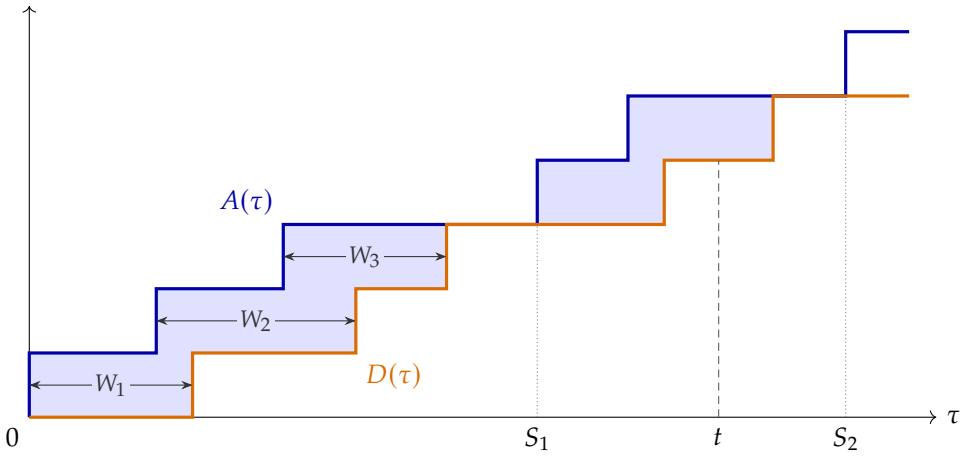


Figure 19: Schematic arrival and departure processes. The shaded region between $A(\tau)$ and $D(\tau)$ has height $L(\tau)$; customer i contributes a horizontal strip of length W_i. Renewals S_1, S_2 occur at arrivals to an empty system.

$L(\tau)$ depends on the departure process, but its value is a function of the interarrival and service times within the current inter-renewal period, so it is a generalized renewal-reward function. The total reward in an inter-renewal period is the area between the two staircases over that period. Counted by horizontal strips, as in Figure 19, it is the total time in system of the customers arriving in that period, all of whom also leave in it:

$$R_n = \int L(\tau)\, d\tau = \sum_i W_i,$$

the sum running over the arrivals in that inter-renewal period.

Theorem 13.3 [Little's theorem]. *Let $\lambda = \lim_t A(t)/t$ be the arrival rate. Under the assumptions above, the time-average number in system $\overline{L} = \lim_t \frac{1}{t}\int_0^t L(\tau)\,d\tau$ and the time-average customer wait $\overline{W} = \lim_t \frac{1}{A(t)}\sum_{i=1}^{A(t)} W_i$ satisfy*

$$\overline{L} = \lambda\,\overline{W} \qquad \text{WP1}.$$

Proof. Over each inter-renewal period the integral of L equals the sum of the W_i for the customers arriving in it, so applying the renewal-reward sandwich to both,

$$\overline{L} = \lim_{t\to\infty} \frac{1}{t}\sum_{i=1}^{A(t)} W_i = \lim_{t\to\infty} \frac{\sum_{i=1}^{A(t)} W_i}{A(t)} \cdot \lim_{t\to\infty} \frac{A(t)}{t} = \overline{W}\lambda. \qquad \square$$

In essence Little's theorem is an accounting identity: over a busy period and the following idle period, $\int L(\tau)\,d\tau = \sum_i W_i$. Renewal theory is what lets us pass to the limit of many renewals. The question is not whether $L = \lambda W$, but whether these quantities exist as sensible time averages WP1, or as limiting ensemble averages.

13.5 The Pollaczek-Khinchin formula for M/G/1

Let the interarrival intervals $X_1, X_2, \ldots$ be IID exponential with rate λ, and the service times $V_1, V_2, \ldots$ IID with first and second moments $\overline{V} = \mathbb{E}[V]$ and $\overline{V^2} = \mathbb{E}[V^2]$. Let $\rho = \lambda\overline{V} < 1$.

Theorem 13.4 [Pollaczek-Khinchin formula]. *The expected delay in queue, between arrival and entering service, is*

$$\overline{W}^q = \frac{\lambda\overline{V^2}}{2(1-\rho)}.$$

The expected total delay, number in system, and number in queue are

$$\overline{W} = \frac{\lambda\overline{V^2}}{2(1-\rho)} + \overline{V}, \qquad \overline{N} = \frac{\lambda^2\overline{V^2}}{2(1-\rho)} + \rho, \qquad \overline{N}^q = \frac{\lambda^2\overline{V^2}}{2(1-\rho)}.$$

The last three follow from the first: $\overline{W} = \overline{W}^q + \overline{V}$, and Little's theorem gives $\overline{N} = \lambda\overline{W}$ and $\overline{N}^q = \lambda\overline{W}^q$.

Example: *Effect of the service distribution.* For deterministic service (M/D/1), $\overline{V^2} = (\overline{V})^2$ and $\overline{W}^q = \rho\overline{V}/(2(1-\rho))$. For exponential service (M/M/1), $\overline{V^2} = 2(\overline{V})^2$ and $\overline{W}^q = \rho\overline{V}/(1-\rho)$. For $p_V(\epsilon) = 1-\epsilon$, $p_V(1/\epsilon) = \epsilon$, we have $\overline{V} \approx 1$, so $\rho \approx \lambda$, and $\overline{V^2} \approx 1/\epsilon$, giving $\overline{W}^q \approx \rho/(2\epsilon(1-\rho))$.

Why does $\overline{W}^q$ grow with $\overline{V^2}$? Let $R(t)$ be the remaining service time of the customer in service at t, with $R(t) = 0$ when the server is idle. Each service contributes a triangle: R jumps to V_i when service i starts and falls with slope -1 to 0. As with residual life, the customer in service is often

one with an unusually long service, and the time average is

$$\mathbb{E}[R] = \lim_{\tau\to\infty} \frac{1}{\tau} \int_0^\tau R(t)\,dt = \lim_{\tau\to\infty} \frac{1}{\tau} \sum_{i=1}^{A(\tau)} \frac{V_i^2}{2} = \frac{\lambda \overline{V^2}}{2}.$$

An arriving customer waits for the end of the current service and then for the services of everyone already in queue. Poisson arrivals see the time-average state of the system, so

$$\overline{W}^q = \mathbb{E}[R] + \overline{N}^q\,\overline{V} = \frac{\lambda \overline{V^2}}{2} + \lambda \overline{W}^q\,\overline{V},$$

using Little's theorem for the queue (a little Little). Solving gives $\overline{W}^q = \lambda \overline{V^2}/(2(1 - \lambda \overline{V}))$.

Source. Robert Gallager, MIT 6.262, *Discrete Stochastic Processes*, Lecture 13, Spring 2011. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 14: Review

The first half of the course rests on a few unifying statements: limit theorems for sums of IID rv's, the structure of arrival processes, convergence of finite-state Markov chains, and sample-path averages for renewal processes.

14.1 Models and laws of large numbers

A probability model has a sample space, a set of events, and a probability measure, each satisfying the axioms. In practice there is a basic countable set of rv's that are IID, Markov, and so on, and a sample point is a collection of sample values, one for each rv. Uncountable families, such as $\{N(t);\, t \geq 0\}$, can usually be defined in terms of a basic countable set.

For IID $X_1, X_2, \ldots$ (as in Poisson and renewal processes), the sample average S_n/n, with $S_n = X_1 + \cdots + X_n$, has mean $\overline{X}$ and variance σ^2/n. The laws of large numbers specify its long-term behavior.

- *Weak law.* If $\mathbb{E}[|X|] < \infty$, then $\lim_n \Pr\{|S_n/n - \overline{X}| \geq \epsilon\} = 0$ for every $\epsilon > 0$: the distribution function of S_n/n approaches a unit step at $\overline{X}$ (convergence in probability and in distribution).
- *Strong law.* If $\mathbb{E}[|X|] < \infty$, then $\lim_n S_n/n = \overline{X}$ WP1: except for a set of sample points of probability zero, every sample sequence has limiting sample average $\overline{X}$. Also $\lim_n f(S_n/n) = f(\overline{X})$ WP1 for f continuous at $\overline{X}$.
- *Central limit theorem.* The most useful result on the speed of convergence: if $\sigma^2 < \infty$, then for every y,

$$\lim_{n\to\infty} \Pr\left\{\frac{S_n - n\overline{X}}{\sqrt{n}\,\sigma} \leq y\right\} = \lim_{n\to\infty} \Pr\left\{\frac{S_n}{n} - \overline{X} \leq \frac{y\sigma}{\sqrt{n}}\right\} = \int_{-\infty}^{y} \frac{1}{\sqrt{2\pi}} e^{-x^2/2}\, dx.$$

  So S_n/n converges to $\overline{X}$ at rate $1/\sqrt{n}$ and becomes Gaussian as an extra benefit.

14.2 Arrival processes and the Poisson process

An arrival process is an increasing sequence of rv's $0 < S_1 < S_2 < \cdots$, with interarrival times $X_1 = S_1$ and $X_i = S_i - S_{i-1}$; it is specified by the joint distribution of either. Its counting process $N(t) = \max\{n : S_n \leq t\}$ satisfies

$$\{S_n \leq t\} = \{N(t) \geq n\} \quad \text{for all } n, t, \qquad S_n = \min\{t : N(t) \geq n\},$$

so $\{N(t);\, t \geq 0\}$ also specifies the process. A renewal process has IID interarrivals; a Poisson process (PP) is a renewal process with exponential interarrivals. A MEMORYLESS rv is a nonnegative, nondeterministic rv with $\Pr\{X > t + x\} = \Pr\{X > x\} \Pr\{X > t\}$ for all $x, t \geq 0$; the exponential is the only one.

For a PP of rate λ and any $t > 0$, the interval Z_1 from t to the next arrival has $F_{Z_1}(z) = 1 - e^{-\lambda z}$, and Z_1 and the later intervals $Z_2, Z_3, \ldots$ are independent of $\{N(\tau);\, \tau \leq t\}$ and form the interarrivals of a PP starting at t. Hence $\widetilde{N}(t, \tau) = N(\tau) - N(t)$ has the distribution of $N(\tau - t)$ (STATIONARY INCREMENTS), and $N(t_1), \widetilde{N}(t_1, t_2), \ldots, \widetilde{N}(t_{k-1}, t_k)$ are independent for $0 < t_1 < \cdots < t_k$ (INDEPENDENT INCREMENTS). A PP can be defined by these two properties plus either the Poisson PMF for $N(t)$ or $\Pr\{\widetilde{N}(t, t + \delta) = 1\} = \lambda\delta + o(\delta)$ and $\Pr\{\widetilde{N}(t, t + \delta) > 1\} = o(\delta)$.

The joint density $f_{S_1 \cdots S_n}(s_1, \ldots, s_n) = \lambda^n e^{-\lambda s_n}$ on $0 \leq s_1 \leq \cdots \leq s_n$ shows that the intermediate epochs are equally likely to be anywhere. Integrating gives the Erlang density, and since arrival n falls in $(t, t + \delta)$ exactly when $N(t) = n - 1$ and an arrival occurs in that interval,

$$f_{S_n}(t) = \frac{\lambda^n t^{n-1} e^{-\lambda t}}{(n-1)!}, \qquad \Pr\{N(t) = n - 1\}\lambda\delta = \delta f_{S_n}(t) + o(\delta), \qquad p_{N(t)}(n) = \frac{(\lambda t)^n e^{-\lambda t}}{n!}.$$

- *Combining and splitting.* The sum of independent PPs of rates $\lambda_1, \ldots, \lambda_k$ is a PP of rate $\sum_j \lambda_j$. One may view it as generating each process and merging the arrivals, or as generating the combined process and allocating each arrival to subprocess i by IID rv's with PMF $\lambda_i / \sum_j \lambda_j$. This is the workhorse of Poisson queueing problems.
- *Order statistics.* Given $N(t) = n$, the density of $(S_1, \ldots, S_n)$ is $n!/t^n$ on $0 < s_1 < \cdots < s_n < t$, the joint distribution of n ordered uniform rv's; in particular $\Pr\{S_1 > \tau \mid N(t) = n\} = \Pr\{S_n < t - \tau \mid N(t) = n\} = ((t - \tau)/t)^n$ for $0 < \tau \leq t$.

14.3 Finite-state Markov chains

$\{X_n;\, n \geq 0\}$ is a Markov chain if $\Pr\{X_n = j \mid X_{n-1} = i, X_{n-2} = k, \ldots, X_0 = m\} = P_{ij}$, depending only on i, j, with p_{X_0} arbitrary. It is finite-state with states $\mathcal{S} = \{1, \ldots, \mathsf{M}\}$; the chain is usually identified with $\{P_{ij}\}$, with p_{X_0} a parameter. The directed graph, with an edge (i, j) only if $P_{ij} > 0$, shows connectivity; the matrix $[P]$ serves algebraic and asymptotic questions.

- *Classes.* Walks, paths (no repeated node), and cycles define accessibility, $i \to j$ if $P^n_{ij} > 0$ for some $n > 0$, which is transitive. Communicating states ($i \leftrightarrow j$) partition $\mathcal{S}$ into classes. A state i is transient if some j has $i \to j$ but $j \not\to i$, and recurrent otherwise; all states of a class are of the same type, and there is at least one recurrent class.
- *Periods.* $d(i) = \gcd\{n : P^n_{ii} > 0\}$ is constant on a class; a recurrent class of period $d > 1$ splits into subclasses $\mathcal{S}_1, \ldots, \mathcal{S}_d$ visited cyclically. An ergodic class is recurrent and aperiodic.

Theorem 14.1 [ERGODIC UNICHAINS]. *For an ergodic unichain (one ergodic recurrent class plus possibly*

transient states), $\lim_n P_{ij}^n = \pi_j$ *for all* i, j*, independent of* i*, where* $\sum_i \pi_i P_{ij} = \pi_j$*,* $\sum_i \pi_i = 1$*,* $\pi_i > 0$ *for recurrent* i*, and* $\pi_i = 0$ *otherwise. Equivalently* $\lim_n [P^n] = \boldsymbol{e}\boldsymbol{\pi}$ *with* $\boldsymbol{e} = (1, \ldots, 1)^\mathsf{T}$*,* $\boldsymbol{\pi}[P] = \boldsymbol{\pi}$*, and* $\boldsymbol{\pi}\boldsymbol{e} = 1$*.*

The proof for an ergodic chain has two substeps. First, $P_{ij}^n > 0$ for all i, j once $n \geq (\mathsf{M}-1)^2 + 1$. The bound must be this large: in a six-state chain whose cycles have lengths 5 and 6, a walk from state 2 is deterministic for four steps, then has two choices, and so on. Second, for $[P] > 0$ with $\alpha = \min_{i,j} P_{ij}$,

$$\max_i P_{ij}^{n+1} - \min_i P_{ij}^{n+1} \leq \Big(\max_\ell P_{\ell j}^n - \min_\ell P_{\ell j}^n\Big)(1-2\alpha), \qquad \text{so} \qquad \max_\ell P_{\ell j}^n - \min_\ell P_{\ell j}^n \leq (1-2\alpha)^n,$$

and $\lim_n \max_\ell P_{\ell j}^n = \lim_n \min_\ell P_{\ell j}^n > 0$: the limit is independent of ℓ and approached exponentially. The ergodic case follows by applying this to $[P^h]$ with $h = (\mathsf{M}-1)^2 + 1$.

Eigenvalues. λ is an eigenvalue of $[P]$ iff $\det[P - \lambda I] = 0$, iff $[P]\boldsymbol{\nu} = \lambda\boldsymbol{\nu}$ for some $\boldsymbol{\nu} \neq 0$, iff $\boldsymbol{\pi}[P] = \lambda\boldsymbol{\pi}$ for some $\boldsymbol{\pi} \neq 0$. Since $[P]\boldsymbol{e} = \boldsymbol{e}$, there is always a left eigenvector for $\lambda = 1$.

- All eigenvalues satisfy $|\lambda| \leq 1$. The multiplicity of $\lambda = 1$ equals the number of recurrent classes; for each recurrent class C there is a left eigenvector equal to the steady state on C and zero elsewhere, and a right eigenvector with $\nu_i = \lim_n \Pr\{X_n \in C \mid X_0 = i\}$.
- A recurrent class of period d contributes d eigenvalues equally spaced on the unit circle; there are no other eigenvalues with $|\lambda| = 1$.
- If the M eigenvalues are distinct, the right eigenvectors form the columns of an invertible $[U]$, the rows of $[U^{-1}]$ are the left eigenvectors, and

  $$[P^n] = [U][\Lambda^n][U^{-1}] = \sum_{i=1}^{\mathsf{M}} \lambda_i^n \boldsymbol{\nu}^{(i)}\boldsymbol{\pi}^{(i)}.$$

  If the eigenvectors span $\mathbb{R}^\mathsf{M}$, then for a unichain $P_{ij}^n \to \pi_j$ as $|\lambda_2|^n$, where λ_2 is the eigenvalue of second-largest magnitude. Otherwise $[P^n] = [U][J^n][U^{-1}]$ with a Jordan form $[J]$.

14.4 Renewal processes

- *Strong law.* If $\overline{X} > 0$, then $\lim_t N(t)/t = 1/\overline{X}$ WP1. This also holds if $\overline{X} = \infty$ (with limit 0), and in both cases $\lim_t N(t) = \infty$ WP1. The elementary renewal theorem gives $\lim_t \mathbb{E}[N(t)/t] = 1/\overline{X}$.
- *Rewards.* The integral of the residual life $Y(t)$ is a sum of terms $X_n^2/2$. The time-average residual life is $\mathbb{E}[X^2]/(2\mathbb{E}[X])$ and the time-average duration is $\mathbb{E}[X^2]/\mathbb{E}[X]$, WP1. For a PP the average duration is $2\mathbb{E}[X]$: big intervals contribute in two ways, since they are both long and more likely to contain a given time. In general, a reward $\mathcal{R}(Z(t), \overline{X}(t))$ depending on the position in the current interval accumulates $R_n = \int_0^{X_n} \mathcal{R}(z, X_n)\, dz$ per interval, and

$\lim_t \frac{1}{t}\int_0^t R(\tau)\,d\tau = \mathbb{E}[R_n]/\overline{X}$ WP1. This also works for ensemble averages.

- *Wald.* For IID X_n, a stopping trial J with $\mathbb{E}[J] < \infty$ satisfies $\mathbb{E}[S_J] = \overline{X}\,\mathbb{E}[J]$. When the X_n are nonnegative, the restriction $\mathbb{E}[J] < \infty$ is not necessary: every term of $\sum_n \mathbb{E}[X_n \mathbb{I}_{\{J \geq n\}}]$ is nonnegative, so the exchange of sum and expectation holds even when both sides are infinite. When X takes both signs it is necessary, as "stop when you're ahead" shows.
- *Little.* With renewals at arrivals to an empty system, $L(t) = A(t) - D(t)$ is a renewal-reward function with $L_n = \sum_i W_i$ over each period. With $\overline{L} = \lim_t \frac{1}{t}\int_0^t L(\tau)\,d\tau$, $\lambda = \lim_t A(t)/t$, and $\overline{W} = \lim_t \frac{1}{A(t)}\sum_{i=1}^{A(t)} W_i$,

$$\overline{W} = \lim_{t\to\infty} \frac{t}{A(t)} \cdot \lim_{t\to\infty} \frac{1}{t}\sum_{i=1}^{A(t)} W_i = \frac{\overline{L}}{\lambda}.$$

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Lecture 15: The last renewal

This unit closes renewal theory: why the sample-path limits behind the strong law, residual life, and Little's theorem are legitimate; how returns to a state make every ergodic Markov chain a renewal process; the expected number of renewals $m(t) = \mathbb{E}[N(t)]$ at finite t, with the renewal equation and Blackwell's theorem; and delayed renewal processes.

15.1 Sample-path averages

Convergence WP1 matters largely because continuous functions preserve it: if $Z_n \to \alpha$ WP1 and f is continuous at α, then $f(Z_n) \to f(\alpha)$ WP1. For a renewal process with $\mathbb{E}[X] < \infty$, $S_n/n \to \overline{X}$ WP1, so $n/S_n \to 1/\overline{X}$ WP1, and the strong law $\lim_t N(t)/t = 1/\overline{X}$ WP1 follows. The strong law also holds if $\overline{X} = \infty$: since X is a rv, each S_n is a rv, so $N(t)$ grows without bound as $t \to \infty$, but $N(t)/t \to 0$.

Thus $N(t)/t$ converges to $1/\overline{X}$ in three senses: WP1 (strong law), hence in probability, and in the mean (elementary renewal theorem):

$$\lim_{t\to\infty} \Pr\left\{\left|\frac{N(t)}{t} - \frac{1}{\overline{X}}\right| > \epsilon\right\} = 0 \text{ for all } \epsilon > 0, \qquad \lim_{t\to\infty} \mathbb{E}\left[\frac{N(t)}{t}\right] = \frac{1}{\overline{X}}.$$

The residual-life limit

$$\lim_{t\to\infty} \frac{\sum_{n=1}^{N(t)} X_n^2}{2t} = \lim_{t\to\infty} \frac{\sum_{n=1}^{N(t)} X_n^2}{N(t)} \cdot \frac{N(t)}{2t} = \frac{\mathbb{E}[X^2]}{2\mathbb{E}[X]} \qquad \text{WP1}$$

is an abbreviation for a sample-path result. For a fixed sample point ω and a given t, the right side is the product of two numbers, so as t increases we take the limit of a product of two numerical functions of t. For ω in a set of probability 1, both functions converge to finite values, so the limit of the product is the product of the limits. The same use of sample-path limits proves the renewal-reward theorem and Little's theorem: there $\overline{L} = \lim_t \frac{1}{t}\sum_{i\le A(t)} W_i$ and $\overline{W} = \lim_t \frac{t}{A(t)} \cdot \lim_t \frac{1}{t}\sum_{i\le A(t)} W_i = \overline{L}/\lambda$, where Little's theorem combines an accounting identity over each busy period with this handling of limits.

15.2 Markov chains and renewal processes

Theorem 15.1 [Mean recurrence time]. *Let $\{X_n;\, n \ge 0\}$ be a finite-state ergodic Markov chain with $X_0 = i$, and let $N_i(t)$ be the number of visits to state i at times $1, \ldots, t$. Then $\{N_i(t);\, t \ge 1\}$ is a renewal counting process, and the mean recurrence time of state i is $\overline{Y} = 1/\pi_i$.*

Proof. By the Markov property the chain restarts at each return to i, so the inter-renewal periods $Y_1, Y_2, \ldots$ between visits are IID. By the elementary renewal theorem, $\lim_t \mathbb{E}[N_i(t)]/t = 1/\overline{Y}$. Since $X_0 = i$,

$$P_{ii}^t = \Pr\{N_i(t) - N_i(t-1) = 1\} = \mathbb{E}[N_i(t) - N_i(t-1)], \qquad \text{so} \qquad \sum_{n=1}^{t} P_{ii}^n = \mathbb{E}[N_i(t)].$$

Since $P_{ii}^n \to \pi_i$ (exponentially), so do the averages:

$$\pi_i = \lim_{t\to\infty} \frac{1}{t} \sum_{n=1}^{t} P_{ii}^n = \lim_{t\to\infty} \frac{\mathbb{E}[N_i(t)]}{t} = \frac{1}{\overline{Y}}. \qquad \square$$

15.3 Expected number of renewals

The elementary renewal theorem says $\lim_t \mathbb{E}[N(t)]/t = 1/\overline{X}$, but for finite t the RENEWAL FUNCTION $m(t) = \mathbb{E}[N(t)]$ can be very messy. If X takes only the values 1 and $\sqrt{2}$, the points at which $N(t)$ can increase become increasingly dense as t increases, and $m(t)$ is nondecreasing but otherwise ugly.

Some progress comes from expressing $m(t)$ through its values at smaller t. Conditioning on $X_1 = x$: if $x > t$ then $N(t) = 0$, while if $x \le t$ the process restarts at x and $N(t)$ is 1 plus the number of renewals of a fresh process in time $t - x$. This gives the RENEWAL EQUATION

$$m(t) = F_X(t) + \int_0^t m(t-x)\, dF_X(x), \qquad m(0) = 0,$$

which equals $\int_0^t [1 + m(t-x)] f_X(x)\, dx$ when the density f_X exists.

The renewal equation is linear in the function m, like the equations of linear systems, and can be solved when f_X has a rational Laplace transform. The solution has the form

$$m(t) = \frac{t}{\overline{X}} + \frac{\sigma^2}{2\overline{X}^2} - \frac{1}{2} + \epsilon(t) \qquad \text{for } t \ge 0, \qquad \text{where } \lim_{t\to\infty} \epsilon(t) = 0.$$

For exponential X, $\sigma^2 = \overline{X}^2$ and the constant vanishes. The dominant term for large t is $t/\overline{X}$, consistent with the elementary renewal theorem; the next two terms say the initial transient never quite dies away. Heavy-tailed distributions pick up extra renewals initially: for $p_X(\epsilon) = 1 - \epsilon$, $p_X(1/\epsilon) = \epsilon$, $\overline{X} \approx 1$ and $\sigma^2/(2\overline{X}^2) \approx 1/(2\epsilon)$, reflecting the many short intervals that typically precede the first long one.

15.4 Blackwell's theorem

Blackwell's theorem essentially says that the expected renewal rate for large t is $1/\overline{X}$. It cannot quite say this: if X is discrete, then every S_n is discrete, and $m(t)$ does not have a derivative. The fundamental distinction is between arithmetic and non-arithmetic distributions.

Definition 15.2 [Arithmetic distribution]. A rv X has an arithmetic distribution if its possible sample values are all integer multiples of some number λ. The largest such λ is the span of the distribution.

If X is arithmetic with span $\lambda > 0$, every S_n is arithmetic with span λ or an integer multiple of it, so $N(t)$ can increase only at multiples of λ. For a non-arithmetic discrete distribution, such as $f_X(1) = f_X(\pi) = 1/2$, the points at which $N(t)$ can increase become dense as $t \to \infty$.

Theorem 15.3 [Blackwell's theorem].

$$\lim_{t\to\infty}[m(t+\lambda) - m(t)] = \frac{\lambda}{\overline{X}} \qquad \textit{for arithmetic } X \textit{ with span } \lambda;$$

$$\lim_{t\to\infty}[m(t+\delta) - m(t)] = \frac{\delta}{\overline{X}} \qquad \textit{for non-arithmetic } X \textit{ and any } \delta > 0.$$

The general proof uses difficult analysis and gives little insight. When Laplace techniques work, the theorem follows from the solution above. The hard case is a non-arithmetic but discrete distribution.

The arithmetic case with a finite set of values is easy, by modeling the renewals as returns to a state of a Markov chain. Take $\lambda = 1$, so X takes values in $\{1, \ldots, K\}$. Let the state at time n be the age, the time since the last renewal, so state 0 means a renewal at time n (Figure 20). From state i we know $X > i$, and the chain moves to $i+1$ if $X > i+1$ and returns to 0 otherwise:

$$P_{i,i+1} = \frac{\Pr\{X > i+1\}}{\Pr\{X > i\}} = \frac{1 - p_X(1) - \cdots - p_X(i+1)}{1 - p_X(1) - \cdots - p_X(i)}, \qquad P_{i,0} = 1 - P_{i,i+1}.$$

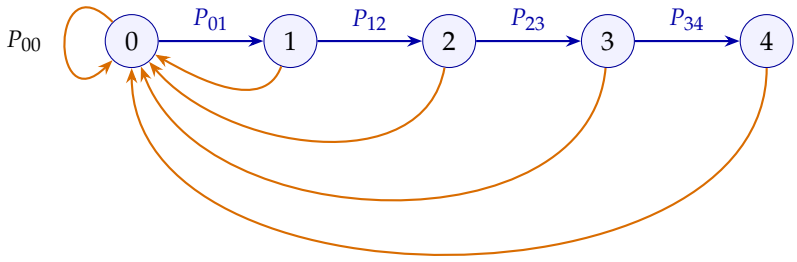


Figure 20: Markov chain for an arithmetic renewal process with span 1 and $X \leq 5$. The state is the age; each forward arc continues the current interval, and each arc back to 0 is a renewal.

The cycles through state 0 have lengths equal to the possible values of X, so the chain is aperiodic exactly when the span is 1. Then $\lim_n P_{00}^n = \pi_0$, and Theorem 15.1 gives $\pi_0 = 1/\overline{X}$. Since at most one renewal occurs at each integer time, $P_{00}^n = \Pr\{\text{renewal at } n\} = m(n) - m(n-1)$, which is Blackwell's theorem for this case. The moral: when doing renewals, think Markov, and when doing Markov, think renewals.

15.5 Delayed renewal processes

Definition 15.4 [Delayed renewal process]. A delayed renewal process is a renewal process modified so that the first inter-renewal interval X_1 has a different distribution from the others, $X_2, X_3, \ldots$; all intervals are still independent.

All the limit theorems remain unchanged, even if $\mathbb{E}[X_1] = \infty$. For the strong law, the renewals after X_1 form an ordinary renewal process N', so $N(t) = N'(t - X_1) + 1$ for $t \geq X_1$; since X_1 is finite WP1, $N(t)/t$ has the same limit as $N'(t)/t$. For Markov chains, this lets us start in one state and count visits to another state.

Source. Robert Gallager, MIT 6.262, *Discrete Stochastic Processes*, Lecture 15, Spring 2011. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 16: Renewals and countable-state Markov

16.1 The major renewal theorems

Let $\{N(t); t > 0\}$ be a renewal process with inter-renewal intervals $X_1, X_2, \ldots$ of mean $\overline{X}$, and let $m(t) = \mathbb{E}[N(t)]$. Three limit theorems describe its long-run behavior.

- *Strong law for renewals* (a sample-path time average):

$$\Pr\Big\{\lim_{t\to\infty} \frac{N(t)}{t} = \frac{1}{\overline{X}}\Big\} = 1.$$

- *Elementary renewal theorem* (an ensemble average of a time average): $\lim_{t\to\infty} \mathbb{E}\Big[\frac{N(t)}{t}\Big] = \frac{1}{\overline{X}}$.
- *Blackwell's theorem* (a pure ensemble average). If X is arithmetic with span λ, then $\lim_{n\to\infty}[m(n\lambda+\lambda) - m(n\lambda)] = \lambda/\overline{X}$. If X is non-arithmetic, then for every $\delta > 0$,

$$\lim_{t\to\infty}[m(t+\delta) - m(t)] = \frac{\delta}{\overline{X}}.$$

In the arithmetic case at most one renewal occurs at each multiple of λ, so $m(n\lambda+\lambda) - m(n\lambda)$ is the probability of a renewal at $(n+1)\lambda$, and Blackwell's theorem reads

$$\lim_{n\to\infty} \Pr\{\text{renewal at } n\lambda\} = \frac{\lambda}{\overline{X}}.$$

If the arithmetic renewal process is modeled as a Markov chain started in a renewal state 0, this says essentially that $P_{00}^n \to \pi_0$. The non-arithmetic statement is the best one could hope for: although

$$\lim_{\delta\to 0}\lim_{t\to\infty} \frac{m(t+\delta) - m(t)}{\delta} = \frac{1}{\overline{X}},$$

the order of the two limits cannot be interchanged, since $m(t)$ need not have a derivative tending to $1/\overline{X}$.

16.2 Age and duration at a given time

The AGE at time t is $Z(t) = t - S_{N(t)}$, the time since the last renewal (with $S_0 = 0$), and the DURATION is $\widetilde{X}(t) = X_{N(t)+1}$, the length of the inter-renewal interval containing t.

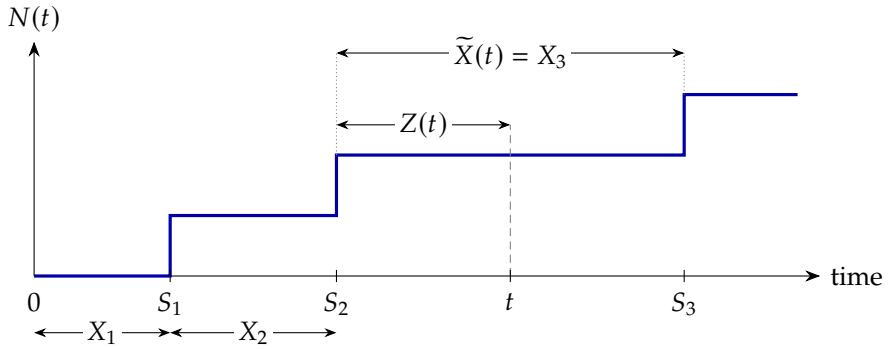


Figure 21: A sample path of $N(t)$. At the time t shown, $N(t) = 2$; the age $Z(t)$ runs back to S_2, and the duration $\widetilde{X}(t) = X_3$ is the whole interval from S_2 to S_3.

Assume an arithmetic renewal process of span 1 and an integer t. Then $Z(t) = i \geq 0$ and $\widetilde{X}(t) = k > i$ if and only if there are successive arrivals at $t - i$ and $t - i + k$. Let $q_j = \Pr\{\text{arrival at } j\} = \sum_{n\geq 1} p_{S_n}(j)$, with $q_0 = 1$ for a nominal arrival at time 0. The arrival at $t - i$ and the next inter-renewal interval are independent, so

$$p_{Z(t),\widetilde{X}(t)}(i,k) = q_{t-i}\, p_X(k), \qquad 0 \leq i \leq t,\ k > i.$$

Since $q_j = m(j) - m(j-1)$, Blackwell's theorem gives $\lim_{j\to\infty} q_j = 1/\overline{X}$. Hence, for $k > i \geq 0$,

$$\lim_{t\to\infty} p_{Z(t),\widetilde{X}(t)}(i,k) = \frac{p_X(k)}{\overline{X}}, \qquad \lim_{t\to\infty} p_{Z(t)}(i) = \frac{\sum_{k=i+1}^{\infty} p_X(k)}{\overline{X}} = \frac{F_X^c(i)}{\overline{X}},$$

$$\lim_{t\to\infty} p_{\widetilde{X}(t)}(k) = \frac{\sum_{i=0}^{k-1} p_X(k)}{\overline{X}} = \frac{k\, p_X(k)}{\overline{X}}, \qquad k \geq 1,$$

where $F_X^c(i) = \Pr\{X > i\}$. The asymptotic expected duration is

$$\lim_{t\to\infty} \mathbb{E}[\widetilde{X}(t)] = \sum_{k=1}^{\infty} k \cdot \frac{k\, p_X(k)}{\overline{X}} = \frac{\mathbb{E}[X^2]}{\overline{X}}.$$

This equals the sample-path time average of the duration, but now it is an ensemble average at a given t, and finite t can be treated as well. It also has a different interpretation: given $\widetilde{X} = k$, there are k equiprobable values $0, \ldots, k-1$ of the age, each with joint probability $p_X(k)/\overline{X}$. The PMF of $\widetilde{X}$ is thus p_X reweighted by $k/\overline{X}$: long intervals are more likely to contain t, so large durations are enhanced relative to inter-renewal intervals.

Averaging the age over these k values gives the expected age,

$$\lim_{t\to\infty} \mathbb{E}[Z(t)] = \sum_{k\geq 1} \frac{p_X(k)}{\overline{X}} \sum_{i=0}^{k-1} i = \sum_{k\geq 1} \frac{p_X(k)\, k(k-1)}{2\overline{X}} = \frac{\mathbb{E}[X^2]}{2\overline{X}} - \frac{1}{2}.$$

This holds at integer values of large t. Between integers the age increases linearly with slope 1, and it drops back by 1 at the next integer.

16.3 Countable-state Markov chains: two examples

The biggest change from finite-state to countable-state Markov chains is in the concept of a recurrent class.

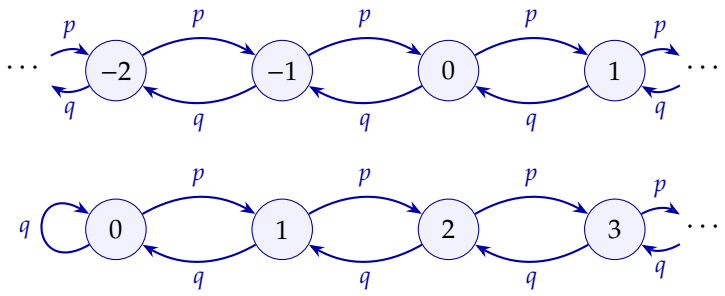


Figure 22: Top: the chain of a Bernoulli ±1 process on the integers. Bottom: a birth-death chain on the nonnegative integers. In both, $q = 1 - p$.

Example: *The Bernoulli ±1 chain.* The top chain of Figure 22 models a Bernoulli ±1 process: the state at time n is $S_n = X_1 + \cdots + X_n$, where the X_i are IID with $\Pr\{X_i = 1\} = p$ and $\Pr\{X_i = -1\} = q$. If k of the n steps are positive, the state is $j = 2k - n$. All states communicate and the period is $d = 2$. The variance $\sigma^2_{S_n} = n[1 - (p - q)^2]$ grows linearly, so the probability mass spreads out and P^n_{0j} approaches 0 at least as fast as $1/\sqrt{n}$ for every j.

Example: *A birth-death chain.* In the bottom chain, if $p > 1/2$ the state drifts to the right and $P^n_{0j} \to 0$ for all j. If $p < 1/2$ it drifts to the left and keeps bumping into state 0. When the chain is truncated at a state k, its steady-state probabilities are proportional to $(p/q)^i$, $0 \le i \le k$. They increase to the right for $p > 1/2$, increase to the left for $p < 1/2$, and are uniform for $p = 1/2$. As the truncation point increases, the steady state remains positive only for $p < 1/2$.

We want *recurrent* to mean that, given $X_0 = i$, there is a future return to i with probability 1 (WP1). The birth-death chain above turns out to be recurrent for $p < 1/2$ and not recurrent for $p > 1/2$; the case $p = 1/2$ is strange and will be called null recurrent. Renewal theory applies to recurrent chains, but first we need first-passage times.

16.4 First-passage times

Definition 16.1 [First-passage-time probability]. The first-passage-time probability is

$$f_{ij}(n) = \Pr\{X_n = j,\ X_{n-1} \neq j, \ldots, X_1 \neq j \mid X_0 = i\},$$

the probability, given $X_0 = i$, that n is the first epoch at which $X_n = j$.

Conditioning on the first step gives

$$f_{ij}(1) = P_{ij}, \qquad f_{ij}(n) = \sum_{k \neq j} P_{ik} f_{kj}(n-1), \quad n > 1.$$

Compare the Chapman-Kolmogorov equation $P_{ij}^n = \sum_k P_{ik} P_{kj}^{n-1}$: the only difference is that the first-passage recursion cuts off the paths that pass through j earlier, as in the computation of expected first-passage times.

Let $F_{ij}(n) = \sum_{m \leq n} f_{ij}(m)$ be the probability of reaching j by time n. If $\lim_{n\to\infty} F_{ij}(n) = 1$, there is a random variable T_{ij}, the FIRST-PASSAGE TIME, with distribution function F_{ij}. Equivalently,

$$F_{ij}(1) = P_{ij}, \qquad F_{ij}(n) = P_{ij} + \sum_{k \neq j} P_{ik} F_{kj}(n-1), \quad n > 1.$$

Since $F_{ij}(n)$ is nondecreasing in n, the limit $F_{ij}(\infty)$ exists, and by monotone convergence it satisfies

$$F_{ij}(\infty) = P_{ij} + \sum_{k \neq j} P_{ik} F_{kj}(\infty).$$

Unfortunately $F_{ij}(\infty) = 1$ for all i, j also satisfies these equations, since $\sum_k P_{ik} = 1$. The correct solution is the smallest nonnegative solution. Indeed, if $\{G_{ij}\}$ is any nonnegative solution, then $F_{ij}(1) = P_{ij} \leq G_{ij}$, and inductively $F_{ij}(n) \leq P_{ij} + \sum_{k \neq j} P_{ik} G_{kj} = G_{ij}$; letting $n \to \infty$ gives $F_{ij}(\infty) \leq G_{ij}$.

16.5 Recurrence and transience

Definition 16.2 [RECURRENT STATE]. State j is RECURRENT if $F_{jj}(\infty) = 1$, and TRANSIENT otherwise.

If j is recurrent, an eventual return occurs WP1 and the recurrence time T_{jj} is a random variable with distribution function F_{jj}. By the Markov property, the successive returns to j are the renewal epochs of a renewal process whose inter-renewal intervals have distribution function F_{jj}. Let $N_{jj}(t)$ be the number of returns to j in steps $1, \ldots, t$, starting from $X_0 = j$.

Theorem 16.3 [TESTS FOR RECURRENCE]. *The following are equivalent:*

1. *state j is recurrent;*
2. $\lim_{t\to\infty} N_{jj}(t) = \infty$ *WP1;*
3. $\lim_{t\to\infty} \mathbb{E}[N_{jj}(t)] = \infty$*;*
4. $\lim_{t\to\infty} \sum_{1 \leq n \leq t} P_{jj}^n = \infty$.

None of these implies that $\mathbb{E}[T_{jj}] < \infty$.

Proof. (1)⇒(2): each inter-renewal interval is finite WP1, so there are infinitely many renewals. (2)⇒(3) by monotone convergence. (3)⇔(4): $N_{jj}(t)$ is the sum of the indicators of $\{X_n = j\}$, $1 \le n \le t$, so $\mathbb{E}[N_{jj}(t)] = \sum_{n \le t} P^n_{jj}$. (3)⇒(1): if j is transient, each return is followed by another with probability $F_{jj}(\infty) < 1$, so the total number of returns is geometric with finite mean $F_{jj}(\infty)/(1 - F_{jj}(\infty))$, and $\mathbb{E}[N_{jj}(t)]$ stays bounded. □

Two states are in the same class if they communicate, as for finite-state chains.

Theorem 16.4 [Recurrence is a class property]. *If states i and j are in the same class, then either both are recurrent or both are transient.*

Proof. Suppose j is recurrent, so $\sum_n P^n_{jj} = \infty$. Since $i \leftrightarrow j$, there are m, ℓ with $P^m_{ij} > 0$ and $P^\ell_{ji} > 0$. Going from i to j in m steps, from j back to j in k steps, and from j to i in ℓ steps gives $P^{m+k+\ell}_{ii} \ge P^m_{ij} P^k_{jj} P^\ell_{ji}$, so

$$\sum_{n=1}^{\infty} P^n_{ii} \ge P^m_{ij} P^\ell_{ji} \sum_{k=1}^{\infty} P^k_{jj} = \infty,$$

and i is recurrent by the previous theorem. Exchanging i and j proves the converse. □

By the same kind of argument, if i and j are recurrent states of the same class, then $F_{ij}(\infty) = 1$.

16.6 Positive and null recurrence

If j is recurrent, T_{jj} might or might not have a finite mean.

Definition 16.5 [Positive and null recurrence]. State j is positive recurrent if $\mathbb{E}[T_{jj}] < \infty$. If T_{jj} is a random variable but $\mathbb{E}[T_{jj}] = \infty$, then j is null recurrent. Otherwise, when $F_{jj}(\infty) < 1$, j is transient.

For $p = 1/2$, every state of both chains in Figure 22 is null recurrent.

Source. Robert Gallager, MIT 6.262, *Discrete Stochastic Processes*, Lecture 16, Spring 2011. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 17: Countable-state Markov chains

17.1 A strong law for renewal rewards

Let $\{Y_i; i \geq 1\}$ be the IID service times of a G/G/∞ queue, and let $\{N(t); t > 0\}$ be the renewal process with interarrival intervals $\{X_i; i \geq 1\}$. Assume $\overline{X} < \infty$ and $\overline{Y} < \infty$. A plausibility argument for the limit of $\frac{1}{t}\sum_{i=1}^{N(t,\omega)} Y_i(\omega)$ runs as follows:

$$\lim_{t\to\infty} \frac{1}{t}\sum_{i=1}^{N(t,\omega)} Y_i(\omega) = \lim_{t\to\infty} \frac{N(t,\omega)}{t} \cdot \frac{\sum_{i=1}^{N(t,\omega)} Y_i(\omega)}{N(t,\omega)} \tag{17.1}$$

$$= \lim_{t\to\infty} \frac{N(t,\omega)}{t} \cdot \lim_{t\to\infty} \frac{\sum_{i=1}^{N(t,\omega)} Y_i(\omega)}{N(t,\omega)} \tag{17.2}$$

$$= \lim_{t\to\infty} \frac{N(t,\omega)}{t} \cdot \lim_{n\to\infty} \frac{\sum_{i=1}^{n} Y_i(\omega)}{n} \tag{17.3}$$

$$= \frac{\overline{Y}}{\overline{X}} \qquad \text{WP1.} \tag{17.4}$$

To make it rigorous, work from the bottom up.

Justification. Let $A_1 = \{\omega : \lim_t N(t,\omega)/t = 1/\overline{X}\}$ and $A_2 = \{\omega : \lim_n \frac{1}{n}\sum_{i=1}^{n} Y_i(\omega) = \overline{Y}\}$. By the strong law for renewals $\Pr\{A_1\} = 1$, and by the SLLN $\Pr\{A_2\} = 1$. Thus (17.3) equals (17.4) for $\omega \in A_1A_2$, and $\Pr\{A_1A_2\} = 1$.

Fix $\omega \in A_1A_2$ and $\epsilon > 0$. Since $\omega \in A_2$, there is $m(\epsilon,\omega)$ with $\left|\frac{1}{n}\sum_{i=1}^{n} Y_i(\omega) - \overline{Y}\right| < \epsilon$ for all $n \geq m(\epsilon,\omega)$. Since $\omega \in A_1$, $N(t,\omega) \to \infty$, so there is $t(\epsilon,\omega)$ with $N(t,\omega) \geq m(\epsilon,\omega)$ for all $t \geq t(\epsilon,\omega)$. Then

$$\left|\frac{\sum_{i=1}^{N(t,\omega)} Y_i(\omega)}{N(t,\omega)} - \overline{Y}\right| < \epsilon \qquad \text{for all } t \geq t(\epsilon,\omega).$$

Since ϵ is arbitrary, (17.2), (17.3), and (17.4) agree on A_1A_2.

Finally, the limit of a product equals the product of the limits when both limits are finite, which gives (17.1)=(17.2). If $f(t) \to a$ and $g(t) \to b$, then

$$f(t)g(t) - ab = (f(t) - a)(g(t) - b) + a(g(t) - b) + b(f(t) - a).$$

Choose $t(\epsilon)$ with $|f(t)-a| \leq \epsilon$ and $|g(t)-b| \leq \epsilon$ for $t \geq t(\epsilon)$. Then $|f(t)g(t)-ab| \leq \epsilon^2+\epsilon|a|+\epsilon|b|$ for $t \geq t(\epsilon)$, so $f(t)g(t) \to ab$. □

17.2 Classes and positive recurrence

Lecture 16 showed that recurrence is a class property: if j is recurrent and $j \leftrightarrow i$, with $P_{ij}^m > 0$ and $P_{ji}^\ell > 0$, then $\sum_n P_{ii}^n \geq P_{ij}^m P_{ji}^\ell \sum_k P_{jj}^k = \infty$. Recall that $\lim_t \mathbb{E}[N_{ii}(t)] = \sum_n P_{ii}^n$. The same argument shows that if $i \leftrightarrow j$ are recurrent, then $\sum_{n=1}^\infty P_{ij}^n = \infty$, so also $\lim_t \mathbb{E}[N_{ij}(t)] = \infty$, where $N_{ij}(t)$ counts the visits to j in steps $1, \ldots, t$ starting from $X_0 = i$.

Positive recurrence is also a class property. Suppose $i \leftrightarrow j$ are recurrent, and consider the renewal process of returns to j, with $X_0 = j$. Earn reward $R(t) = 1$ whenever $X(t) = i$. By the renewal-reward theorem,

$$\lim_{t\to\infty} \frac{1}{t} \int_0^t R(\tau)\, d\tau = \frac{\mathbb{E}[R_n]}{\overline{T}_{jj}} \qquad \text{WP1},$$

where $\mathbb{E}[R_n]$ is the expected number of visits to i during one recurrence of j and $\overline{T}_{jj} = \mathbb{E}[T_{jj}]$. The left side is $\lim_t \frac{1}{t} N_{ji}(t)$. The visits to i form a delayed renewal process with inter-renewal mean $\overline{T}_{ii}$, so this limit is $1/\overline{T}_{ii}$. Hence

$$\frac{1}{\overline{T}_{ii}} = \frac{\mathbb{E}[R_n]}{\overline{T}_{jj}}.$$

There is a path from j to i, so $\mathbb{E}[R_n] > 0$. Thus if $\overline{T}_{jj} < \infty$, then $1/\overline{T}_{ii} > 0$, that is, $\overline{T}_{ii} < \infty$.

Theorem 17.1 [Positive recurrence is a class property]. *If $i \leftrightarrow j$ are recurrent, then either both are positive recurrent or both are null recurrent.*

17.3 Steady state for positive-recurrent chains

Definition 17.2 [Steady-state distribution]. A set $\{\pi_i; i \geq 0\}$ is a steady-state distribution if

$$\pi_j \geq 0, \qquad \pi_j = \sum_i \pi_i P_{ij} \text{ for all } j \geq 0, \qquad \sum_j \pi_j = 1.$$

Definition 17.3 [Irreducible chain]. A Markov chain is irreducible if all pairs of states communicate.

For finite-state chains, irreducible implies recurrent. A countable-state irreducible chain can be positive recurrent, null recurrent, or transient.

If a steady state π exists and $\Pr\{X_0 = i\} = \pi_i$ for each i, then $p_{X_1}(j) = \sum_i \pi_i P_{ij} = \pi_j$. Iterating gives $p_{X_n}(j) = \pi_j$, so steady state is preserved. Let $\widetilde{N}_j(t)$ be the number of visits to j in steps $1, \ldots, t$, starting in steady state. Then

$$\mathbb{E}[\widetilde{N}_j(t)] = \sum_{k=1}^t \Pr\{X_k = j\} = t\pi_j.$$

Renewal arguments use $N_{jj}(t)$, while steady-state arguments use $\widetilde{N}_j(t)$. The two are related as

follows: $N_{ij}(t)$ is 1 for the first visit to j, if any, plus at most $N_{jj}(t) - 1$ for the subsequent recurrences from j to j. Thus

$$\mathbb{E}[N_{ij}(t)] \le 1 + \mathbb{E}[N_{jj}(t)], \qquad \mathbb{E}[\widetilde{N}_j(t)] = \sum_i \pi_i \mathbb{E}[N_{ij}(t)] \le 1 + \mathbb{E}[N_{jj}(t)].$$

Theorem 17.4 [Steady state and positive recurrence]. *For an irreducible Markov chain, the steady-state equations have a solution if and only if the states are positive recurrent. If a solution exists, then $\pi_i = 1/\overline{T}_{ii} > 0$ for all i.*

Proof of the only-if direction. Assume π exists. For each j and t,

$$\pi_j = \frac{\mathbb{E}[\widetilde{N}_j(t)]}{t} \le \frac{1}{t} + \frac{\mathbb{E}[N_{jj}(t)]}{t}.$$

Letting $t \to \infty$, the elementary renewal theorem for the returns to j gives $\pi_j \le 1/\overline{T}_{jj}$ (read as 0 if $\overline{T}_{jj} = \infty$). Since $\sum_j \pi_j = 1$, some $\pi_j > 0$, so $\lim_t \mathbb{E}[N_{jj}(t)]/t > 0$ and $\overline{T}_{jj} < \infty$ for that j. Hence j is positive recurrent, and, by the class property, so are all states. □

The converse, and the fact that the inequality $\pi_j \le 1/\overline{T}_{jj}$ is an equality, take a longer argument that is omitted here.

17.4 Birth-death Markov chains

A birth-death chain on the nonnegative integers moves from i to $i+1$ with probability p_i, from $i+1$ to i with probability q_{i+1}, and otherwise stays put.

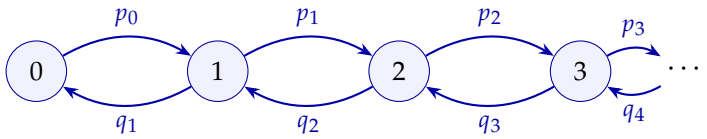


Figure 23: A birth-death chain. Each state i also has a self-loop of probability $1 - p_i - q_i$ (with $q_0 = 0$).

On any sample path, the number of $i \to i+1$ transitions is within 1 of the number of $i+1 \to i$ transitions, since every crossing upward must be followed by a crossing downward before the next upward one. Dividing by the path length and letting it grow suggests

$$\pi_i p_i = \pi_{i+1} q_{i+1}, \qquad \text{so} \qquad \pi_{i+1} = \frac{\pi_i p_i}{q_{i+1}}.$$

With $\rho_i = p_i/q_{i+1}$ this becomes

$$\boxed{\pi_i = \pi_0 \prod_{j=0}^{i-1} \rho_j, \qquad \pi_0 = \frac{1}{1 + \sum_{i=1}^{\infty} \prod_{j=0}^{i-1} \rho_j}}.$$

This agrees with the steady-state equations: the equation for state j reads $\pi_j(p_j + q_j) = \pi_{j-1}p_{j-1} + \pi_{j+1}q_{j+1}$, and the two terms on the right equal $\pi_j q_j$ and $\pi_j p_j$.

The solution depends only on $\rho_0, \rho_1, \ldots$, not on the sizes of the self-loops. The expression for π_0 converges, making the chain positive recurrent, essentially when the ρ_i are asymptotically less than 1. Rather than checking carefully whether counting up- and down-crossings is justified, it is easier to rely on Theorem 17.4: a positive solution of the steady-state equations exists, so the chain is positive recurrent and the solution is the unique steady state. Birth-death chains are particularly useful in queueing, where births are arrivals and deaths are departures.

17.5 Reversibility

For any Markov chain,

$$\Pr\{X_{n+k}, \ldots, X_{n+1} \mid X_n, \ldots, X_0\} = \Pr\{X_{n+k}, \ldots, X_{n+1} \mid X_n\}.$$

Hence, for any event A^+ defined on $X_{n+1}, X_{n+2}, \ldots$ and any event A^- defined on $X_{n-1}, X_{n-2}, \ldots, X_0$,

$$\Pr\{A^+ \mid X_n, A^-\} = \Pr\{A^+ \mid X_n\}, \qquad \Pr\{A^+, A^- \mid X_n\} = \Pr\{A^+ \mid X_n\} \Pr\{A^- \mid X_n\},$$

$$\Pr\{A^- \mid X_n, A^+\} = \Pr\{A^- \mid X_n\}.$$

Given the present, past and future are independent. In particular $\Pr\{X_{n-1} \mid X_n, X_{n+1}, \ldots, X_{n+k}\} = \Pr\{X_{n-1} \mid X_n\}$, so the chain run backward is still Markov. By Bayes,

$$\Pr\{X_{n-1} \mid X_n\} = \frac{\Pr\{X_n \mid X_{n-1}\} \Pr\{X_{n-1}\}}{\Pr\{X_n\}}.$$

If the forward chain is in steady state, then $\Pr\{X_{n-1} = j \mid X_n = i\} = P_{ji}\pi_j/\pi_i$, which does not depend on n. Apart from the inhomogeneity caused by starting at time 0, the backward chain is a homogeneous Markov chain; think of the forward chain as started in steady state at time $-\infty$.

Definition 17.5 [Backward chain and reversibility]. The backward transition probabilities are $P^*_{ij} = \Pr\{X_{n-1} = j \mid X_n = i\}$, related to the forward ones by

$$\pi_i P^*_{ij} = \pi_j P_{ji}.$$

A chain is reversible if $P^*_{ij} = P_{ij}$ for all i, j.

Theorem 17.6 [Birth-death chains are reversible]. *A birth-death Markov chain that has a steady-state distribution is reversible.*

Proof. The only nonzero transitions between distinct states are $i \leftrightarrow i+1$, and the steady state satisfies $\pi_i P_{i,i+1} = \pi_{i+1} P_{i+1,i}$. Hence $P^*_{i,i+1} = \pi_{i+1} P_{i+1,i} / \pi_i = P_{i,i+1}$, similarly $P^*_{i+1,i} = P_{i+1,i}$, and $P^*_{ii} = P_{ii}$ trivially. □

For a queue, this means that a sample path read backward in time is a sample path of the same chain, with every arrival (an up-step) becoming a departure (a down-step) and vice versa.

Source. Robert Gallager, MIT 6.262, *Discrete Stochastic Processes*, Lecture 17, Spring 2011. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 18: Countable-state Markov chains and processes

18.1 Reversibility and the guessing theorem

The Markov condition works in both directions: given X_n, the past and the future are independent (Lecture 17). The forward chain must be in steady state, however, for the backward chain to be homogeneous. For a positive-recurrent chain in steady state, the backward transition probabilities $P^*_{ij} = \Pr\{X_{n-1} = j \mid X_n = i\}$ satisfy

$$\pi_i P^*_{ij} = \pi_j P_{ji} = \Pr\{X_n = i,\ X_{n-1} = j\}.$$

The chain is reversible ($P^*_{ij} = P_{ij}$ for all i, j) if and only if

$$\pi_i P_{ij} = \pi_j P_{ji} \quad \text{for all } i, j,$$

that is, $\Pr\{X_n = i, X_{n-1} = j\} = \Pr\{X_n = j, X_{n-1} = i\}$. In words, the long-term fraction of $i \to j$ transitions equals the long-term fraction of $j \to i$ transitions. All positive-recurrent birth-death chains are reversible.

More generally, suppose the nonzero transitions of a positive-recurrent chain form a tree. Between two crossings of an edge in one direction, the chain must cross it in the other direction, since removing the edge disconnects the graph. So the numbers of crossings in the two directions differ by at most one, and the chain is reversible. A birth-death chain is a very skinny tree.

Theorem 18.1 [Guessing theorem]. *Let $\{P_{ij}; i, j \geq 0\}$ be an irreducible Markov chain. If numbers $\pi_i > 0$ with $\sum_i \pi_i = 1$ satisfy $\pi_i P_{ij} = \pi_j P_{ji}$ for all i, j, then the chain is positive recurrent and reversible, and $\{\pi_i\}$ is its steady-state distribution.*

Proof. Summing over i gives $\sum_i \pi_i P_{ij} = \pi_j \sum_i P_{ji} = \pi_j$. Together with $\sum_i \pi_i = 1$ and $\pi_i \geq 0$, these are the steady-state equations, so by the steady-state theorem of Lecture 17 the chain is positive recurrent and $\{\pi_i\}$ is the unique steady state. The detailed-balance equations then say $P^*_{ij} = P_{ij}$. □

Some sanity checks for reversibility follow from detailed balance:

1. if $P_{ij} > 0$, then $P_{ji} > 0$;
2. if the chain is periodic, the period is 2, since $P_{ij}P_{ji} > 0$ gives a return in two steps;
3. $P_{ij}P_{jk}P_{ki} = P_{ik}P_{kj}P_{ji}$, obtained by multiplying the detailed-balance equations for the pairs (i, j), (j, k), (k, i) and canceling $\pi_i\pi_j\pi_k$.

Theorem 18.2 [Guessing theorem for non-reversible chains]. *If there are $\{\pi_i \geq 0; i \geq 0\}$ with $\sum_i \pi_i = 1$ and transition probabilities $\{P^*_{ij}\}$ such that $\pi_i P_{ij} = \pi_j P^*_{ji}$ for all i, j, then $\{\pi_i\}$ are steady-state probabilities and $\{P^*_{ij}\}$ are the backward transition probabilities.*

Proof. Summing over i gives $\sum_i \pi_i P_{ij} = \pi_j \sum_i P^*_{ji} = \pi_j$, the steady-state equations. The given relation is then exactly the defining relation of the backward probabilities. □

18.2 The sampled-time M/M/1 queue

Sample the state of an M/M/1 queue, with arrival rate λ and service rate $\mu > \lambda$, at increments δ so small that more than one arrival or departure in an increment can be ignored.

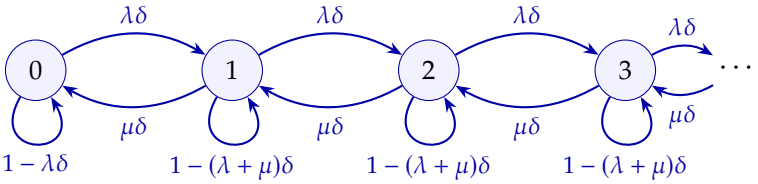


Figure 24: The sampled-time M/M/1 chain, with $(\lambda + \mu)\delta \leq 1$.

By the guessing theorem, or by the general birth-death result, $\pi_{n-1}\lambda\delta = \pi_n \mu\delta$. With $\rho = \lambda/\mu$,

$$\pi_n = \rho\pi_{n-1}, \qquad \pi_n = \rho^n \pi_0, \qquad \boxed{\pi_n = (1-\rho)\rho^n}.$$

Curiously, this does not depend on δ, as long as $(\lambda + \mu)\delta \leq 1$, so these are also the steady-state probabilities as $\delta \to 0$.

In the forward chain the state increases on arrivals and decreases on departures. Since the chain is reversible, each sample path, read from right to left, is a sample path of the same M/M/1 sampled-time chain, in which each departure becomes an arrival and each arrival becomes a departure.

Theorem 18.3 [Burke's theorem, sampled time]. *For the sampled-time M/M/1 chain in steady state:*

1. *the departure process is Bernoulli at rate λ, that is, a departure occurs in each increment with probability $\lambda\delta$, independently;*
2. *the state at $n\delta$ is independent of the departures prior to $n\delta$.*

Proof. In the right-to-left chain the departures prior to $n\delta$ are the arrivals after $n\delta$. That chain is M/M/1, so its arrivals are Bernoulli at rate λ, and its arrivals after $n\delta$ are independent of its state at $n\delta$. Translating back to the forward chain gives both statements. □

Thus everything known about the M/M/1 sampled-time chain has a counterpart with time reversed and arrivals and departures switched.

18.3 Branching processes

A BRANCHING PROCESS is a simple model of organisms that procreate or die away, used for photons in a photomultiplier, cancer cells, insects, and so on. Let X_n be the number of elements in generation n, and let $Y_{k,n}$ be the number of offspring of element k, $1 \le k \le X_n$. Then

$$X_{n+1} = \sum_{k=1}^{X_n} Y_{k,n},$$

where the nonnegative integer random variables $Y_{k,n}$ are IID over both n and k. The initial generation X_0 can be an arbitrary positive random variable, but is usually taken to be 1.

Example: If $Y_{k,n} = 1$ deterministically, then $X_n = X_0$ for all n; if $Y_{k,n} = 2$, then $X_n = 2^n X_0$. If $p_Y(0) = p_Y(2) = 1/2$, then $\{X_n; n \ge 0\}$ is a rather peculiar Markov chain: it can grow explosively or die out, and once dead it stays dead.

State 0 is a trapping state in general. In the last example the even-numbered states all communicate (but turn out to be transient), and each odd-numbered state communicates with no other state.

To find the probability that the process dies out, let $p_k = p_Y(k)$ and $P_{ij} = \Pr\{X_n = j \mid X_{n-1} = i\}$, and let $F_{ij}(n)$ be the probability that j is reached by step n from i. Recall from Lecture 16 that $F_{ij}(n) = P_{ij} + \sum_{k \ne j} P_{ik} F_{kj}(n-1)$ for $n > 1$, and $F_{ij}(1) = P_{ij}$. Starting from k elements, the k lines of descendants evolve independently, and the process is dead by step $n-1$ only if every line is, so $F_{k0}(n-1) = [F_{10}(n-1)]^k$. Hence

$$F_{10}(n) = p_0 + \sum_{k=1}^{\infty} p_k [F_{10}(n-1)]^k = \sum_{k=0}^{\infty} p_k [F_{10}(n-1)]^k.$$

With $h(z) = \sum_k p_k z^k$, this is $F_{10}(n) = h(F_{10}(n-1))$, starting from $F_{10}(1) = p_0 = h(0)$.

Assume $p_0 > 0$. The function h is increasing and convex on $[0,1]$, with $h(0) = p_0$ and $h(1) = 1$. The iterates increase to the smallest root of $h(z) = z$ in $[0,1]$, which is $F_{10}(\infty)$.

(a) If $h'(1) = \overline{Y} > 1$, the curve crosses the diagonal below 1, and $F_{10}(\infty) < 1$: the process explodes with probability $1 - F_{10}(\infty)$ and dies out with probability $F_{10}(\infty)$.

(b) If $h'(1) = \overline{Y} \le 1$, the curve stays above the diagonal on $[0,1)$, and $F_{10}(\infty) = 1$: the process dies out WP1.

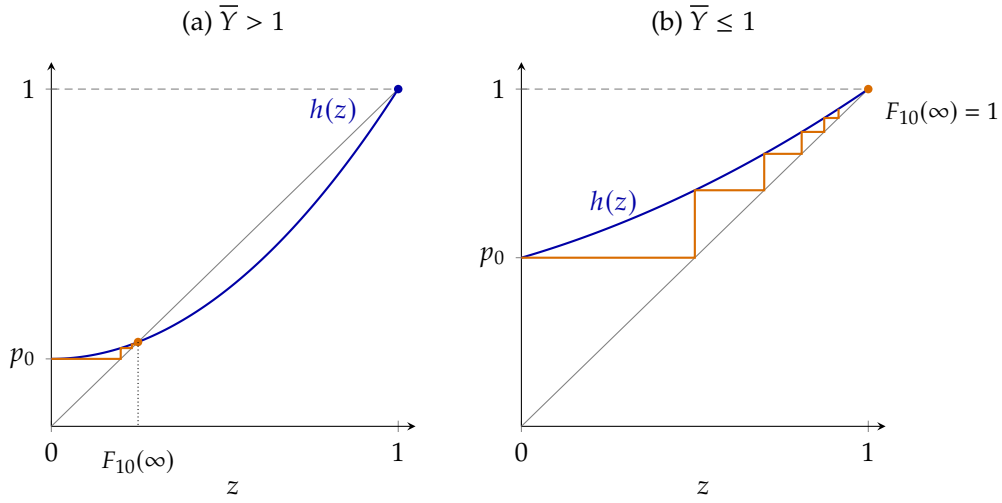


Figure 25: Iterating $F_{10}(n) = h(F_{10}(n-1))$ from $F_{10}(1) = p_0$. The offspring distributions are illustrative: (a) $p_0 = 0.2$, $p_2 = 0.8$, with $\overline{Y} = 1.6$; (b) $p_0 = 0.5$, $p_1 = 0.3$, $p_2 = 0.2$, with $\overline{Y} = 0.7$.

18.4 Markov processes with countable state spaces

A countable-state MARKOV PROCESS extends a countable-state Markov chain $X_0, X_1, \ldots$, called its EMBEDDED CHAIN. With each step, from X_{n-1} to X_n, there is an exponential HOLDING TIME U_n before X_n is entered. The rate ν_i of U_n is determined by $X_{n-1} = i$; conditional on X_{n-1}, U_n is independent of all other states and holding times.

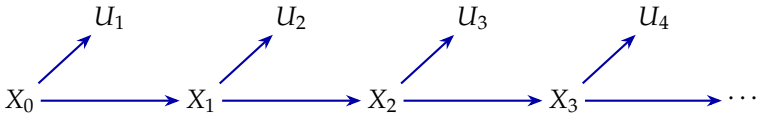


Figure 26: Each holding time U_n depends only on the preceding state X_{n-1}; the epochs of the transitions are $S_n = U_1 + \cdots + U_n$.

The process itself is $\{X(t); t \ge 0\}$ with $X(t) = X_n$ for $t \in [S_n, S_{n+1})$, so in state i the process waits an exponential time of rate ν_i before moving. We usually assume that the embedded chain has no self-transitions, since they are hidden in a sample path of the process. A sample path of $\{X_n; n \ge 0\}$ and $\{U_n; n \ge 1\}$ specifies $\{X(t); t \ge 0\}$, and vice versa. By the memorylessness of the exponential holding times,

$$\Pr\{X(t) = j \mid X(\tau) = i,\ \{X(s); s < \tau\}\} = \Pr\{X(t-\tau) = j \mid X(0) = i\}.$$

18.5 The M/M/1 queue

In an M/M/1 queue with arrival rate λ and service rate μ, the empty state is left only by an arrival, at rate λ. From state $i \ge 1$, the next event is an arrival or a departure, the first of two independent

exponentials of rates λ and μ. So $\nu_0 = \lambda$, $\nu_i = \lambda + \mu$ for $i \geq 1$, and the embedded chain has $P_{01} = 1$, $P_{i,i+1} = \lambda/(\lambda + \mu)$, and $P_{i,i-1} = \mu/(\lambda + \mu)$.

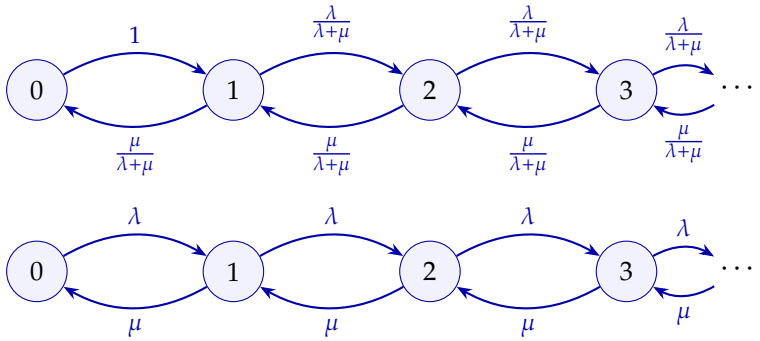


Figure 27: The M/M/1 queue. Top: the embedded Markov chain. Bottom: the process represented by the rates of transitions out of each state.

The second diagram gives the rate of each transition given the current state, and it is the natural description of the real queue.

Source. Robert Gallager, MIT 6.262, *Discrete Stochastic Processes*, Lecture 18, Spring 2011. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 19: Countable-state Markov processes

19.1 Dependence structure and sample paths

Recall from Lecture 18 that a countable-state Markov process consists of an embedded Markov chain $X_0, X_1, \ldots$ and exponential holding times $U_1, U_2, \ldots$: the rate of U_n is ν_i when $X_{n-1} = i$, and conditional on X_{n-1}, U_n is independent of all other states and holding times.

These dependencies form a tree: $X_0 \to X_1 \to X_2 \to \cdots$, with each U_n hanging off X_{n-1}. In a directed tree of dependencies, each random variable, conditional on its parent, is independent of all earlier ones, but the direction in the tree is not needed. For example,

$$\begin{aligned}\Pr\{X_0X_1X_2U_2\} &= \Pr\{X_0\}\Pr\{X_1|X_0\}\Pr\{X_2|X_1\}\Pr\{U_2|X_1\}\\ &= \Pr\{X_1\}\Pr\{X_0|X_1\}\Pr\{X_2|X_1\}\Pr\{U_2|X_1\}.\end{aligned}$$

Conditioning on any node breaks the tree into independent subtrees. Given X_2, for instance, (X_0, X_1, U_1, U_2), (U_3), and (X_3, U_4) are statistically independent.

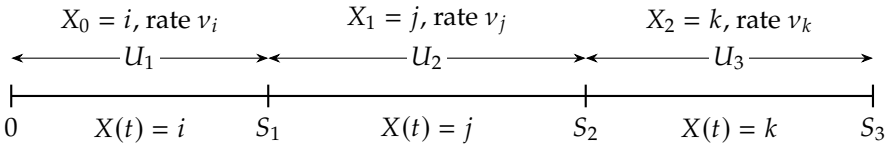


Figure 28: A sample path: the process stays in $X_0 = i$ for a holding time U_1 of rate ν_i, then in $X_1 = j$ for U_2 of rate ν_j, and so on, with transition epochs $S_n = U_1 + \cdots + U_n$.

The process itself, $\{X(t); t \geq 0\}$, is usually called the Markov process:

$$X(t) = X_n \qquad \text{for } t \in [S_n, S_{n+1}).$$

Self-transitions in the embedded chain would not change $X(t)$, so we assume there are none.

19.2 Transition rates

A transition out of state i can be visualized in three equivalent ways:

1. choose the next state j with probability P_{ij}, then the transition time, exponential with rate ν_i;
2. choose the transition time first, then the next state; the two are independent;
3. run an independent Poisson process of rate $q_{ij} = \nu_i P_{ij}$ for each pair i, j; on entry to i, the next

state is the j whose Poisson process has the next arrival.

The third view agrees with the first because the minimum of independent exponentials of rates q_{ij} is exponential with rate $\sum_j q_{ij} = \nu_i$, and the index achieving it is j with probability $q_{ij}/\nu_i = P_{ij}$, independently of the minimum. In particular, U_1 given $X_0 = i$ and $X_1 = j$ is still exponential with rate ν_i. Thus

$$\nu_i = \sum_j q_{ij}, \qquad P_{ij} = \frac{q_{ij}}{\nu_i},$$

so $[q]$ specifies $[P]$ and $\vec{\nu}$, and conversely. A Markov process can be drawn either as the graph of its embedded chain with the rates ν_i on the nodes, or as a graph with the rates q_{ij} on the edges. Both contain the same information. For the M/M/1 queue of Lecture 18 the second form, with $q_{i,i+1} = \lambda$ and $q_{i+1,i} = \mu$, corresponds more closely to the real-world interpretation.

19.3 Sampled-time approximation

Quantize time into increments of δ and replace each Poisson process of the Markov process by a Bernoulli process with $P_{ij}(\delta) = \delta q_{ij}$. Since shrinking Bernoulli processes converge to Poisson, the sampled chain $\{X'_n\}$ should approximate the process as $\delta \to 0$, in the sense $X(t) \approx X'(t/\delta)$ at $t = n\delta$. Self-transitions are needed to model increments in which nothing happens:

$$P_{ii} = 1 - \delta\nu_i, \qquad P_{ij} = \delta q_{ij} \quad (j \neq i).$$

This requires $\delta \leq 1/\max_i \nu_i$, so it is only possible when the holding-time rates are bounded.

Example: *M/M/1.* The sampled-time chain of the M/M/1 queue is the one of Lecture 18, with $P_{i,i+1} = \lambda\delta$, $P_{i+1,i} = \mu\delta$, and self-loops $1 - \lambda\delta$ at 0 and $1 - (\lambda + \mu)\delta$ elsewhere. Its steady state does not depend on δ: $\pi'_i = (1 - \rho)\rho^i$ with $\rho = \lambda/\mu$, the fraction of time in state i. The embedded chain has $P_{01} = 1$, $P_{i,i+1} = \lambda/(\lambda + \mu)$, $P_{i,i-1} = \mu/(\lambda + \mu)$, and its birth-death equations $\pi_0 = \pi_1\mu/(\lambda + \mu)$ and $\pi_i\lambda = \pi_{i+1}\mu$ give

$$\pi_0 = \frac{1 - \rho}{2}, \qquad \pi_i = \tfrac{1}{2}(1 - \rho^2)\rho^{i-1}, \quad i \geq 1.$$

Here π_i is the fraction of transitions going into state i.

19.4 Renewals for Markov processes

Definition 19.1 [Irreducible Markov process]. A Markov process is irreducible if its embedded chain is irreducible, that is, all states are in the same class.

Irreducible Markov chains can be transient: the state wanders off, never to return. Irreducible Markov processes can behave even more bizarrely, with infinitely many transitions in finite time or a transition rate decaying to 0.

Theorem 19.2 [Transitions keep occurring]. *Let $M_i(t)$ be the number of transitions in $(0, t]$ of a Markov process starting in state i. Then $\lim_{t\to\infty} M_i(t) = \infty$ WP1.*

Essentially, from any state a transition occurs within a finite time, then another, and so on.

Theorem 19.3 [Visits form a renewal process]. *Let $M_{ij}(t)$ be the number of transitions into j in $(0, t]$, starting in state i. If the embedded chain is recurrent, then $\{M_{ij}(t); t > 0\}$ is a delayed renewal process.*

Transitions keep occurring, and the embedded chain returns to j WP1, so entries into j keep occurring; by the Markov property the times between successive entries are IID.

19.5 Steady state for irreducible processes

Recall that an irreducible countable-state chain is positive recurrent if and only if the steady-state equations

$$\pi_j = \sum_i \pi_i P_{ij} \text{ for all } j, \qquad \pi_j \geq 0, \qquad \sum_j \pi_j = 1$$

have a solution; the solution is then unique, $\pi_i > 0$ for all i, and the number $N_{ij}(n)$ of visits to j in the first n transitions, starting from i, satisfies $\lim_n \frac{1}{n} N_{ij}(n) = \pi_j$ WP1. The process spends a mean time $1/\nu_j$ in j per visit, so we guess that the fraction of time in state j is

$$p_j = \frac{\pi_j/\nu_j}{\sum_i \pi_i/\nu_i}.$$

To prove this, let $p_j(i)$ be the time-average fraction of time in state j for the delayed renewal process $\{M_{ij}(t); t > 0\}$. Let $R_j(t) = 1$ when $X(t) = j$ and 0 otherwise, and let $\overline{W}(j)$ be the mean time between successive entries to j. During one renewal interval the reward is the holding time in j, with mean $\overline{U}(j) = 1/\nu_j$. By the delayed renewal-reward theorem,

$$p_j(i) = \lim_{t\to\infty} \frac{1}{t} \int_0^t R_j(\tau)\, d\tau = \frac{\overline{U}(j)}{\overline{W}(j)} = \frac{1}{\nu_j \overline{W}(j)} \qquad \text{WP1.}$$

This relates the time-average state probabilities to the mean recurrence times, and shows that $p_j(i) = p_j$ does not depend on the starting state.

To find $\overline{W}(j)$, use the strong law for renewals, $\lim_t M_{ij}(t)/t = 1/\overline{W}(j)$ WP1. The entries into j are the visits to j of the embedded chain, so $M_{ij}(t) = N_{ij}(M_i(t))$, and since $M_i(t) \to \infty$,

$$\lim_{t\to\infty} \frac{M_{ij}(t)}{M_i(t)} = \lim_{n\to\infty} \frac{N_{ij}(n)}{n} = \pi_j \qquad \text{WP1.}$$

Therefore

$$\frac{1}{\overline{W}(j)} = \lim_{t\to\infty} \frac{M_{ij}(t)}{M_i(t)} \frac{M_i(t)}{t} = \pi_j \lim_{t\to\infty} \frac{M_i(t)}{t}, \qquad p_j = \frac{1}{\nu_j \overline{W}(j)} = \frac{\pi_j}{\nu_j} \lim_{t\to\infty} \frac{M_i(t)}{t} \quad \text{WP1.}$$

This also shows that $\lim_t M_i(t)/t$ does not depend on i.

Theorem 19.4 [Time averages of a Markov process]. *If the embedded chain is positive recurrent, then WP1*

$$p_j = \frac{\pi_j/\nu_j}{\sum_k \pi_k/\nu_k}, \qquad \lim_{t\to\infty} \frac{M_i(t)}{t} = \frac{1}{\sum_k \pi_k/\nu_k}.$$

If $\sum_k \pi_k/\nu_k < \infty$, this is almost obvious from the previous display, apart from mathematical details: $\lim_t M_i(t)/t$ is the transition rate of the process, and it must have the given value for $\sum_j p_j = 1$. It is possible, however, to have $\sum_k \pi_k/\nu_k = \infty$, and then the rate of transitions is 0.

Example: *A rattled server.* Consider the birth-death chain with $P_{01} = 1$, $P_{i,i+1} = 0.4$ and $P_{i,i-1} = 0.6$ for $i \geq 1$, and holding rates $\nu_j = 2^{-j}$ (top of Figure 29). This is a queue whose server becomes increasingly rattled and whose customers become increasingly discouraged as the state grows. The embedded chain is positive recurrent: $\pi_{j+1} = \rho\pi_j$ with $\rho = 2/3$ for $j \geq 1$, and solving gives $\pi_0 = 1/6$ and $\pi_j = \frac{5}{12}(2/3)^j$ for $j \geq 1$. Thus

$$\frac{\pi_j}{\nu_j} = \frac{5}{12} 2^j \left(\frac{2}{3}\right)^j = \frac{5}{12}\left(\frac{4}{3}\right)^j, \qquad j \geq 1,$$

and $\sum_k \pi_k/\nu_k = \infty$. Truncating the chain shows that the service rate approaches 0 as more states are added.

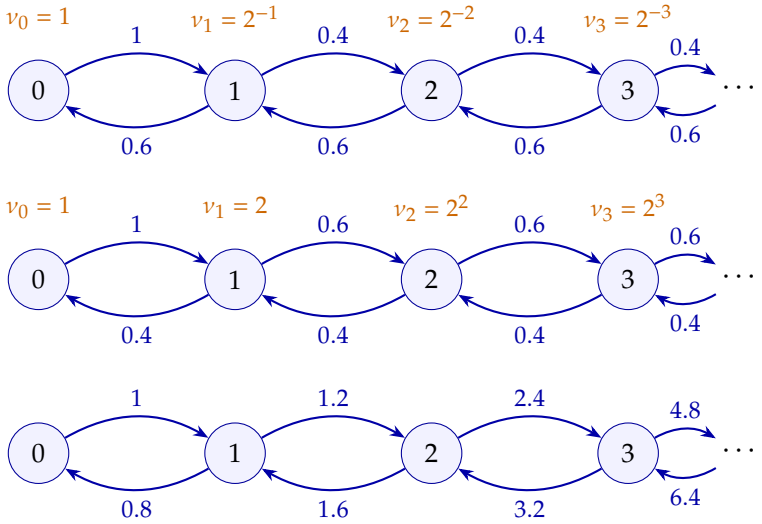


Figure 29: Top: the rattled server, an embedded chain with holding rates $\nu_j = 2^{-j}$ above the nodes. Middle: the embedded chain of the hyperactive birth-death process, with $\nu_j = 2^j$. Bottom: the same hyperactive process in terms of the rates q_{ij}.

19.6 Solving for the time averages directly

Assume the typical case of a positive-recurrent embedded chain with $\sum_i \pi_i/\nu_i < \infty$. Then

$$p_j = \frac{\pi_j/\nu_j}{\sum_k \pi_k/\nu_k}, \tag{19.1}$$

$$p_j\nu_j = \sum_i p_i q_{ij}, \qquad p_j > 0, \qquad \sum_j p_j = 1, \tag{19.2}$$

$$\pi_j = \frac{p_j\nu_j}{\sum_i p_i\nu_i}. \tag{19.3}$$

Substituting $\pi_i \propto p_i\nu_i$ and $P_{ij} = q_{ij}/\nu_i$ into $\pi_j = \sum_i \pi_i P_{ij}$ gives (19.2): in steady state, the rate $p_j\nu_j$ of transitions out of j equals the rate $\sum_i p_i q_{ij}$ of transitions into j.

Theorem 19.5 [Balance equations]. *If the embedded chain is positive recurrent and $\sum_i \pi_i/\nu_i < \infty$, then (19.2) has a unique solution, $\{p_j\}$ and $\{\pi_j\}$ are related by (19.1) and (19.3), and*

$$\sum_i \frac{\pi_i}{\nu_i} = \Big(\sum_j p_j\nu_j\Big)^{-1}.$$

The last identity follows by summing $p_j\nu_j = \pi_j/\sum_k(\pi_k/\nu_k)$ over j. We can also go the opposite way: if (19.2) has a solution and $\sum_j p_j\nu_j < \infty$, then $\pi_j = p_j\nu_j/\sum_j p_j\nu_j$ solves the steady-state equations of the embedded chain, which is therefore positive recurrent. If ν_j is bounded over j, then $\sum_j p_j\nu_j < \infty$ automatically; the sampled-time chain then exists and has the same steady-state solution. For a birth-death process, the balance equations reduce to $p_i q_{i,i+1} = p_{i+1}q_{i+1,i}$.

If $\sum_j p_j\nu_j = \infty$, then $\pi_j = 0$ for all j and the embedded chain is transient or null recurrent. In the transient case there can be infinitely many transitions in finite time, so the notion of steady state does not make much sense.

Example: *A hyperactive birth-death process.* In the middle and bottom of Figure 29, the embedded chain moves up with probability 0.6 and down with 0.4, while the rates $\nu_j = 2^j$ grow. The balance equations have a nice solution $\{p_j\}$, but the embedded chain is transient. Such processes are called irregular: the expected number of transitions per unit time is infinite, and they do not make much sense.

Source. Robert Gallager, MIT 6.262, *Discrete Stochastic Processes*, Lecture 19, Spring 2011. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 20: Markov processes and random walks

20.1 Steady state: the hyperactive example

By Lecture 19, if the embedded chain of a Markov process is positive recurrent and $\sum_k \pi_k/\nu_k < \infty$, then the sample-path fraction of time p_j in state j (WP1, independent of the starting state) and the sample-path transition rate are

$$p_j = \frac{\pi_j/\nu_j}{\sum_k \pi_k/\nu_k}, \qquad \lim_{t\to\infty} \frac{M_i(t)}{t} = \frac{1}{\sum_k \pi_k/\nu_k}.$$

If $\sum_k \pi_k/\nu_k = \infty$, the transition rate tends to 0 and the process has no meaningful steady state. Otherwise the steady state is the unique solution of

$$p_j\nu_j = \sum_i p_i q_{ij}, \qquad p_j > 0 \text{ for all } j, \qquad \sum_j p_j = 1,$$

which says that the rate into each state equals the rate out; for a birth-death process, $p_j q_{j,j+1} = p_{j+1}q_{j+1,j}$. Conversely, for an irreducible process whose balance equations have a solution with $\sum_i \nu_i p_i < \infty$, the embedded chain is positive recurrent with $\pi_j = p_j\nu_j/\sum_i p_i\nu_i$; if $\sum_i \nu_i p_i = \infty$, each $\pi_j = 0$ and steady state makes no sense.

Example: *The hyperactive birth-death process of Lecture 19.* Its rates are $q_{01} = 1$, $q_{j,j+1} = 0.6 \cdot 2^j$, and $q_{j+1,j} = 0.4 \cdot 2^{j+1}$. The birth-death equations give $p_1 = p_0/0.8$ and $p_{j+1} = \frac{0.6\cdot 2^j}{0.8\cdot 2^j}\, p_j = \frac{3}{4}p_j$ for $j \geq 1$. Normalizing,

$$p_0 = \frac{1}{6}, \qquad p_j = \frac{5}{18}\Big(\frac{3}{4}\Big)^j \quad (j \geq 1), \qquad \sum_j p_j\nu_j = \frac{1}{6} + \frac{5}{18}\sum_{j\geq 1}\Big(\frac{3}{2}\Big)^j = \infty.$$

If the process is truncated to the states $0, \ldots, k$, the p_j keep the same ratios, and $\sum_j p_j\nu_j$ grows like $(3/2)^k$: the transition rate is unbounded as $k \to \infty$, while the embedded chain's steady state piles up near the truncation point.

20.2 Reversibility for Markov processes

For any Markov chain in steady state, the backward transition probabilities are defined by $\pi_i P^*_{ij} = \pi_j P_{ji}$. There is nothing mysterious here; this is just

$$\Pr\{X_n = j, X_{n+1} = i\} = \Pr\{X_{n+1} = i\}\Pr\{X_n = j \mid X_{n+1} = i\} = \Pr\{X_n = j\}\Pr\{X_{n+1} = i \mid X_n = j\},$$

and it holds for the embedded chain of a Markov process.

Now consider the holding intervals. Moving to the right, after the process enters state j the exit rate is ν_j: it leaves in each small interval of length δ with probability $\nu_j\delta$. The same holds moving to the left, since a Poisson process is clearly reversible from its incremental definition. Thus $\{\pi_i\}$ and $\{\nu_i\}$ are the same for the left-going process as for the right-going one.

The probability of a right-going transition from j to k in $(t, t+\delta)$ is $p_j q_{jk}\delta$. If q^*_{kj} is the transition rate of the left-going process, the same event has probability $p_k q^*_{kj}\delta$, so

$$p_j q_{jk} = p_k q^*_{kj}.$$

With $c = 1/\sum_i(\pi_i/\nu_i)$ we have $p_j\nu_j = c\pi_j$, so $p_j q_{jk} = c\pi_j P_{jk} = c\pi_k P^*_{kj} = p_k\nu_k P^*_{kj}$. Hence $q^*_{kj} = \nu_k P^*_{kj}$.

Definition 20.1 [Reversible Markov process**].** A Markov process is reversible if $q^*_{ij} = q_{ij}$ for all i, j.

Since $q_{ij} = \nu_i P_{ij}$ and $q^*_{ij} = \nu_i P^*_{ij}$ with the same ν_i, a process with positive-recurrent embedded chain and $\sum_i \pi_i/\nu_i < \infty$ is reversible if and only if its embedded chain is.

Theorem 20.2 [Guessing theorem for Markov processes**].** *Suppose a Markov process is irreducible and $\{p_i\}$ is a set of probabilities with $p_i q_{ij} = p_j q_{ji}$ for all i, j and $\sum_i p_i\nu_i < \infty$. Then*

1. *$p_i > 0$ for all i;*
2. *p_i is the sample-path fraction of time in state i, WP1;*
3. *the process is reversible;*
4. *the embedded chain is positive recurrent.*

Proof. Summing $p_i q_{ij} = p_j q_{ji}$ over i gives $\sum_i p_i q_{ij} = p_j\nu_j$, the balance equations. Since $\sum_i p_i\nu_i < \infty$, the converse result above makes the embedded chain positive recurrent with $\pi_j = p_j\nu_j/\sum_i p_i\nu_i$, and $\pi_j > 0$ gives $p_j > 0$. The relation $\pi \propto p\nu$ is the relation of Lecture 19, so p_j is the fraction of time in j. Finally, $p_i q_{ij} = p_i\nu_i P_{ij}$, so detailed balance for p is $\pi_i P_{ij} = \pi_j P_{ji}$: the embedded chain, and hence the process, is reversible. □

In particular, all birth-death processes with $\sum_j p_j\nu_j < \infty$ are reversible, and so is any process whose

Markov graph is a tree and which has $\sum_j p_j \nu_j < \infty$.

20.3 Burke's theorem and tandem queues

Theorem 20.3 [Burke's theorem]. *Given an M/M/1 queue in steady state with arrival rate λ less than the service rate μ:*

1. *the departure process is Poisson with rate λ;*
2. *the state $X(t)$ is independent of the departures before t;*
3. *for FCFS service, a customer's arrival time, given that it departs at t, is independent of the departures before t.*

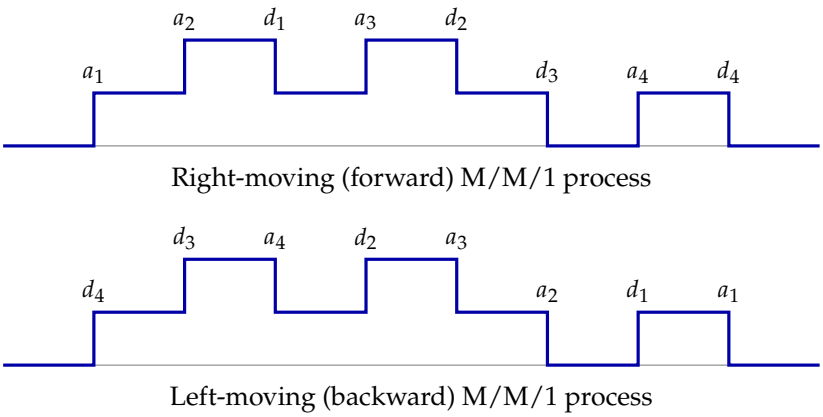


Figure 30: One sample path of an M/M/1 queue, labeled as the right-moving (forward) process and as the left-moving (backward) process. Each forward arrival is a backward departure and vice versa; customers are numbered FCFS in the direction of motion.

Proof. The M/M/1 process is a birth-death process with bounded rates, hence reversible, and the left-moving process is again an M/M/1 queue with arrival rate λ. Departures of the right-moving process are its arrivals, which are Poisson of rate λ, proving (1). The departures before t are the left-moving arrivals after t, which are independent of the left-moving state at t; this is (2).
For (3), a departure at t in the right-moving path is an arrival at t in the left-moving path. Time runs leftward in that path, so the customers ahead of this one are the arrivals to the right of t. In the FCFS left-moving queue, the departure time of the arrival at t depends only on those arrivals and on the service requirements; it is independent of the left-moving arrivals to the left of t. Translated back, the arrival time of the forward customer that departs at t is independent of the forward departures before t. □

Example: *Tandem M/M/1 queues.* Poisson arrivals of rate λ pass through an M/M/1 queue with service rate μ_1 and then through a second with service rate μ_2; service times are independent from

queue to queue and independent of the arrivals at each. By Burke's theorem the arrivals at queue 2 are Poisson at rate λ, and they are independent of the service times at queue 2, so the second queue is also M/M/1. Moreover, the states of the two systems are independent, and the time a customer spends in system 1 is independent of its time in system 2.

20.4 Random walks

Definition 20.4 [RANDOM WALK]. Let $\{X_i; i \geq 1\}$ be IID random variables and $S_n = X_1 + \cdots + X_n$ for $n \geq 1$. The integer-time process $\{S_n; n \geq 1\}$ is the RANDOM WALK based on $\{X_i; i \geq 1\}$.

The interest is now in the whole process rather than in a single sum. Typical questions are:

1. *Threshold crossing.* For given $\alpha > 0$, what is the probability that $S_n \geq \alpha$ for at least one $n \geq 1$, what is the smallest n at which this happens, and what is the overshoot $S_n - \alpha$?
2. *Two thresholds.* For given $\alpha > 0 > \beta$, what is the probability that $\{S_n\}$ crosses α before it crosses β, and at which n does the first such crossing occur?

These problems arise in studying overflow in queues, errors in digital communication, hypothesis testing, ruin, and other catastrophes. Often the relevant probabilities are very small (large deviation problems), and moment generating functions give upper bounds on them.

Example: *Simple random walks.* A random walk is SIMPLE if X is binary with $p_X(1) = p$ and $p_X(-1) = q = 1 - p$; it is a scaling of a Bernoulli process, and the Bernoulli ±1 chain of Lecture 16 is its Markov chain. If $X_i = 1$ for m of the n trials, then $S_n = 2m - n$, so

$$\Pr\{S_n = 2m - n\} = \frac{n!}{m!\,(n-m)!}\, p^m (1-p)^{n-m}.$$

As in "stop when you're ahead,"

$$\Pr\Big\{\bigcup_{n=1}^{\infty} \{S_n \geq k\}\Big\} = \left(\frac{p}{1-p}\right)^k \qquad \text{if } p \leq 1/2.$$

To see this, let θ be the probability of ever reaching +1. Either the first step is +1, or it is −1 and the walk must then climb two levels, each climb independent with probability θ; so $\theta = p + q\theta^2$, whose roots are 1 and p/q. As for the first-passage equations of Lecture 16, θ is the smallest root, p/q when $p \leq 1/2$. Reaching k requires k independent unit climbs, giving $(p/q)^k$.

Integer random walks, where X is integer valued, are similar and can also be modeled as Markov chains, but a threshold may be crossed with an overshoot, and the analysis is much harder. Renewal processes are also random walks, with $X > 0$. When sketching sample paths the axes are usually swapped: a random walk plots S_n against the trial n, while a renewal process plots $n = N(t)$ against the time S_n.

20.5 Queueing delay in a G/G/1 queue

Let $\{X_i; i \geq 1\}$ be the IID interarrival intervals of a G/G/1 queue and $\{Y_i; i \geq 0\}$ the IID service requirements, where customer 0 arrives at time 0 and customer n arrives X_n after customer $n-1$. Let w_n be the time customer n spends in queue before service, with $w_0 = 0$.

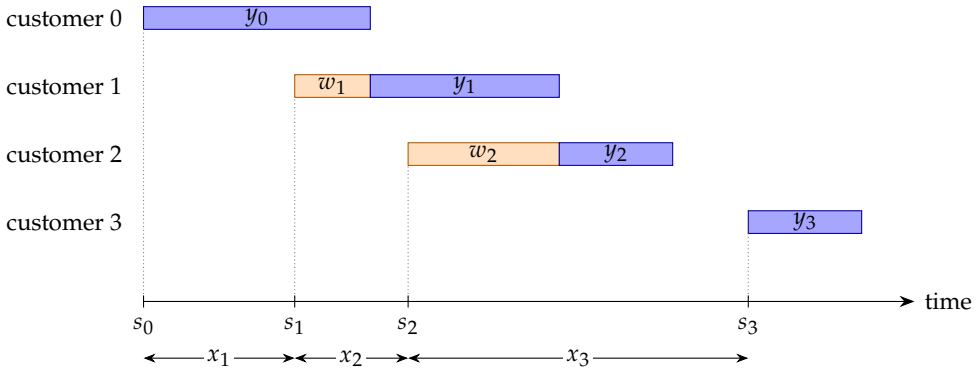


Figure 31: Schematic sample path of a G/G/1 queue. Each row shows one customer's wait (light) and service (dark). Customer 2 is queued, so $x_2 + w_2 = w_1 + y_1$; customer 3 finds the system empty, so $w_3 = 0$.

If customer n is queued, it starts service when customer $n-1$ departs, and measuring from the arrival of customer $n-1$ gives $x_n + w_n = w_{n-1} + y_{n-1}$. If customer n sees an empty system, then $w_n = 0$. So $w_n = w_{n-1} + y_{n-1} - x_n$ if $w_{n-1} + y_{n-1} \geq x_n$, and $w_n = 0$ otherwise:

$$w_n = \max[w_{n-1} + y_{n-1} - x_n,\, 0].$$

Since this holds for every sample path, $W_n = \max[W_{n-1} + Y_{n-1} - X_n,\, 0]$. Defining $U_n = Y_{n-1} - X_n$,

$$\boxed{W_n = \max[W_{n-1} + U_n,\, 0]}.$$

Without the max, $\{W_n; n \geq 1\}$ would be the random walk based on $\{U_i; i \geq 1\}$. With the max, it behaves like that random walk but resets to 0 every time it would go negative.

20.6 Detection, decisions, and hypothesis testing

Detection, decision making, and hypothesis testing are different names for the same problem: given observations, decide among a set of alternatives. Consider only binary decisions. The sample space contains a random variable H, the HYPOTHESIS, with values 0 and 1; its PMF $p_H(0) = p_0$, $p_H(1) = p_1$ gives the A PRIORI PROBABILITIES. Observations $Y_1, \ldots, Y_n$ are IID conditional on $H = 0$ and IID conditional on $H = 1$, with

$$f_{\vec{Y}|H}(\vec{y} \mid \ell) = \prod_{i=1}^{n} f_{Y|H}(y_i \mid \ell), \qquad \ell = 0, 1.$$

By Bayes' law,

$$\Pr\{H = \ell \mid \vec{y}\} = \frac{p_\ell f_{\vec{Y}|H}(\vec{y} \mid \ell)}{p_0 f_{\vec{Y}|H}(\vec{y} \mid 0) + p_1 f_{\vec{Y}|H}(\vec{y} \mid 1)}, \qquad \frac{\Pr\{H = 0 \mid \vec{y}\}}{\Pr\{H = 1 \mid \vec{y}\}} = \frac{p_0 f_{\vec{Y}|H}(\vec{y} \mid 0)}{p_1 f_{\vec{Y}|H}(\vec{y} \mid 1)}.$$

Given the observation, the probability that $H = \ell$ is the correct hypothesis is $\Pr\{H = \ell \mid \vec{Y}\}$. Choosing the ℓ that maximizes it maximizes the a posteriori probability of choosing correctly. This is the MAP RULE (maximum a posteriori probability); it requires knowing p_0 and p_1.

Source. Robert Gallager, MIT 6.262, *Discrete Stochastic Processes*, Lecture 20, Spring 2011. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 21: Hypothesis testing and random walks

21.1 Random walks

Definition 21.1 [Random walk]. Let $\{X_i; i \geq 1\}$ be IID rv's and $S_n = X_1 + \cdots + X_n$ for $n \geq 1$. The integer-time process $\{S_n; n \geq 1\}$ is the random walk based on $\{X_i; i \geq 1\}$.

The focus is on threshold crossings. For the binary walk with $p_X(1) = p$ and $p_X(-1) = q = 1 - p$, the probability of ever reaching a positive integer k is

$$\Pr\Big\{\bigcup_{n=1}^{\infty}\{S_n \geq k\}\Big\} = \left(\frac{p}{1-p}\right)^k \qquad \text{if } p \leq 1/2.$$

21.2 Detection and the MAP rule

The model contains a discrete, usually binary, hypothesis rv H. Its sample values 0 and 1 are the alternative hypotheses, with a priori probabilities $p_0 = \Pr\{H = 0\}$ and $p_1 = \Pr\{H = 1\}$. Among the other rv's is the observation $\vec{Y}^{(m)} = (Y_1, \ldots, Y_m)$. Usually $Y_1, Y_2, \ldots$ are IID conditional on $H = 0$ and IID conditional on $H = 1$, so for continuous observations

$$f_{\vec{Y}^{(m)}|H}(\vec{y} \mid \ell) = \prod_{n=1}^{m} f_{Y|H}(y_n \mid \ell), \qquad \ell = 0, 1.$$

On observing a sample value $\vec{y}$ we must decide, or detect, $\hat{h} = 0$ or $\hat{h} = 1$. Decisions can be wrong, so we need a criterion, such as the probability of a correct choice or an expected cost. The probability experiment includes both the sample value of the hypothesis and the measured data.

By Bayes' rule, with $f(\vec{y}) = p_0 f(\vec{y} \mid 0) + p_1 f(\vec{y} \mid 1)$,

$$\Pr\{H = \ell \mid \vec{y}\} = \frac{p_\ell f_{\vec{Y}|H}(\vec{y} \mid \ell)}{p_0 f_{\vec{Y}|H}(\vec{y} \mid 0) + p_1 f_{\vec{Y}|H}(\vec{y} \mid 1)}, \qquad \frac{\Pr\{H = 0 \mid \vec{y}\}}{\Pr\{H = 1 \mid \vec{y}\}} = \frac{p_0 f_{\vec{Y}|H}(\vec{y} \mid 0)}{p_1 f_{\vec{Y}|H}(\vec{y} \mid 1)}.$$

Given the observation, $\Pr\{H = \ell \mid \vec{y}\}$ is the probability that ℓ is correct. Choosing the ℓ that maximizes it maximizes the probability of a correct decision; this is the MAP rule (maximum a posteriori probability). It requires p_0 and p_1.

Definition 21.2 [Likelihood ratio and threshold test]. The likelihood ratio is

$$\Lambda(\vec{y}) = \frac{f_{\vec{Y}|H}(\vec{y} \mid 0)}{f_{\vec{Y}|H}(\vec{y} \mid 1)}.$$

A THRESHOLD TEST with threshold $\eta > 0$ selects $\hat{h} = 0$ if $\Lambda(\vec{y}) > \eta$ and $\hat{h} = 1$ if $\Lambda(\vec{y}) \le \eta$.

The MAP rule is the threshold test with $\eta = p_1/p_0$. Many decision rules, including the most common and sensible ones, are threshold tests and differ only in how η is chosen. For example, MAXIMUM LIKELIHOOD uses $\eta = 1$: it is MAP for $p_0 = p_1$, but is also used in other ways.

21.3 Log-likelihood ratios as random walks

For IID observations the log-likelihood ratio is a sum,

$$\ln \Lambda(\vec{y}^{(m)}) = \sum_{n=1}^{m} z_n, \qquad z_n = \ln \frac{f_{Y|H}(y_n \mid 0)}{f_{Y|H}(y_n \mid 1)}.$$

Each z_n is the same real-valued function of y_n. Since $Y_1, Y_2, \ldots$ are IID conditional on $H = 0$ (or on $H = 1$), so are $Z_1, Z_2, \ldots$. Hence $\{\ln \Lambda(\vec{Y}^{(m)}); m \ge 1\}$ is a random walk conditional on $H = 0$ and another conditional on $H = 1$. The two walks have the same sample values but different probability measures. Sequential detection, where observations continue until a threshold is crossed, uses exactly these walks.

21.4 Threshold tests and the error curve

A general test maps each sample sequence $\vec{y}$ to 0 or 1, so it is a set A of sequences mapped to hypothesis 1. Its conditional error probabilities are

$$q_0(A) = \Pr\{\vec{Y} \in A \mid H = 0\}, \qquad q_1(A) = \Pr\{\vec{Y} \in A^c \mid H = 1\}.$$

With priors p_0, p_1 and $\eta = p_1/p_0$,

$$\Pr\{e(A)\} = p_0 q_0(A) + p_1 q_1(A) = p_0[q_0(A) + \eta q_1(A)].$$

Write $q_0(\eta), q_1(\eta)$ for the threshold test with threshold η.

Theorem 21.3 [THRESHOLD TESTS ARE OPTIMAL]**.** *For every test A and every $\eta > 0$,*

$$q_0(\eta) + \eta q_1(\eta) \le q_0(A) + \eta q_1(A).$$

Proof. Choose priors $p_0 = 1/(1+\eta)$ and $p_1 = \eta/(1+\eta)$, so that $p_1/p_0 = \eta$. The threshold test with threshold η is then MAP and minimizes the error probability, so $p_0[q_0(\eta) + \eta q_1(\eta)] \le p_0[q_0(A) + \eta q_1(A)]$. The points $(q_1(A), q_0(A))$ do not depend on the priors, which were used only to prove the inequality. □

Geometrically, the line of slope $-\eta$ through $(q_1(\eta), q_0(\eta))$ meets the vertical axis at $q_0(\eta) + \eta q_1(\eta)$.

The theorem says that every point $(q_1(A), q_0(A))$ lies on or to the northeast of this line, for every η. So it lies northeast of the upper envelope of these lines, the ERROR CURVE

$$u(\alpha) = \sup_{\eta>0} \big[q_0(\eta) + \eta q_1(\eta) - \eta\alpha\big], \qquad 0 \le \alpha \le 1,$$

and any test with $q_1(A) = \alpha$ has $q_0(A) \ge u(\alpha)$. As η increases, the tangent point moves up and to the left along the curve. With the vertical axis inverted, the error curve is the RECEIVER OPERATING CURVE (ROC) of radar design.

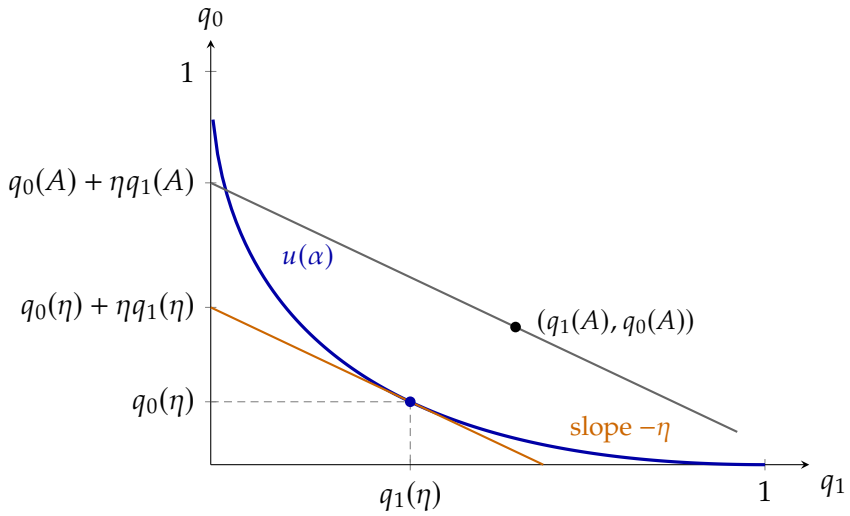


Figure 32: Schematic error curve $u(\alpha)$. The line of slope $-\eta$ through the threshold test's point $(q_1(\eta), q_0(\eta))$ supports the curve; a parallel line through any other test's point $(q_1(A), q_0(A))$ has a higher intercept.

The NEYMAN–PEARSON TEST chooses A to minimize $q_1(A)$ subject to a constraint on $q_0(A)$. Typically it is a threshold test; sometimes, especially when Y is discrete, it is a randomized threshold test.

21.5 Chernoff bounds in terms of the semi-invariant MGF

Proposition 21.4 [CHERNOFF BOUND]. *For any real b and any r such that $g_Z(r) = \mathbb{E}[e^{rZ}]$ exists,*

$$\Pr\{Z \ge b\} \le g_Z(r)e^{-rb} \quad (r > 0), \qquad \Pr\{Z \le b\} \le g_Z(r)e^{-rb} \quad (r < 0).$$

The first is useful for $b > \overline{Z}$ and the second for $b < \overline{Z}$.

Proof. For $r > 0$, $\{Z \ge b\} = \{e^{rZ} \ge e^{rb}\}$, and Markov's inequality gives $\Pr\{e^{rZ} \ge e^{rb}\} \le g_Z(r)/e^{rb}$. For $r < 0$, $\{Z \le b\} = \{e^{rZ} \ge e^{rb}\}$, and the same argument applies. □

For $S_n = X_1 + \cdots + X_n$ with IID X_i and $g_X(r) = \mathbb{E}[e^{rX}]$, independence gives $\mathbb{E}[e^{rS_n}] = \mathbb{E}\big[\prod_{i=1}^n e^{rX_i}\big] =$

$g_X^n(r)$. In terms of the SEMI-INVARIANT MGF $\gamma_X(r) = \ln g_X(r)$, so that $g_X^n(r) = e^{n\gamma_X(r)}$,

$$\Pr\{S_n \geq na\} \leq \exp\big(n[\gamma_X(r) - ra]\big) \qquad \text{for } a > \overline{X},\ r > 0,$$
$$\Pr\{S_n \leq na\} \leq \exp\big(n[\gamma_X(r) - ra]\big) \qquad \text{for } a < \overline{X},\ r < 0.$$

The function γ_X is convex with $\gamma_X(0) = 0$ and $\gamma_X'(0) = \overline{X}$. The exponent $\gamma_X(r) - ra$ is minimized at the r_o with $\gamma_X'(r_o) = a$; the minimum

$$\mu_X(a) = \gamma_X(r_o) - r_o a, \qquad \Pr\{S_n \geq na\} \leq \exp(n\mu_X(a)),$$

is the vertical-axis intercept of the tangent to γ_X of slope a (Figure 33). The optimized Chernoff bound is essentially exponentially tight: for any $\epsilon > 0$, $\Pr\{S_n \geq na\} \geq \exp(n[\mu_X(a) - \epsilon])$ for all large enough n.

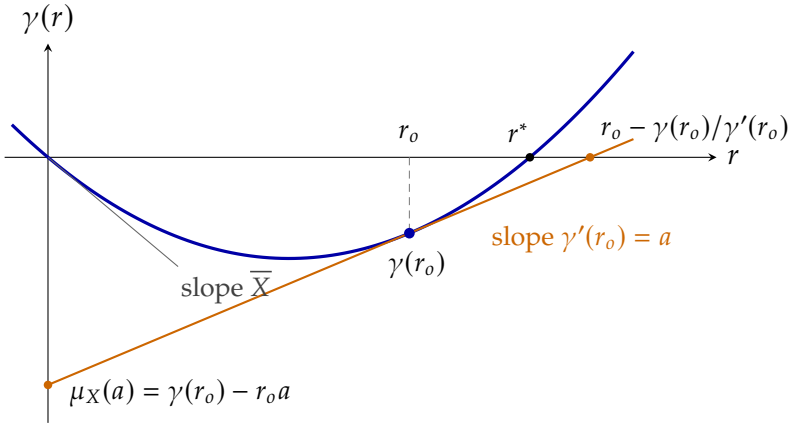


Figure 33: Schematic semi-invariant MGF with $\overline{X} < 0$ and root $r^* > 0$. The tangent at r_o has slope $a = \gamma'(r_o)$; it meets the vertical axis at the exponent $\mu_X(a)$ and the horizontal axis at $r_o - \gamma(r_o)/\gamma'(r_o)$.

21.6 Threshold crossings: fixed threshold, varying n

For threshold problems we want the probability that $S_n \geq \alpha$ for *some* n, so we need a bound focused on how $\Pr\{S_n \geq \alpha\}$ varies with n for fixed α, that is, on when the threshold is crossed if it is crossed. Set $\alpha = na$ and choose r_o with $\gamma_X'(r_o) = \alpha/n$. Substituting $n = \alpha/\gamma_X'(r_o)$,

$$\Pr\{S_n \geq \alpha\} \leq \exp\big(n\gamma_X(r_o) - r_o\alpha\big) = \exp\Big(-\alpha\Big[r_o - \frac{\gamma_X(r_o)}{\gamma_X'(r_o)}\Big]\Big).$$

The bracket is the horizontal-axis intercept of the tangent of slope α/n. For very large n the slope is close to 0 and the intercept, the negative exponent, is very large. As n decreases the intercept decreases to r^*, the positive root of $\gamma_X(r^*) = 0$, and then increases again. Indeed, the derivative of $h(r) = r - \gamma(r)/\gamma'(r)$ is $h'(r) = \gamma(r)\gamma''(r)/\gamma'(r)^2$, which is negative for $r < r^*$ (where $\gamma < 0$) and

positive for $r > r^*$. Hence

$$\Pr\{S_n \geq \alpha\} \leq e^{-\alpha r^*} \quad \text{for all } n, \qquad \Pr\Big\{\bigcup_n \{S_n \geq \alpha\}\Big\} \approx e^{-\alpha r^*},$$

where the nature of the approximation remains to be explained.

Example: *The binary walk.* Let $p_X(1) = p$ and $p_X(-1) = 1 - p$ with $p < 1/2$. Then $g_X(r) = pe^r + (1-p)e^{-r}$ and $\gamma_X(r) = \ln[pe^r + (1-p)e^{-r}]$. The root satisfies $pe^{r^*} + (1-p)e^{-r^*} = 1$; with $z = e^{r^*}$ this is $pz + (1-p)/z = 1$, whose roots are $z = 1$ and $z = (1-p)/p$. Thus $r^* = \ln[(1-p)/p]$ and

$$\Pr\Big\{\bigcup_n \{S_n \geq \alpha\}\Big\} \approx e^{-\alpha r^*} = \left(\frac{1-p}{p}\right)^{-\alpha},$$

which is exact for integer α, in agreement with the formula of the first subsection.

Source. Robert Gallager, MIT 6.262, *Discrete Stochastic Processes*, Lecture 21, Spring 2011. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 22: Random walks and thresholds

22.1 Chernoff bounds for a fixed threshold

Let a rv Z have an MGF $g_Z(r)$ for $0 \le r < r_+$ and mean $\overline{Z} < 0$. By the Chernoff bound, for any $\alpha > 0$ and $r \in (0, r_+)$,

$$\Pr\{Z \ge \alpha\} \le g_Z(r)e^{-r\alpha} = \exp(\gamma_Z(r) - r\alpha), \qquad \gamma_Z(r) = \ln g_Z(r).$$

For $Z = S_n = X_1 + \cdots + X_n$ with IID X_i, $\gamma_{S_n}(r) = n\gamma_X(r)$, so $\Pr\{S_n \ge na\} \le \min_r \exp(n[\gamma_X(r) - ra])$, which is exponential in n for fixed a, with minimizing r given by $\gamma'(r) = a$. For threshold crossings, $\Pr\{\bigcup_n \{S_n \ge \alpha\}\}$, first fix α and let n vary:

$$\Pr\{S_n \ge \alpha\} \le \min_{0<r<r_+} \exp\Big(-\alpha\Big[r - \frac{n}{\alpha}\gamma_X(r)\Big]\Big).$$

For each r, the bracket $r - \gamma(r)n/\alpha$ is the horizontal-axis intercept of the line of slope α/n through $(r, \gamma(r))$. The best r_o makes this line tangent to γ, with $\gamma'(r_o) = \alpha/n$. As in Lecture 21, the intercept decreases to r^*, the positive root of $\gamma(r^*) = 0$, as n decreases, and then increases again. So $\Pr\{S_n \ge \alpha\} \le e^{-\alpha r^*}$ for every n, and the n at which crossing is essentially most likely is $n = \alpha/\gamma'(r^*)$. The approximation $\Pr\{\bigcup_n \{S_n \ge \alpha\}\} \approx e^{-\alpha r^*}$ is explained by Wald's identity.

22.2 Wald's identity with two thresholds

Let $\{S_n; n \ge 1\}$ be a random walk with $S_n = X_1 + \cdots + X_n$, where X is not identically zero and has a semi-invariant MGF $\gamma(r)$ for $r \in (r_-, r_+)$ with $r_- < 0 < r_+$. Let $\alpha > 0$ and $\beta < 0$ be two thresholds, and let J be the smallest n for which $S_n \ge \alpha$ or $S_n \le \beta$.

Then J is a stopping trial: the indicator $\mathbf{1}_{\{J=n\}}$ is a function of $S_1, \ldots, S_n$, and J is a rv. That J is finite with probability 1 is almost obvious. Since X is not identically zero, there is $\epsilon > 0$ with, say, $\delta = \Pr\{X \ge \epsilon\} > 0$ (the case $\Pr\{X \le -\epsilon\} > 0$ is symmetric). Any $k = \lceil (\alpha - \beta)/\epsilon \rceil$ consecutive steps of size at least ϵ force an exit from (β, α), so the walk is still inside after mk steps with probability at most $(1 - \delta^k)^m \to 0$.

Theorem 22.1 [Wald's identity]. *For every $r \in (r_-, r_+)$,*

$$\mathbb{E}\big[\exp(rS_J - J\gamma(r))\big] = 1.$$

With a fixed n in place of J this just says $\mathbb{E}[e^{rS_n}] = e^{n\gamma(r)}$, so it is not implausible. The proof, by tilted probabilities, is at the end of this lecture. First, its use for threshold crossings.

Corollary 22.2 [Threshold-crossing bound]. *Assume further that $\overline{X} < 0$ and that $r^* > 0$ satisfies $\gamma(r^*) = 0$. Then* $\Pr\{S_J \geq \alpha\} \leq \exp(-r^*\alpha)$.

Proof. At r^*, Wald's identity is $\mathbb{E}[\exp(r^*S_J)] = 1$. Since $\exp(r^*S_J) \geq 0$,

$$\Pr\{S_J \geq \alpha\}\, \mathbb{E}\big[\exp(r^*S_J) \mid S_J \geq \alpha\big] \leq \mathbb{E}\big[\exp(r^*S_J)\big] = 1.$$

On $\{S_J \geq \alpha\}$, $\exp(r^*S_J) \geq \exp(r^*\alpha)$, so $\Pr\{S_J \geq \alpha\} \exp(r^*\alpha) \leq 1$. □

The bound holds for every $\beta < 0$, and hence without a lower threshold:

$$\Pr\Big\{\bigcup_{n\geq 1}\{S_n \geq \alpha\}\Big\} \leq \exp(-r^*\alpha).$$

Indeed, the event $\{S_J \geq \alpha\}$ grows as β decreases, and a sample path that ever reaches α belongs to it once β is below the minimum of the path before that first crossing; continuity of probability then gives the claim. This is stronger and cleaner than the Chernoff statement that $\Pr\{S_n \geq \alpha\} \leq \exp(-\alpha r^*)$ for each n. The same argument, applied at a root $r^* < 0$ when $\overline{X} > 0$, gives $\Pr\{S_J \leq \beta\} \leq \exp(-r^*\beta)$.

22.3 The Kingman bound for G/G/1

Apply the corollary to the queueing time W_n of arrival n in a G/G/1 queue. Let X_i be the interarrival time between arrivals $i-1$ and i, let Y_{i-1} be the service time of arrival $i-1$, with arrival 0 at time 0 and $W_0 = 0$, and let

$$U_i = Y_{i-1} - X_i,$$

the previous service time minus the ith interarrival time. Then $W_n = \max(0, W_{n-1} + U_n)$, and unrolling the recursion,

$$W_n = \max\big(0,\ U_n,\ U_n + U_{n-1},\ \ldots,\ U_n + U_{n-1} + \cdots + U_1\big).$$

The U_i are IID, so $(U_n, \ldots, U_1)$ has the same distribution as $(U_1, \ldots, U_n)$: $\{U_i\}$ is a random walk looking backward, and W_n has the distribution of $\max(0, S_1, \ldots, S_n)$ for the walk $S_i = U_1 + \cdots + U_i$. Hence $\Pr\{W_n \geq \alpha\}$ is nondecreasing in n, with limit $\Pr\{W \geq \alpha\} = \Pr\big\{\bigcup_i \{S_i \geq \alpha\}\big\}$.

Theorem 22.3 [Kingman bound]. *Let $\gamma(r)$ be the semi-invariant MGF of U_i, with $\overline{U} < 0$ and $r^* > 0$ satisfying $\gamma(r^*) = 0$. Then for all $n \geq 1$ and $\alpha > 0$,*

$$\Pr\{W_n \geq \alpha\} \leq \Pr\{W \geq \alpha\} \leq \exp(-r^*\alpha).$$

22.4 Large deviations for hypothesis tests

Let $\vec{Y} = (Y_1, \ldots, Y_n)$ be IID conditional on H_0 and also IID conditional on H_1. Then

$$\ln \Lambda(\vec{y}) = \ln \frac{f(\vec{y} \mid H_0)}{f(\vec{y} \mid H_1)} = \sum_{i=1}^{n} z_i, \qquad z_i = \ln \frac{f(y_i \mid H_0)}{f(y_i \mid H_1)}.$$

A threshold test compares $\sum_i z_i$ with $\ln \eta = \ln(p_1/p_0)$, choosing $\hat{h} = 0$ above it. Conditional on H_1, an error occurs when $\sum_i Z_i > \ln \eta$, where the Z_i are IID given H_1. Conditional on H_0, an error occurs when $\sum_i Z_i \le \ln \eta$.

Given H_1, the semi-invariant MGF of Z is

$$\gamma_1(r) = \ln \int f(y \mid H_1) \exp\Big[r \ln \frac{f(y \mid H_0)}{f(y \mid H_1)}\Big] dy = \ln \int f^{1-r}(y \mid H_1) f^{r}(y \mid H_0)\, dy.$$

At $r = 1$ this is $\ln \int f(y \mid H_0)\, dy = 0$, so $r^* = 1$. Given H_0,

$$\gamma_0(s) = \ln \int f^{-s}(y \mid H_1) f^{1+s}(y \mid H_0)\, dy = \gamma_1(s+1),$$

which vanishes at $s = -1$. Write $q_\ell(\eta) = \Pr\{e \mid H = \ell\}$. The Chernoff bound with $r > 0$ for the upper tail given H_1, and with $s = r - 1 < 0$ for the lower tail given H_0, gives for $0 < r < 1$

$$q_1(\eta) \le \exp n\big[\gamma_1(r) - r \ln(\eta)/n\big], \qquad q_0(\eta) \le \exp n\big[\gamma_1(r) + (1-r)\ln(\eta)/n\big].$$

Both exponents are values of the line of slope $\ln(\eta)/n$ through $(r, \gamma_1(r))$, at $r = 0$ and at $r = 1$. Each is minimized by the same r_o, the one with $\gamma_1'(r_o) = \ln(\eta)/n$, where the line is tangent to γ_1 (Figure 34).

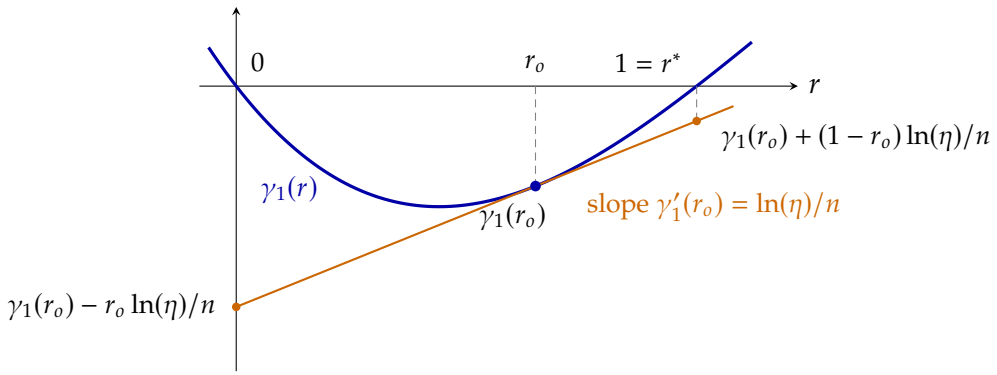


Figure 34: Schematic $\gamma_1(r)$, which vanishes at 0 and 1. The tangent of slope $\ln(\eta)/n$ gives the exponent for $q_1(\eta)$ at $r = 0$ and the exponent for $q_0(\eta)$ at $r = 1$; moving the tangent point trades one exponent against the other.

These are the exponents for the two kinds of errors, a large-deviation form of Neyman–Pearson:

choose one exponent, and the other is given by the see-saw of tangents. The a priori probabilities are usually not the essential characteristic here; they enter only through $\ln(\eta)/n$.

22.5 Sequential detection

This problem calls for a variable number of trials. There are two coupled random walks, $\{S_n = Z_1 + \cdots + Z_n\}$ under H_0 and under H_1, with $\mathbb{E}[Z \mid H_0] > 0$ and $\mathbb{E}[Z \mid H_1] < 0$. Use two thresholds $\alpha > 0$ and $\beta < 0$, stop at the first n with $S_n \geq \alpha$ or $S_n \leq \beta$, and decide $\hat{h} = 0$ if $S_J \geq \alpha$ and $\hat{h} = 1$ if $S_J \leq \beta$. Crossing α is a rare event under H_1, and crossing β is rare under H_0. Since $r^* = 1$ for the H_1 walk and $r^* = -1$ for the H_0 walk, Corollary 22.2 and its lower-threshold version give

$$\Pr\{e \mid H_1\} = \Pr\{S_J \geq \alpha \mid H_1\} \leq e^{-\alpha}, \qquad \Pr\{e \mid H_0\} = \Pr\{S_J \leq \beta \mid H_0\} \leq e^{\beta}.$$

This is not surprising. For $p_0 = p_1 = 1/2$, a path that stops with $\sum_i z_i = \alpha$ has $\ln[\Pr\{H = 0 \mid \vec{y}\}/\Pr\{H = 1 \mid \vec{y}\}] = \alpha$, so the a posteriori probability that deciding $\hat{h} = 0$ is wrong is $e^{-\alpha}/(1 + e^{-\alpha}) < e^{-\alpha}$.

The coupling between the two error probabilities is weaker than for fixed n. Increasing α lowers $\Pr\{e \mid H_1\}$ exponentially and increases $\mathbb{E}[J \mid H_0] \approx \alpha/\mathbb{E}[Z \mid H_0]$, by Wald's equality, since $\mathbb{E}[S_J \mid H_0] \approx \alpha$. Thus

$$\Pr\{e \mid H = 1\} \sim \exp\bigl(-\mathbb{E}[J \mid H = 0]\,\mathbb{E}[Z \mid H = 0]\bigr),$$

and similarly $\Pr\{e \mid H = 0\} \sim \exp\bigl(\mathbb{E}[J \mid H = 1]\,\mathbb{E}[Z \mid H = 1]\bigr)$. Each error probability is essentially exponential in the expected number of trials under the *other* hypothesis, with exponents $\mathbb{E}[Z \mid H = 0]$ and $-\mathbb{E}[Z \mid H = 1]$ per trial. These are the intercepts of the tangents to γ_1 at its two roots: the tangent at $r = 0$ has slope $\gamma_1'(0) = \mathbb{E}[Z \mid H_1]$, and the tangent at $r = 1$ has slope $\gamma_1'(1) = \int f(y \mid H_0) z(y)\, dy = \mathbb{E}[Z \mid H_0]$ (Figure 35). With fixed n, a single tangent gives both exponents, and making one approach its extreme value drives the other to 0; sequential testing obtains both extremes at once.

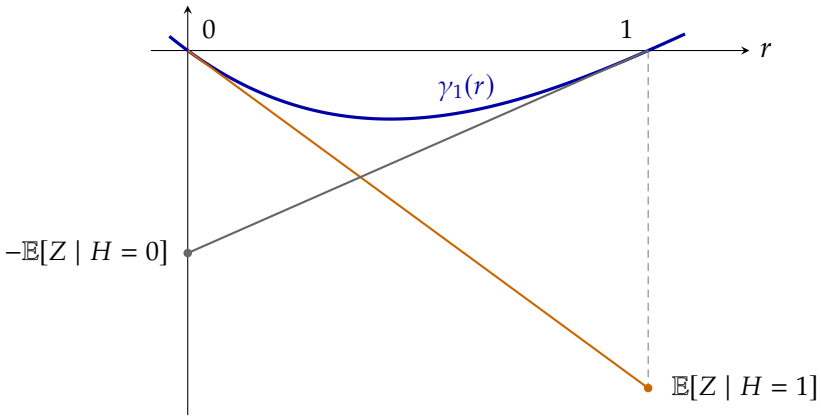


Figure 35: Schematic $\gamma_1(r)$ with its tangents at $r = 0$ and $r = 1$. They reach $\mathbb{E}[Z \mid H = 1]$ at $r = 1$ and $-\mathbb{E}[Z \mid H = 0]$ at $r = 0$; the magnitudes are the per-trial error exponents of sequential detection.

22.6 Tilted probabilities

Let $\{X_n; n \geq 1\}$ be IID discrete rv's with an MGF at a given r. The TILTED PMF of X is

$$q_{X,r}(x) = p_X(x) \exp[rx - \gamma(r)].$$

Summing over x, $\sum_x q_{X,r}(x) = g_X(r)e^{-\gamma_X(r)} = 1$. In the new probability space, the X_n are IID with PMF $q_{X,r}$, and their mean is

$$\mathbb{E}_r[X] = \sum_x x p_X(x) \exp[rx - \gamma(r)] = \frac{1}{g_X(r)} \sum_x \frac{d}{dr} p_X(x) e^{rx} = \frac{g'_X(r)}{g_X(r)} = \gamma'(r).$$

The joint tilted PMF of $\vec{X}^n = (X_1, \ldots, X_n)$ is

$$q_{\vec{X}^n,r}(x_1, \ldots, x_n) = p_{\vec{X}^n}(x_1, \ldots, x_n) \exp\Big(\sum_{i=1}^n [rx_i - \gamma(r)]\Big) = p_{\vec{X}^n}(x_1, \ldots, x_n) \exp[rs_n - n\gamma(r)],$$

where $s_n = x_1 + \cdots + x_n$. Summing over the set $A(s_n)$ of n-tuples with this sum, and using $p_{S_n}(s_n) = \Pr\{A(s_n)\}$,

$$q_{S_n,r}(s_n) = p_{S_n}(s_n) \exp[rs_n - n\gamma(r)].$$

This is the key to much of large deviation theory. For $r > 0$ it tilts the distribution of S_n toward large values, and the laws of large numbers can be applied in the tilted measure.

22.7 Proof of Wald's identity

Proof of Theorem 22.1. Fix $r \in (r_-, r_+)$. The tilted PMF has the same support as p_X, so J is a rv in both measures, by the argument above. Let

$$\mathcal{T}_n = \{\vec{x}^n : s_n \notin (\beta, \alpha);\ s_i \in (\beta, \alpha) \text{ for } 1 \le i < n\}$$

be the set of n-tuples for which stopping occurs at trial n. The tilted PMF of J is

$$\begin{aligned} q_{J,r}(n) &= \sum_{\vec{x}^n \in \mathcal{T}_n} q_{\vec{X}^n,r}(\vec{x}^n) = \sum_{\vec{x}^n \in \mathcal{T}_n} p_{\vec{X}^n}(\vec{x}^n) \exp[rs_n - n\gamma(r)] \\ &= \mathbb{E}\big[\exp(rS_n - n\gamma(r)) \mid J = n\big] \Pr\{J = n\}. \end{aligned}$$

Summing over n, the left side gives 1 because J is a rv in the tilted measure, and the right side gives $\mathbb{E}[\exp(rS_J - J\gamma(r))]$ because J is a rv in the original measure. □

Source. Robert Gallager, MIT 6.262, *Discrete Stochastic Processes*, Lecture 22, Spring 2011. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 23: Martingales (plain, sub and super)

23.1 Wald's identity and sequential tests revisited

Let $\{X_i; i \geq 1\}$ be IID with semi-invariant MGF $\gamma(r) = \ln \mathbb{E}[\exp(rX)]$ on (r_-, r_+), $r_- < 0 < r_+$, and let J be the trial at which $S_n = X_1 + \cdots + X_n$ first crosses $\alpha > 0$ or $\beta < 0$. Wald's identity (Lecture 22) states $\mathbb{E}[\exp(rS_J - J\gamma(r))] = 1$ for $r \in (r_-, r_+)$. If $\overline{X} < 0$ and $\gamma(r^*) = 0$ with $r^* > 0$, then $\mathbb{E}[\exp(r^* S_J)] = 1$, and the Markov inequality applied to $\exp(r^* S_J)$ gives

$$\Pr\{S_J \geq \alpha\} \leq \exp(-\alpha r^*), \qquad \text{and hence} \qquad \Pr\Big\{\bigcup_n \{S_n \geq \alpha\}\Big\} \leq \exp(-r^*\alpha).$$

The bound holds for every lower threshold and for none. For threshold crossing it is stronger than the Chernoff bound, which gives $\Pr\{S_n \geq \alpha\} \leq \exp(-r^*\alpha)$ only for each n separately.

In binary hypothesis testing, H has $p_H(0) = p_0$ and $p_H(1) = p_1$, and given $H = \ell$ the observations $\{Y_n; n \geq 1\}$ are IID with density $f_{Y|H}(y \mid \ell)$. With

$$\Lambda(\vec{y}^{\,n}) = \prod_{i=1}^{n} \frac{f_{Y|H}(y_i \mid 0)}{f_{Y|H}(y_i \mid 1)}, \qquad \frac{\Pr\{H = 0 \mid \vec{y}^{\,n}\}}{\Pr\{H = 1 \mid \vec{y}^{\,n}\}} = \frac{p_0}{p_1}\Lambda(\vec{y}^{\,n}),$$

the MAP rule selects $\hat{h} = 0$ if $\Lambda(\vec{y}^{\,n}) > p_1/p_0$ and $\hat{h} = 1$ otherwise. The log-likelihood ratio is $s_n = \ln \Lambda(\vec{y}^{\,n}) = \sum_{i=1}^{n} z_i$ with $z_i = \ln[f_{Y|H}(y_i \mid 0)/f_{Y|H}(y_i \mid 1)]$. Conditional on $H = 1$, $\{S_n; n \geq 1\}$ is a random walk whose increments have semi-invariant MGF

$$\gamma_1(r) = \ln \int f_{Y|H}^{1-r}(y \mid 1)\, f_{Y|H}^{r}(y \mid 0)\, dy,$$

with $\gamma_1(1) = 0$, so $r^* = 1$.

For fixed n, a threshold rule chooses $\hat{H} = 0$ if $S_n \geq \ln \eta$, so given $H = 1$ an error occurs if $S_n \geq \ln \eta$. With r_o chosen as in Lecture 22, the Chernoff bound gives

$$\Pr\{e \mid H = 1\} \leq \exp\big(n\gamma_1(r_o) - r_o \ln \eta\big), \qquad \Pr\{e \mid H = 0\} \leq \exp\big(n\gamma_1(r_o) + (1 - r_o)\ln \eta\big).$$

Sequential decisions do better: continue trials until $S_n \geq \alpha$ or $S_n \leq \beta$, and decide $\hat{H} = 0$ if $S_J \geq \alpha$ and $\hat{H} = 1$ if $S_J \leq \beta$. Given $H = 1$ an error means $S_J \geq \alpha$, so $\Pr\{e \mid H = 1\} \leq \exp(-\alpha r^*) = e^{-\alpha}$.

The same bound appears sample path by sample path. With $p_0 = p_1$, at the end of trial n,

$$\frac{\Pr\{H = 0 \mid S_n\}}{\Pr\{H = 1 \mid S_n\}} = \frac{1 - \Pr\{H = 1 \mid S_n\}}{\Pr\{H = 1 \mid S_n\}} = \exp(S_n), \qquad \Pr\{H = 1 \mid S_n\} = \frac{\exp(-S_n)}{1 + \exp(-S_n)}.$$

This is the probability of error if $\hat{h} = 0$ is decided at the end of trial n, so deciding $\hat{h} = 0$ on crossing α makes it at most $e^{-\alpha}$. The cost of a large α is many trials under $H = 0$; by Wald's equality,

$$\mathbb{E}[J \mid H = 0] = \frac{\mathbb{E}[S_J \mid H = 0]}{\mathbb{E}[Z \mid H = 0]} \approx \frac{\alpha + \mathbb{E}[\text{overshoot} \mid H = 0]}{\mathbb{E}[Z \mid H = 0]}.$$

23.2 Wald's identity with zero-mean rv's

Differentiating Wald's identity at $r = 0$ (assuming the derivatives may be taken inside the expectation) gives two useful equalities. Since $\gamma(0) = 0$, $\gamma'(0) = \overline{X}$, and $\gamma''(0) = \sigma_X^2$,

$$\frac{d}{dr}\mathbb{E}\big[\exp(rS_J - J\gamma(r))\big] = \mathbb{E}\big[(S_J - J\gamma'(r))\exp(rS_J - J\gamma(r))\big],$$
$$\frac{d^2}{dr^2}\mathbb{E}\big[\exp(rS_J - J\gamma(r))\big] = \mathbb{E}\big[\big((S_J - J\gamma'(r))^2 - J\gamma''(r)\big)\exp(rS_J - J\gamma(r))\big].$$

Both derivatives of the constant 1 vanish. At $r = 0$,

$$\mathbb{E}[S_J - J\overline{X}] = 0 \quad \text{(Wald's equality)}, \qquad \mathbb{E}\big[S_J^2 - \sigma_X^2 J\big] = 0 \quad \text{if } \overline{X} = 0.$$

Example: *Zero-mean simple random walk.* Let $X = \pm 1$ with probability 1/2 each, and let $\alpha > 0$ and $\beta < 0$ be integers. There is no overshoot, so $S_J \in \{\alpha, \beta\}$. Wald's equality gives $\mathbb{E}[S_J] = 0$, hence $\Pr\{S_J = \alpha\} = -\beta/(\alpha - \beta)$. With $\sigma_X^2 = 1$,

$$\mathbb{E}[J] = \mathbb{E}[S_J^2] = \frac{\alpha^2(-\beta) + \beta^2\alpha}{\alpha - \beta} = -\beta\alpha.$$

23.3 Martingales

Definition 23.1 [Martingale]. A sequence $\{Z_n; n \geq 1\}$ of rv's is a martingale if $\mathbb{E}[|Z_n|] < \infty$ for all $n \geq 1$ and

$$\mathbb{E}[Z_n \mid Z_{n-1}, Z_{n-2}, \ldots, Z_1] = Z_{n-1}. \tag{23.1}$$

The condition $\mathbb{E}[|Z_n|] < \infty$ is almost a mathematical fine point and is mostly ignored here. Condition (23.1) looks very weak, yet it leads to surprising applications. In times of doubt, read it as

$$\mathbb{E}[Z_n \mid Z_{n-1} = z_{n-1}, \ldots, Z_1 = z_1] = z_{n-1} \qquad \text{for all sample values } z_{n-1}, \ldots, z_1.$$

Lemma 23.2 [Conditioning on the earlier past]. *For a martingale $\{Z_n; n \geq 1\}$ and $n > i \geq 1$, $\mathbb{E}[Z_n \mid Z_i, Z_{i-1}, \ldots, Z_1] = Z_i$. In particular $\mathbb{E}[Z_n \mid Z_1] = Z_1$ and $\mathbb{E}[Z_n] = \mathbb{E}[Z_1]$.*

Proof. Using $\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y]]$ in its conditional form,

$$\mathbb{E}[Z_3 \mid Z_1] = \mathbb{E}\big[\mathbb{E}[Z_3 \mid Z_2, Z_1] \,\big|\, Z_1\big] = \mathbb{E}[Z_2 \mid Z_1] = Z_1.$$

In the same way,

$$\mathbb{E}[Z_{i+2} \mid Z_i, \ldots, Z_1] = \mathbb{E}\big[\mathbb{E}[Z_{i+2} \mid Z_{i+1}, \ldots, Z_1] \,\big|\, Z_i, \ldots, Z_1\big] = \mathbb{E}[Z_{i+1} \mid Z_i, \ldots, Z_1] = Z_i,$$

and induction on $n - i$ gives the general case. Taking expectations of $\mathbb{E}[Z_n \mid Z_1] = Z_1$ gives $\mathbb{E}[Z_n] = \mathbb{E}[Z_1]$. □

23.4 Simple examples of martingales

1. *Zero-mean random walk.* Let $Z_n = X_1 + \cdots + X_n$ with $\{X_i\}$ IID and zero mean. Since X_n is independent of $Z_1, \ldots, Z_{n-1}$,

$$\mathbb{E}[Z_n \mid Z_{n-1}, \ldots, Z_1] = \mathbb{E}[X_n + Z_{n-1} \mid Z_{n-1}, \ldots, Z_1] = \mathbb{E}[X_n] + Z_{n-1} = Z_{n-1}.$$

2. *Sums of arbitrary dependent rv's.* If $\mathbb{E}[X_i \mid X_{i-1}, \ldots, X_1] = 0$ for each i, then $Z_n = X_1 + \cdots + X_n$ is a martingale, since $Z_1, \ldots, Z_{n-1}$ and $X_1, \ldots, X_{n-1}$ determine each other. This can serve as an alternate definition: one can start either with the sums Z_n or with the differences between successive sums.

3. Let $X_i = U_i Y_i$, where $\{U_i\}$ are IID and equiprobable ± 1, and the Y_i are nonnegative and independent of the U_i but otherwise arbitrary. Since U_n has mean 0 and is independent of $Y_n, X_{n-1}, \ldots, X_1$, we get $\mathbb{E}[X_n \mid X_{n-1}, \ldots, X_1] = 0$, so $Z_n = X_1 + \cdots + X_n$ is a martingale.

4. *Product-form martingales.* If $\{X_i\}$ are IID with unit mean, then $Z_n = X_1 X_2 \cdots X_n$ is a martingale, because X_n is independent of $Z_1, \ldots, Z_{n-1}$:

$$\mathbb{E}[Z_n \mid Z_{n-1}, \ldots, Z_1] = \mathbb{E}[X_n Z_{n-1} \mid Z_{n-1}, \ldots, Z_1] = \mathbb{E}[X_n]\, Z_{n-1} = Z_{n-1}.$$

5. *A special case.* Let the X_i be IID and equiprobably 2 or 0. Then

$$\Pr\{Z_n = 2^n\} = 2^{-n}, \qquad \Pr\{Z_n = 0\} = 1 - 2^{-n}, \qquad \mathbb{E}[Z_n] = 1.$$

   Since Z_n stays 0 once some $X_i = 0$ and $\Pr\{Z_n > 0\} = 2^{-n} \to 0$, $\lim_n Z_n = 0$ with probability 1, but $\mathbb{E}[Z_n] = 1$ for all n.

6. *Branching processes.* Let X_n be the number of elements in generation n, with $X_{n+1} = \sum_{i=1}^{X_n} Y_{i,n}$ for IID offspring counts $Y_{i,n}$ of mean $\overline{Y}$. The scaled process $Z_n = X_n / \overline{Y}^n$ is a martingale:

$$\mathbb{E}[Z_n \mid Z_{n-1}, \ldots, Z_1] = \mathbb{E}\left[\frac{X_n}{\overline{Y}^n} \;\middle|\; X_{n-1}, \ldots, X_1\right] = \frac{\overline{Y} X_{n-1}}{\overline{Y}^n} = Z_{n-1}.$$

23.5 Submartingales and supermartingales

Definition 23.3 [Submartingale and supermartingale]. A sequence $\{Z_n; n \geq 1\}$ with $\mathbb{E}[|Z_n|] < \infty$ for all n is a submartingale if $\mathbb{E}[Z_n \mid Z_{n-1}, \ldots, Z_1] \geq Z_{n-1}$ for all n, and a supermartingale if $\mathbb{E}[Z_n \mid Z_{n-1}, \ldots, Z_1] \leq Z_{n-1}$ for all n.

Replacing Z_n by $-Z_n$ exchanges the two, so only submartingales are treated. The argument of Lemma 23.2, with inequalities, gives for submartingales

$$\mathbb{E}[Z_n \mid Z_i, \ldots, Z_1] \geq Z_i, \qquad \mathbb{E}[Z_n] \geq \mathbb{E}[Z_i] \qquad \text{for all } n > i > 0.$$

A function $h : \mathbb{R} \to \mathbb{R}$ is convex if each tangent to its curve lies on or below the curve; at a corner, any line through the point that stays below the curve serves as a tangent. The condition $h''(x) \geq 0$ is sufficient but not necessary, as $h(x) = |x|$ shows.

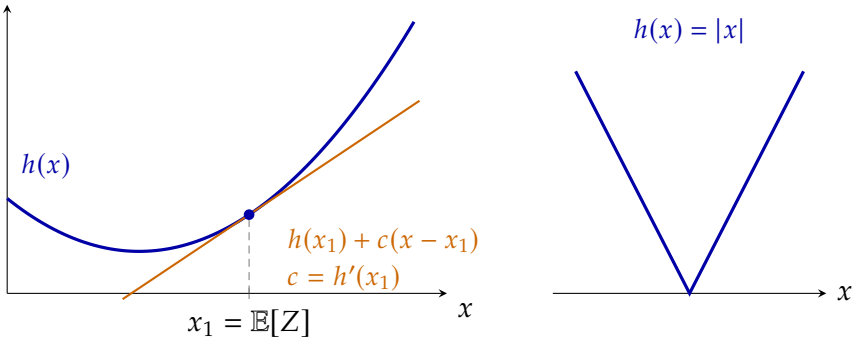


Figure 36: Left: a convex function lies above its tangent at $x_1 = \mathbb{E}[Z]$, which gives Jensen's inequality. Right: $|x|$ is convex although h'' does not exist at 0. Schematic.

Lemma 23.4 [Jensen's inequality]. *If h is convex and Z is a rv with finite expectation, then $h(\mathbb{E}[Z]) \leq \mathbb{E}[h(Z)]$.*

Proof. Let $x_1 = \mathbb{E}[Z]$ and let $h(x_1) + c(x - x_1)$ be a tangent at x_1 (for differentiable h, $c = h'(x_1)$). Convexity gives $h(Z) \geq h(x_1) + c(Z - x_1)$. Taking expectations, $\mathbb{E}[h(Z)] \geq h(x_1) + c(\mathbb{E}[Z] - x_1) = h(\mathbb{E}[Z])$. □

Theorem 23.5 [Convex functions of martingales]. *Let h be convex and $\mathbb{E}[|h(Z_n)|] < \infty$ for all n. If $\{Z_n; n \geq 1\}$ is a martingale, or a submartingale and h is also nondecreasing, then $\{h(Z_n); n \geq 1\}$ is a submartingale.*

Proof. Jensen's inequality applied to the conditional distribution of Z_n given $Z_{n-1}, \ldots, Z_1$ gives

$$\mathbb{E}[h(Z_n) \mid Z_{n-1}, \ldots, Z_1] \geq h\big(\mathbb{E}[Z_n \mid Z_{n-1}, \ldots, Z_1]\big) \geq h(Z_{n-1}).$$

The last step is an equality for a martingale and uses monotonicity of h for a submartingale. Since $h(Z_1), \ldots, h(Z_{n-1})$ are functions of $Z_1, \ldots, Z_{n-1}$, conditioning the inequality further on them preserves it. □

Monotonicity is needed for submartingales: $Z_n = n$ is a submartingale, but $h(Z_n) = -n$ is not. For a martingale $\{Z_n\}$, the processes $\{|Z_n|\}$, $\{Z_n^2\}$, and $\{e^{rZ_n}\}$ are therefore submartingales whenever the relevant expectations are finite.

23.6 Stopped martingales

The definition of a stopping trial applies to any process: for $\{Z_n; n \geq 1\}$, J must be a rv and each event $\{J = n\}$ must be determined by $Z_1, \ldots, Z_n$. It extends to a POSSIBLY DEFECTIVE STOPPING TRIAL, for which J may be infinite with positive probability, as for a random walk with a single threshold.

Definition 23.6 [STOPPED PROCESS]. For a possibly defective stopping trial J on $\{Z_n; n \geq 1\}$, the STOPPED PROCESS $\{Z_n^*; n \geq 1\}$ has $Z_n^* = Z_n$ if $n \leq J$ and $Z_n^* = Z_J$ if $n > J$.

For example, if Z_n is the net worth at time n under a gambling strategy, the strategy can be modified to stop when Z_n reaches a given value. Then Z_n^* stays at that value forever after, while Z_n follows the original strategy.

Theorem 23.7 [STOPPED MARTINGALES]. *If $\{Z_n; n \geq 1\}$ is a martingale (submartingale) and J is a possibly defective stopping trial for it, then the stopped process $\{Z_n^*; n \geq 1\}$ is a martingale (submartingale).*

The intuition: before stopping, $Z_n^* = Z_n$, so Z_n^* satisfies the martingale (submartingale) condition; afterwards Z_n^* is constant, so it again satisfies the condition. The careful proof is in Lecture 24.

Source. Robert Gallager, MIT 6.262, *Discrete Stochastic Processes*, Lecture 23, Spring 2011. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 24: Martingales: stopping and converging

24.1 Martingales and conditional expectations

Recall that $\{Z_n; n \geq 1\}$ is a martingale if $\mathbb{E}[|Z_n|] < \infty$ and $\mathbb{E}[Z_n \mid Z_{n-1}, \ldots, Z_1] = Z_{n-1}$ for all $n \geq 1$. Here $\mathbb{E}[Z_n \mid Z_{n-1}, \ldots, Z_1]$ is a rv: it maps each sample point ω to the conditional expectation of Z_n given $Z_1(\omega), \ldots, Z_{n-1}(\omega)$, and for a martingale this rv must be Z_{n-1}. As shown in Lecture 23, $\mathbb{E}[Z_n \mid Z_i, \ldots, Z_1] = Z_i$ and $\mathbb{E}[Z_n] = \mathbb{E}[Z_i]$ for $n > i \geq 1$, while for $m \geq n$, $\mathbb{E}[Z_n \mid Z_m, \ldots, Z_1] = Z_n$, because Z_n is then a function of the conditioning rv's. The examples of Lecture 23 (zero-mean random walks, sums of increments with zero conditional mean, products of IID unit-mean rv's, scaled branching processes) and the facts about submartingales and convex functions are used below.

24.2 Stopped martingales

A possibly defective rv J is a mapping from Ω to the extended reals in which $\{J = \infty\}$ and $\{J = -\infty\}$ may have positive probability; the other provisos of rv's still hold. A possibly defective stopping trial is thus a stopping rule in which stopping may never happen, such as a random walk with a single threshold. The stopped process has $Z_n^* = Z_n$ for $n \leq J$ and $Z_n^* = Z_J$ for $n > J$.

Theorem 24.1 [Stopped martingales]. *If J is a possibly defective stopping trial for a martingale (submartingale) $\{Z_n; n \geq 1\}$, then the stopped process $\{Z_n^*; n \geq 1\}$ is a martingale (submartingale).*

Proof. Write $\mathbf{1}_{\{.\}}$ for indicators. Then

$$Z_n^* = \sum_{m=1}^{n-1} Z_m \mathbf{1}_{\{J=m\}} + Z_n \mathbf{1}_{\{J \geq n\}},$$

so $|Z_n^*| \leq \sum_{m<n} |Z_m| + |Z_n|$ and $\mathbb{E}[|Z_n^*|] < \infty$, being bounded by a sum of n finite numbers. Let $\vec{Z}^{(n-1)}$ denote $Z_{n-1}, \ldots, Z_1$. For $m < n$ the indicator $\mathbf{1}_{\{J=m\}}$ is a function of $\vec{Z}^{(n-1)}$, and so is $\mathbf{1}_{\{J \geq n\}} = 1 - \sum_{m<n} \mathbf{1}_{\{J=m\}}$. Hence $\mathbb{E}[Z_m \mathbf{1}_{\{J=m\}} \mid \vec{Z}^{(n-1)}] = Z_m \mathbf{1}_{\{J=m\}}$ and $\mathbb{E}[Z_n \mathbf{1}_{\{J \geq n\}} \mid \vec{Z}^{(n-1)}] = \mathbf{1}_{\{J \geq n\}} \mathbb{E}[Z_n \mid \vec{Z}^{(n-1)}] = Z_{n-1} \mathbf{1}_{\{J \geq n\}}$. Adding,

$$\begin{aligned} \mathbb{E}\big[Z_n^* \mid \vec{Z}^{(n-1)}\big] &= \sum_{m<n} Z_m \mathbf{1}_{\{J=m\}} + Z_{n-1} \mathbf{1}_{\{J \geq n\}} \\ &= \sum_{m<n-1} Z_m \mathbf{1}_{\{J=m\}} + Z_{n-1}\big[\mathbf{1}_{\{J=n-1\}} + \mathbf{1}_{\{J \geq n\}}\big] = Z_{n-1}^*. \end{aligned}$$

For a submartingale, the same steps give $\mathbb{E}[Z_n^* \mid \vec{Z}^{(n-1)}] \geq Z_{n-1}^*$, since $\mathbb{E}[Z_n \mid \vec{Z}^{(n-1)}] \geq Z_{n-1}$ and $\mathbf{1}_{\{J \geq n\}} \geq 0$.
It remains to condition on $\vec{Z}^{*(n-1)} = Z_{n-1}^*, \ldots, Z_1^*$ rather than on $\vec{Z}^{(n-1)}$. Since $\vec{Z}^{*(n-1)}$ is a function of $\vec{Z}^{(n-1)}$, every sample value $\vec{z}^{(n-1)}$ leading to a given $\vec{z}^{*(n-1)}$ has $\mathbb{E}[Z_n^* \mid \vec{Z}^{(n-1)} = \vec{z}^{(n-1)}] = z_{n-1}^*$, and averaging over these sample values gives $\mathbb{E}[Z_n^* \mid \vec{Z}^{*(n-1)} = \vec{z}^{*(n-1)}] = z_{n-1}^*$. □

Corollary 24.2 [Expectations of stopped processes]. *Under the same assumptions,*

$$\mathbb{E}[Z_1] \leq \mathbb{E}[Z_n^*] \leq \mathbb{E}[Z_n] \quad \textit{(submartingale)}, \qquad \mathbb{E}[Z_1] = \mathbb{E}[Z_n^*] = \mathbb{E}[Z_n] \quad \textit{(martingale)}.$$

Proof. The stopped process is a (sub)martingale with $Z_1^* = Z_1$, so $\mathbb{E}[Z_n^*] \geq \mathbb{E}[Z_1^*] = \mathbb{E}[Z_1]$. For the other side, $Z_n - Z_n^* = \sum_{m<n}(Z_n - Z_m)\mathbf{1}_{\{J=m\}}$, and since $\{J = m\}$ is determined by $Z_1, \ldots, Z_m$,

$$\mathbb{E}\big[(Z_n - Z_m)\mathbf{1}_{\{J=m\}}\big] = \mathbb{E}\big[\mathbf{1}_{\{J=m\}}\big(\mathbb{E}[Z_n \mid Z_m, \ldots, Z_1] - Z_m\big)\big] \geq 0,$$

with equality for a martingale. □

Example: *Wald's identity from martingales.* Let $\{S_n; n \geq 1\}$ be a random walk with IID steps X_n, and for r such that $\gamma(r)$ exists let

$$Z_n = \exp[rS_n - n\gamma(r)] = \prod_{i=1}^{n} \exp[rX_i - \gamma(r)].$$

This is the generating-function product martingale: the factors are IID with unit mean, so $\mathbb{E}[Z_n] = 1$ for all n. Let J be the nondefective stopping trial at which S_n first crosses $\alpha > 0$ or $\beta < 0$. Then $\mathbb{E}[Z_n^*] = 1$ for all n, and since J is finite with probability 1, $\lim_n Z_n^* = Z_J$ with probability 1. Passing to the limit gives

$$\mathbb{E}[Z_J] = \mathbb{E}\big[\exp(rS_J - J\gamma(r))\big] = 1,$$

which is Wald's identity, now as a consequence of the stopped-martingale theorem. The interchange of limit and expectation in the last step needs more care.

24.3 Kolmogorov's submartingale inequality

Theorem 24.3 [Kolmogorov's submartingale inequality]. *Let $\{Z_n; n \geq 1\}$ be a nonnegative submartingale. For any positive integer m and any $a > 0$,*

$$\Pr\Big\{\max_{1 \leq i \leq m} Z_i \geq a\Big\} \leq \frac{\mathbb{E}[Z_m]}{a}.$$

With Z_m in place of the maximum, this is the lowly but useful Markov inequality.

Proof. Let J be the smallest $n \le m$ with $Z_n \ge a$, and $J = m$ if $Z_n < a$ for all $n \le m$. This is a stopping trial, and the process stops by time m, so $Z_J = Z_m^*$. Moreover $Z_J \ge a$ if and only if $Z_n \ge a$ for some $n \le m$. By the Markov inequality and Corollary 24.2,

$$\Pr\Big\{\max_{1\le n\le m} Z_n \ge a\Big\} = \Pr\{Z_J \ge a\} \le \frac{\mathbb{E}[Z_J]}{a} = \frac{\mathbb{E}[Z_m^*]}{a} \le \frac{\mathbb{E}[Z_m]}{a}. \qquad \square$$

The Chebyshev inequality is strengthened in the same way.

Corollary 24.4 [Kolmogorov version of Chebyshev]. *Let $\{Z_n; n \ge 1\}$ be a martingale with $\mathbb{E}[Z_n^2] < \infty$ for all n. Then for every integer $m \ge 1$ and every $b > 0$,*

$$\Pr\Big\{\max_{1\le n\le m} |Z_n| \ge b\Big\} \le \frac{\mathbb{E}[Z_m^2]}{b^2}.$$

In particular, if $S_n = X_1 + \cdots + X_n$ is a random walk whose steps have mean $\overline{X}$ and variance σ^2, then for every positive integer m and every $\epsilon > 0$,

$$\Pr\Big\{\max_{1\le n\le m} |S_n - n\overline{X}| \ge m\epsilon\Big\} \le \frac{\sigma^2}{m\epsilon^2}.$$

Proof. By the convex-function theorem of Lecture 23 with $h(x) = x^2$, $\{Z_n^2\}$ is a nonnegative submartingale, and $\max_n |Z_n| \ge b$ exactly when $\max_n Z_n^2 \ge b^2$; apply Theorem 24.3 with $a = b^2$. For the random walk, $Z_n = S_n - n\overline{X}$ is a zero-mean random walk, hence a martingale, with $\mathbb{E}[Z_m^2] = m\sigma^2$; take $b = m\epsilon$. $\square$

24.4 The strong law of large numbers for IID rv's with a variance

Theorem 24.5 [Strong law of large numbers]. *Let $\{X_i; i \ge 1\}$ be IID with mean $\overline{X}$ and standard deviation $\sigma < \infty$, and let $S_n = X_1 + \cdots + X_n$. Then*

$$\Pr\Big\{\lim_{n\to\infty} \frac{S_n}{n} = \overline{X}\Big\} = 1.$$

Proof. Fix $\epsilon > 0$. Corollary 24.4 with m replaced by 2^m and the union bound give

$$\Pr\Big\{\bigcup_{m=k}^{\infty}\Big\{\max_{1\le n\le 2^m} |S_n - n\overline{X}| \ge 2^m\epsilon\Big\}\Big\} \le \sum_{m=k}^{\infty} \frac{\sigma^2}{2^m\epsilon^2} = \frac{2\sigma^2}{2^k\epsilon^2}.$$

Now lower bound the left side. If $2^{m-1} \le n \le 2^m$ and $|S_n - n\overline{X}| \ge 2n\epsilon$, then $|S_n - n\overline{X}| \ge 2^m\epsilon$.

Hence

$$\Pr\Big\{\bigcup_{m=k}^{\infty}\bigcup_{2^{m-1}\le n\le 2^m}\Big\{\Big|\frac{S_n}{n}-\overline{X}\Big|\ge 2\epsilon\Big\}\Big\}\le\frac{2\sigma^2}{2^k\epsilon^2}.$$

The event on the left is the event that $|S_n/n-\overline{X}|\ge 2\epsilon$ for some $n\ge 2^{k-1}$. It contains the event that this happens for infinitely many n, which therefore has probability at most $2\sigma^2/(2^k\epsilon^2)$ for every k, hence probability 0. The union of these null events over $\epsilon=1/(2j)$, $j=1,2,\ldots$, is null, so with probability 1, $|S_n/n-\overline{X}|<1/j$ for all large n, for every j. □

24.5 The martingale convergence theorem

Theorem 24.6 [Martingale convergence]. *Let $\{Z_n;n\ge 1\}$ be a martingale, and assume that there is a finite M with $\mathbb{E}[|Z_n|]\le M$ for all n. Then there is a random variable Z such that $\lim_{n\to\infty}Z_n=Z$ with probability 1.*

Bounded $\mathbb{E}[|Z_n|]$, or the stronger condition of bounded $\mathbb{E}[Z_n^2]$, is a strong constraint, but the theorem is still very powerful. For a branching process $\{X_n;n\ge 1\}$ whose offspring count Y has $\overline{Y}>1$, the martingale $Z_n=X_n/\overline{Y}^n$ of Lecture 23 is nonnegative, so $\mathbb{E}[|Z_n|]=\mathbb{E}[Z_n]=\mathbb{E}[Z_1]$ is bounded. Hence $X_n/\overline{Y}^n\to Z$ with probability 1.

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Lecture 25: Putting it all together

The final lecture draws together martingales, countable- or finite-state Markov chains, countable-state Markov processes, and random walks.

25.1 Martingales

A sequence $\{Z_n; n \geq 1\}$ is a martingale if $\mathbb{E}[|Z_n|] < \infty$ and $\mathbb{E}[Z_n \mid Z_{n-1}, \ldots, Z_1] = Z_{n-1}$; then $\mathbb{E}[Z_n \mid Z_i, \ldots, Z_1] = Z_i$ and $\mathbb{E}[Z_n] = \mathbb{E}[Z_i]$ for $n > i \geq 1$. Equivalently, the increments $X_n = Z_n - Z_{n-1}$ satisfy $\mathbb{E}[X_n \mid X_{n-1}, \ldots, X_1] = 0$. A good special case is $X_n = U_n Y_n$ with $\{U_n\}$ IID and $p_U(1) = p_U(-1) = 1/2$; zero-mean random walks and products of IID unit-mean rv's are the basic examples. For a submartingale, $\mathbb{E}[Z_n \mid Z_i, \ldots, Z_1] \geq Z_i$ and $\mathbb{E}[Z_n] \geq \mathbb{E}[Z_i]$. If h is convex and $\mathbb{E}[|h(Z_n)|] < \infty$, then $\{h(Z_n)\}$ is a submartingale when $\{Z_n\}$ is a martingale, and also when $\{Z_n\}$ is a submartingale and h is nondecreasing.

The stopped process $\{Z_n^*\}$ of a (sub)martingale under a possibly defective stopping rule is a (sub)martingale, with $\mathbb{E}[Z_1] \leq \mathbb{E}[Z_n^*] \leq \mathbb{E}[Z_n]$ for submartingales and equality throughout for martingales. For the product martingale $Z_n = \exp[rS_n - n\gamma(r)]$ and a stopping rule such as threshold crossing, this leads, subject to some small mathematical points, to Wald's identity $\mathbb{E}[\exp(rS_J - J\gamma(r))] = 1$.

Theorem 25.1 [Kolmogorov's inequality, supremum form]. *For a nonnegative submartingale, any positive integer m, and $a > 0$, $\Pr\{\max_{1\leq i\leq m} Z_i \geq a\} \leq \mathbb{E}[Z_m]/a$. For a nonnegative martingale,*

$$\Pr\Big\{\sup_{n\geq 1} Z_n \geq a\Big\} \leq \frac{\mathbb{E}[Z_1]}{a}.$$

Proof. The first statement is Kolmogorov's strengthening of the Markov inequality (Lecture 24). For a martingale, $\mathbb{E}[Z_m] = \mathbb{E}[Z_1]$ for every m. For $0 < a' < a$, the events $\{\max_{i\leq m} Z_i > a'\}$ increase with m to $\{\sup_n Z_n > a'\} \supseteq \{\sup_n Z_n \geq a\}$, so by continuity of probability $\Pr\{\sup_n Z_n \geq a\} \leq \mathbb{E}[Z_1]/a'$. Let $a' \uparrow a$. □

The Kolmogorov version of Chebyshev's inequality states that for a martingale with $\mathbb{E}[Z_n^2] < \infty$, $\Pr\{\max_{1\leq n\leq m} |Z_n| \geq b\} \leq \mathbb{E}[Z_m^2]/b^2$ for $m \geq 1$ and $b > 0$. It is often more useful to maximize only over the upper half of each interval and take a union over intervals of doubling length:

$$\Pr\Big\{\bigcup_{j\geq k}\Big\{\max_{2^{j-1}<n\leq 2^j} |Z_n| \geq b_j\Big\}\Big\} \leq \sum_{j=k}^{\infty} \frac{\mathbb{E}[Z_{2^j}^2]}{b_j^2}.$$

For a zero-mean random walk $S_n = X_1 + \cdots + X_n$ with $\mathbb{E}[X^2] = \sigma^2$, $\mathbb{E}[S_{2^j}^2] = 2^j\sigma^2$, and the choice $b_j = (3/2)^j$ gives

$$\Pr\Big\{\bigcup_{j\geq k}\Big\{\max_{2^{j-1}<n\leq 2^j} |S_n| \geq \Big(\frac{3}{2}\Big)^j\Big\}\Big\} \leq \sum_{j=k}^{\infty}\Big(\frac{8}{9}\Big)^j\sigma^2 = \Big(\frac{8}{9}\Big)^k 9\sigma^2. \tag{25.1}$$

Theorem 25.2 [Strong law of large numbers]. *Let $\{X_n; n \geq 1\}$ be IID with $\overline{X} = 0$ and $\mathbb{E}[X^2] = \sigma^2 < \infty$, and $S_n = X_1 + \cdots + X_n$. Then* $\Pr\{\omega : \lim_{n\to\infty} S_n/n = 0\} = 1$.

Proof. For $2^{j-1} < n \leq 2^j$, $|S_n|/n \geq 2(3/4)^j$ implies $|S_n| \geq 2(3/4)^j 2^{j-1} = (3/2)^j$. So by (25.1),

$$\Pr\Big\{\bigcup_{j\geq k}\Big\{\max_{2^{j-1}<n\leq 2^j} \frac{|S_n|}{n} \geq 2\Big(\frac{3}{4}\Big)^j\Big\}\Big\} \leq \Big(\frac{8}{9}\Big)^k 9\sigma^2.$$

A sample sequence outside this union has $|S_n|/n < 2(3/4)^j$ for all n in $(2^{j-1}, 2^j]$ and all $j \geq k$, hence $\lim_n |S_n|/n = 0$. Thus $\Pr\{\omega : \lim_n |S_n|/n = 0\} \geq 1 - (8/9)^k 9\sigma^2$ for every k. □

The argument uses only the martingale property and the growth of $\mathbb{E}[S_n^2]$, so it applies to other martingales as well.

25.2 Markov chains, countable or finite state

Definition 25.3 [First passage time; recurrence]. The first passage time T_{ij} from i to j is the smallest $n \geq 1$ at which $X_n = j$, given $X_0 = i$. It is a possibly defective rv with PMF $f_{ij}(n)$ and distribution function $F_{ij}(n)$:

$$f_{ij}(1) = P_{ij}, \qquad f_{ij}(n) = \sum_{k\neq j} P_{ik} f_{kj}(n-1), \qquad F_{ij}(n) = \sum_{m=1}^{n} f_{ij}(m), \qquad n > 1.$$

State j is recurrent if T_{jj} is nondefective and transient otherwise. A recurrent j is positive recurrent if $\mathbb{E}[T_{jj}] < \infty$ and null recurrent otherwise.

For each recurrent j, the visits to j starting in j form an integer renewal counting process $\{N_{jj}(t); t > 0\}$ with inter-renewal distribution F_{jj}, and visits to j starting in i form a delayed renewal counting process $\{N_{ij}(t); t > 0\}$. All states in a class are positive recurrent, or all are null recurrent, or all are transient. A chain is irreducible if all pairs of states communicate. The name reflects that getting from transient classes into an irreducible class is largely separable from the analysis of that class, which then becomes the entire chain; an irreducible class can be positive recurrent, null recurrent, or transient.

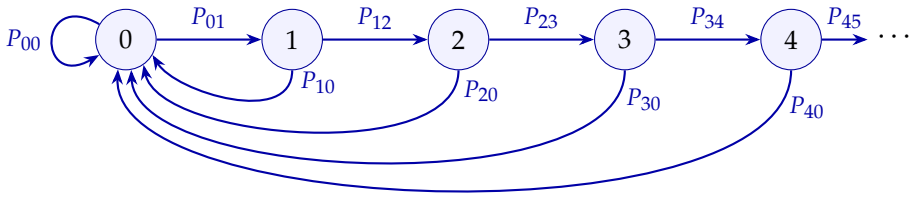
Theorem 25.4 [Steady state of an irreducible chain]. *For an irreducible Markov chain, if the steady-state*

equations

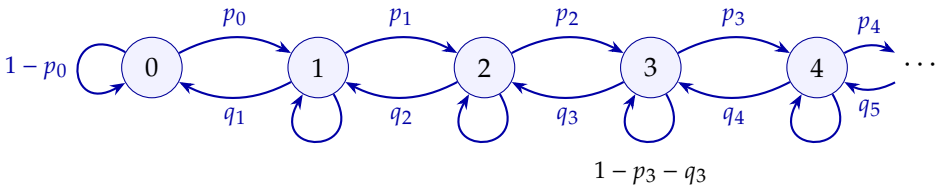
$$\pi_j = \sum_i \pi_i P_{ij}, \qquad \pi_j \geq 0 \text{ for all } j, \qquad \sum_j \pi_j = 1$$

have a solution, then it is unique, $\pi_j = 1/\overline{T}_{jj} > 0$ for all j, and the states are positive recurrent. Conversely, if the states are positive recurrent, the equations have a solution. Under positive recurrence, $\lim_{n\to\infty} N_{ij}(n)/n = \pi_j$ with probability 1.

This is an infinite set of equations, so it is not necessarily solvable by computer. Two chains give useful examples (Figure 37).



Age of a renewal process



Birth-death chain

Figure 37: Top: the age of an integer renewal process with inter-renewal time Z; from age n the chain moves to $n+1$ or returns to 0. Bottom: a birth-death chain, with self-loop probability $1 - p_i - q_i$ at state $i \geq 1$ (labeled at state 3).

For the age chain, $P_{n,n+1} = \Pr\{Z > n+1\}/\Pr\{Z > n\}$, so

$$\pi_n = \pi_0 P_{01} P_{12} \cdots P_{n-1,n} = \pi_0 \Pr\{Z > n\}, \qquad 1 = \sum_i \pi_i = \pi_0 \sum_{n=0}^{\infty} \Pr\{Z > n\} = \pi_0 \overline{Z}.$$

If $\overline{Z} = \infty$ while Z is finite, there is no solution and the recurrent chain is null recurrent, which makes it a convenient source of examples. The birth-death chain is another very useful model for questions of recurrence and for sampled-time queueing systems. Its steady-state equations are almost trivial:

$$\pi_i p_i = \pi_{i+1} q_{i+1}, \qquad \frac{\pi_{i+1}}{\pi_i} = \rho_i, \qquad \rho_i = \frac{p_i}{q_{i+1}}.$$

25.3 Countable-state Markov processes

A MARKOV PROCESS $\{X(t); t \geq 0\}$ combines a countable-state embedded Markov chain $\{X_n; n \geq 0\}$ with an exponential holding time in each state:

$$\Pr\{U_{n+1} \leq \tau \mid X_n = j, \text{past}\} = 1 - \exp(-\nu_j \tau).$$

The process is in state X_n during $[S_n, S_{n+1})$, where $S_n = U_1 + \cdots + U_n$. It is specified by the embedded transition probabilities P_{ij} and the rates ν_i; the TRANSITION RATES are $q_{ij} = \nu_i P_{ij}$. Figure 38 shows three equivalent representations of the M/M/1 queue.

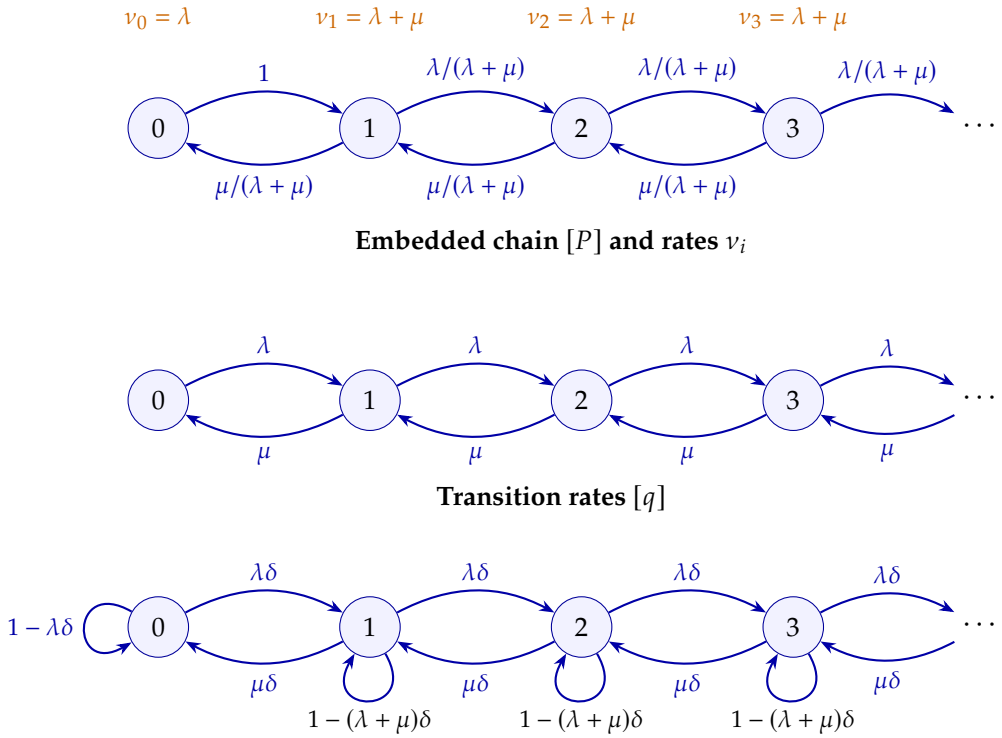


Figure 38: The M/M/1 queue with arrival rate λ and service rate μ: by its embedded chain and holding rates, by its transition rates, and in sampled time.

Let the embedded chain be positive recurrent with steady state $\{\pi_j\}$. Successive entries to j form a delayed renewal process with mean inter-renewal time $\overline{W}(j)$; give reward $R_j(t) = 1$ when $X(t) = j$. Each renewal interval earns the holding time in j, of mean $\overline{U}(j) = 1/\nu_j$, so by the renewal reward theorem the time-average probability of j is

$$p_j = \lim_{t\to\infty} \frac{1}{t} \int_0^t R_j(\tau)\, d\tau = \frac{\overline{U}(j)}{\overline{W}(j)} = \frac{1}{\nu_j \overline{W}(j)} \qquad \text{W.P.1.}$$

Assigning instead unit reward to each transition, and letting $M_i(t)$ be the number of transitions in

$(0, t]$ given $X(0) = i$, each renewal interval for j has $\overline{T}_{jj} = 1/\pi_j$ transitions on average, so

$$\overline{M} = \lim_{t\to\infty} \frac{M_i(t)}{t} = \frac{1}{\pi_j \overline{W}(j)} \quad \text{W.P.1}, \qquad \text{hence} \qquad p_j = \frac{\pi_j}{\nu_j}\overline{M}.$$

If $0 < \overline{M} < \infty$, then every $p_j > 0$ and $\sum_j p_j = 1$, and summing p_j and π_j gives

$$\overline{M} = \frac{1}{\sum_i \pi_i/\nu_i}, \qquad p_j = \frac{\pi_j/\nu_j}{\sum_i \pi_i/\nu_i}, \qquad \overline{M} = \sum_i p_i\nu_i, \qquad \pi_j = \frac{p_j\nu_j}{\sum_i p_i\nu_i}.$$

A sampled-time Markov process exists if $\nu_i \le A$ for some A and all i. Its steady-state probabilities are the time averages $\{p_i\}$, which satisfy

$$\nu_j p_j = \sum_i p_i q_{ij}, \qquad p_j \ge 0, \qquad \sum_i p_i = 1.$$

The strange cases are $\overline{M} = 0$ and $\overline{M} = \infty$. $\overline{M} = 0$ occurs for the queue with a rattled server and discouraged customers: the embedded chain is positive recurrent, but all p_j are 0 and the sampled-time chain is null recurrent. The case $\overline{M} = \infty$ is impossible for a positive-recurrent embedded chain, but is possible when the equations $\nu_j p_j = \sum_i p_i q_{ij}$ have a solution; such processes are irregular and allow infinitely many transitions in finite time.

Definition 25.5 [Reversibility]. For an embedded chain, the backward transition probabilities satisfy $\pi_i P^*_{ij} = \pi_j P_{ji}$. Moving left in time, after entering state j the exit rate is again ν_j, so $\{\pi_i\}$ and $\{\nu_i\}$ are the same in both directions. A transition from j to k in $(t, t+\delta)$ has probability $p_j q_{jk}\delta$; if q^*_{kj} is the transition rate of the left-going process, the same event has probability $p_k q^*_{kj}\delta$. Hence

$$p_j q_{jk} = p_k q^*_{kj}, \qquad q^*_{kj} = \nu_k P^*_{kj}.$$

A Markov process is reversible if $q^*_{ij} = q_{ij}$ for all i, j.

Assuming positive recurrence and $\sum_i \pi_i/\nu_i < \infty$, the process is reversible if and only if its embedded chain is.

Theorem 25.6 [Guessing theorem]. *Suppose a Markov process is irreducible, and $\{p_i\}$ is a set of probabilities with $p_i q_{ij} = p_j q_{ji}$ for all i, j and $\sum_i p_i\nu_i < \infty$. Then $p_i > 0$ for all i, p_i is the steady-state time-average probability of state i, the process is reversible, and the embedded chain is positive recurrent.*

In particular, all birth-death processes with $\sum_j p_j\nu_j < \infty$ are reversible.

25.4 Random walks

A random walk is the sequence $S_n = X_1 + \cdots + X_n$ of sums of IID rv's. The questions are exponential bounds on S_n for large n (Chernoff bounds) and threshold crossings: for thresholds $\alpha > 0$ and $\beta < 0$, the stopping time at which S_n first crosses one of them, the probability of crossing each, and the distribution of the overshoot. The Chernoff bound with $\gamma(r) = \ln \mathbb{E}[e^{rX}]$ gives $\Pr\{S_n \geq na\} \leq \min_r \exp(n[\gamma(r) - ra])$, and for fixed α, $\Pr\{S_n \geq \alpha\} \leq \min_{0<r<r_+} \exp(-\alpha[r - n\gamma(r)/\alpha])$, whose best exponent over n is r^* (Lectures 21 and 22).

Wald's identity $\mathbb{E}[\exp(rS_J - J\gamma(r))] = 1$ for $r \in (r_-, r_+)$ holds for the two-threshold rule and, more generally, whenever the stopping time is a rv under both the given and the tilted probability measure: the proof just sums the probabilities of the stopping nodes under both measures. Its consequences include

$$\Pr\{S_J \geq \alpha\} \leq \exp(-\alpha r^*) \quad (\overline{X} < 0,\ \gamma(r^*) = 0), \qquad \mathbb{E}[S_J] = \mathbb{E}[J]\overline{X}, \qquad \mathbb{E}[S_J^2] = \mathbb{E}[J]\sigma_X^2 \quad \text{if } \overline{X} = 0.$$

These bounds are all exponentially tight: if any $\epsilon > 0$ is added to such an exponent, the upper bound becomes a lower bound for sufficiently large α with α/n fixed. The slack comes partly from the overshoot and partly from the lower threshold. The lower threshold is unimportant when both thresholds are far from 0. The overshoot resembles the residual life of a renewal process; it does not exist for simple random walks and is easy to calculate when the positive part of the density of X is exponential.

In hypothesis testing with $Z_i = \ln[f_{Y|H}(Y_i \mid 0)/f_{Y|H}(Y_i \mid 1)]$ and $S_n = Z_1 + \cdots + Z_n$, the MAP rule selects $\hat{h} = 0$ when $S_n > \ln(p_1/p_0)$. Conditional on $H = 1$, $\{S_n\}$ is a random walk with negative drift, $\gamma_1(r) = \ln \mathbb{E}[e^{rZ} \mid H = 1]$, and $r^* = 1$. With thresholds $\alpha > 0$ and $\beta < 0$, $\Pr\{S_J \geq \alpha\} \leq e^{-\alpha}$; this is the probability of error given $H = 1$, and is essentially the probability of error conditional on any sample path crossing α. Conditional on $H = 0$, the walk has positive drift and $\Pr\{S_J \leq \beta\} \leq e^{\beta}$, which also essentially holds for each sample path.

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